OPERATIONAL REQUIREMENTS DOCUMENT

FOR

DEPARTMENT OF DEFENSE (DOD) TELEPORT

Generation Two Update
9 July 04

Current as of 29 May 2007

THIS PAGE IS UNCLASSIFIED

OPERATIONAL REQUIREMENTS DOCUMENT, CHANGE 2
FOR
DEPARTMENT OF DEFENSE (DOD) TELEPORT, GENERATION 2
ACAT I-AM

Serial Number (*): 726-6F-07

SUBMITTED:

(N6F) (Program Sponsor) — 5 Jun 07 (DATE)

ENDORSED and FORWARDED:

(N81D) — 1 Aug 07 (DATE)

(N8F) — 7 Aug 07 (DATE)

([illegible]) — 30 July 07 (DATE)

(N8) — 8/10/07 (DATE)

REVIEWED: (JROC INTEREST)

For
(CFFC N00) — 6/28/07 (DATE)

APPROVED and VALIDATED: (JROC INTEREST)

Patrick M. Walsh
(VCNO) — 4 Sept 07 (DATE)

(CNO) — 9.9.07 (DATE)

N/A
(JROC) — N/A (DATE)

UNCLASSIFIED

THE JOINT STAFF
WASHINGTON, DC

Reply ZIP Code:
20318-6000

29 May 2007

MEMORANDUM FOR DIRECTOR, WARFARE INTEGRATION (N6F)
2000 Navy Pentagon
Washington D.C. 20350-2000

Subject: Withdrawal of USTRANSCOM's DOD Teleport UHF-DSN Requirement

1. At the request of USTRANSCOM[1], the Joint Staff J-6 has validated the withdrawal of USTRANSCOM's legacy UHF-to-DSN interface requirement from the Generation 1 and 2 Operational Requirements Document (ORD).

2. Per the official Joint Staff Action Process, all Services, combatant commands, and agencies have agreed that this requirement should be removed from the ORD by the Navy (DOD Teleport Executive Agent for documentation) as a non-KPP requirement.[2] With the deletion of this requirement, JITC should remove UHF-to-DSN testing requirements accordingly.

3. If you have any questions regarding this memorandum or require further information, feel free to contact my POC for this issue, Lt Col Rick Folks at commercial (703) 697-8073 or DSN 227-8073.

E-Signed by Executive Automatio
VERIFY authenticity with ApproveIt

DENNIS C. MORAN
Major General, USA
Vice Director for Command, Control, Communications and Computer Systems

References:
1. USTRANSCOM/TJ6 Draft Memo, "Withdrawal of USTRANSCOM DOD Teleport Requirements"
2. Joint Requirements Oversight Council Memorandum 116-04

UNCLASSIFIED

THE JOINT STAFF
WASHINGTON, D.C. 20318-8000

JOINT REQUIREMENTS
OVERSIGHT COUNCIL

JROCM 091-05
2 May 2005

MEMORANDUM FOR: Assistant Secretary of Defense for Networks and Information Integration
Vice Chief of Naval Operations

Subject: DoD Teleport Generation 2 Operational Requirements Document (ORD)

The Joint Requirements Oversight Council (JROC) approved changing the DoD Teleport ORD to shift the Generation 1 Initial Operational Capability 2 (IOC 2) timeline for providing ultra high frequency (UHF) to Defense Information System Network (DISN) services over UHF to FY 2002 – 2007. The JROC also approved shifting the Generation 2 timeline to FY 2004 - 2008. The Navy is designated lead Service. The ORD is assigned the Joint Potential Designator of "JROC Interest".

Peter Pace

PETER PACE
General, United States Marine Corps
Vice Chairman
of the Joint Chiefs of Staff

Copy to:
- Commander, Joint Forces Command
- Commander, Strategic Command
- Vice Chief of Staff, US Army
- Vice Chief of Staff, US Air Force
- Assistant Commandant of the Marine Corps
- Director for Command, Control, Communications, and Computer Systems, Joint Staff

THE JOINT STAFF
WASHINGTON, DC

Reply ZIP Code:
20318-6000

24 January 2005

MEMORANDUM FOR USN

Subject: J6 Interoperability Certification of the Net Readiness-Key Performance Parameter (NR-KPP) for the Department of Defense (DOD) Teleport Generation Two Update, Operational Requirements Document (ORD), 5 January 2005

1. In accordance with the references (Enclosure A), the Joint Staff J-6 and appropriate Department of Defense agencies (Enclosure B), conducted an assessment of the Net Readiness-Key Performance Parameter (NR-KPP) for the Department of Defense (DOD) Teleport Generation Two Update ORD, 5 January 2005. Based on this assessment, interoperability certification is granted.

2. To ensure compliance with interoperability standards and cited directives, a plan for interoperability test certification must be coordinated with the DISA Joint Interoperability Test Command (JITC).

3. The Joint Staff J-6 must review subsequent modifications to the program that change the interoperability key performance parameters.

4. For coordination, contact LtCol Weydan Flax, (703) 695-0314/DSN 225-0314, weydan.flax@js.pentagon.smil.mil, CDR Charles C. Moore II, (703) 697-4232/DSN 227-4232, charles.moore@js.pentagon.smil.mil or CDR Ron Greiff, (703) 614-5290/DSN 224-5290, ronald.greiff@js.pentagon.smil.mil .

FOR

ROBERT A. GEARHART, JR.
Colonel, USMC
Chief, Integration and
Information Assurance
Division, J6I

Enclosures

THE JOINT STAFF
WASHINGTON, D.C. 20318-8000

JOINT REQUIREMENTS OVERSIGHT COUNCIL

JROCM 116-04
9 July 2004

MEMORANDUM FOR: Assistant Secretary of Defense for Networks and Information Integration
Vice Chief of Naval Operations
Director, Defense Information Systems Agency
Director for Command, Control, Communications, and Computers, Joint Staff

Subject: DOD Teleport Generation Two Operational Requirements Document (ORD)

1. The Joint Requirements Oversight Council (JROC) has approved the DOD Teleport Generation Two ORD and validated the enclosed Key Performance Parameters (KPPs). The DOD Teleports will provide the Warfighter with Netcentric Internet Protocol (IP) access to the Global Information Grid. The JROC approved shifting the Generation Two timeline to FY 04-07 and shifting the Generation Three timeline to FY 08-12. ORD approval authority for non-KPP changes is delegated to the Navy.

2. In order to satisfy the capability-based transformation to Netcentric IP from legacy transmission systems, the JROC requests the DOD Teleport Program Manager to develop an implementation plan for providing threshold IP capability based upon the requirements specified in the FY06-11 Joint Programming Guidance.

3. The Joint Staff will lead a study of emerging network centric Teleport IP capacity requirements, including Intelligence, Surveillance, and Reconnaissance, for Generation Two and update the Joint Staff Satellite Throughput and DISN Services Network Requirements Memorandum to include network centric IP requirements. This study will determine the actual mix of IP and legacy capacity required during Generation Two. The Navy will return to the JROC in July 2004 to present the findings of the study and recommend changes, if any, to the Teleport ORD. In conjunction with these findings, the Navy will also present the Net Ready KPP and applicable component compliance.

PETER PACE
General, United States Marine Corps
Vice Chairman
of the Joint Chiefs of Staff

Enclosure

THE JOINT STAFF
WASHINGTON, D.C. 20318-8000

JOINT REQUIREMENTS
OVERSIGHT COUNCIL

JROCM 140-00
28 Aug 00

MEMORANDUM FOR THE UNDER SECRETARY OF DEFENSE FOR ACQUISITION, TECHNOLOGY, AND LOGISTICS

SUBJECT: Department of Defense (DoD) Teleport Operational Requirements Document (ORD)

The Joint Requirements Oversight Council (JROC) reviewed and approved the DoD Teleport ORD and validated the key performance parameters (KPPs). The JROC considers the KPPs as essential to meet the DoD Teleport mission need. The JROC retained ORD approval authority, and approved the Analysis of Alternatives Architecture Alternative #5.

RICHARD B. MYERS
General, USAF
JROC Chairman

Enclosure

cc: ASD(C3I)
DISA

TABLE OF CONTENTS

LIST OF TABLES

Summary of Changes

The DOD Teleport ORD was approved by the Joint Requirements Oversight Council (JROC) on 31 Jul 00. As a multi-phased program with three generations of implementation, the ORD for each generation will be updated to capture new or evolving requirements and technology advancements not previously incorporated. The changes to this document from the previously approved 31 July 2000 ORD are based on a Generation Two review of requirements and were approved by the JROC 9 July 2004 (JROCM 116-04). Specifically, the following new requirements were added to the Systems Requirements, Table 4-2:

- *Capacity – The DOD Teleport system must have the constant capacity to provide DISN, legacy Tactical C4I services and SATCOM throughput in any given geographical region*
    - *Provide 100% of the projected 2006 required DISN services for one MCO (Gen Two, Threshold KPP)*
- *Capacity – The DOD Teleport system must have the constant capacity to provide DISN, legacy Tactical C4I services and SATCOM throughput in any given geographical region*
    - *Provide 100% of the projected 2006 required SATCOM throughput for one MCO (Gen Two, Threshold KPP)*
- *Interoperability – The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DISN and Legacy Tactical C4I*
    - *Access to DISN POPs for on demand transport and COI services (Gen Two, Threshold KPP)*
- *Interoperability – The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DISN and Legacy Tactical C4I*
    - *Access to Legacy Tactical C4I systems (Gen Two, Objective KPP)*
- *Interoperability – The DOD Teleport must provide common access into satellite system and be able to cross-band between them*
    - *Access to EHF (MDR) (Gen Two, Threshold* KPP)
- *Interoperability – Teleports shall support the migration to IP services*
    - *Teleport shall be interoperable with IP routing requirements for tactical users in accordance with a DOD-wide standard IP routing architecture (Gen Three, Threshold KPP)*
- *Interoperability – Teleports shall support an architecture that uses policy – based (Quality of Service) IP routing and dynamically allocated satellite bandwidth to provide a responsive network centric capability.*
    - *Teleport shall provide the Warfighter access to network centric IP architecture (Gen Two, Threshold KPP)*
    - *Objective = Threshold*
- *Interoperability – Teleports shall support the migration to IP services*

  - *Teleport architecture shall conform to the Government Reference Architecture (GRA) for Transformational Communications (Gen Three, Objective KPP)*
- *Interoperability – Provide users the ability to access commercial phone services from any Teleport*
  - *Teleports shall provide capability to access Public Switched Network via DSN (Gen Three, Threshold non-KPP)*
- *Interoperability – The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DCGS services through standard DISN networks (e.g. JWICS, SIPR, etc) or through user provided and funded alternate connectivity*
  - *Teleport shall provide the capability to receive ISR data from appropriate ISR airborne assets outfitted with military or commercial satellite relay capability (e.g. Predator, Global Hawk, F/A-22) and appropriate forward deployed ISR ground stations that have or use commercial and/or military satellite reachback (e.g. Predator GCS, Global Hawk MCE). These ISR airborne assets and/or ground stations may transmit simultaneously from any operational theater. (Gen Three, Threshold non-KPP)*
- *Interoperability – Teleports will be interoperable with MUOS*
  - *Must interoperate with MUOS waveform when developed (Gen Three, Threshold non-KPP)*
- *Protection – DOD Teleport must support Information Assurance*
  - *System shall meet and maintain minimum IA Defense in Depth Standards, including certification and accreditation in accordance with the DITSCAP process CJCSI 6510.01C and DoDI 5200.40 (Gen Two, Threshold KPP)*
- *Quality of Service – The system shall maintain and guarantee during transport the integrity of all information elements exchanged throughout the GIG to enable user confidence; information integrity.*
  - *99.99% (Gen Two, Threshold non-KPP); 99.999% (Gen Two, Objective non-KPP)*
- *Quality of Service – All transport elements (e.g. switches, routers, etc.) shall be capable of providing status changes to network management devices by means of an automated capability in near real time.*
  - *99% (Gen Two, Threshold non-KPP); 99.9% (Gen Two, Objective non-KPP)*
- *Quality of Service - The system shall have the NM capability to obtain status of networks and associated assets in near real time*
  - *99% (Gen Two, Threshold non-KPP); 99.9% (Gen Two, Objective non-KPP)*
- *Technology Insertion – DOD Teleports must provide the warfighter the capability to terminate VOIP services*
  - *The Teleport must be capable of terminating VOIP services and provide an interface into DRSN and DSN (Gen Three, Threshold non-KPP)*

The following Systems Requirements in Table 4-2 were previously approved by the JROC in July 2000 but were changed during the Generation Two review. Note, only those requirements that changed Generational implementation or requirement level (i.e. threshold or objective) are reflected below; acronym updates or requirement clarifications are not included.

- *Coverage- DOD Teleport will provide warfighter coverage on a global, regional, interregional and theater basis*
    - *Jul 00 – Incorporate planned polar SATCOM capabilities (Gen Two and Three, Objective non-KPP)*
    - *May 03 - Incorporate planned polar SATCOM capabilities (Gen Two and Three, Threshold non-KPP)*

- *Interoperability- The DOD Teleport must provide common access into satellite system and be able to cross-band between them.*
    - *Jul 00 – Access to Ka Commercial Band (Gen Two, Threshold KPP)*
    - *May 03 – Access to Ka Commercial Band (Gen Three, Threshold KPP)*

- *Interoperability-The DOD Teleport must be capable of providing hub connectivity.*
    - *Jul 00 – Access to UHF and EHF (LDR) (Gen One IOC2, Threshold KPP)*
    May 03 – Requirement removed in total

In addition to the aforementioned changes to Table 4-2, the timeframes of the Teleport generations were updated to reflect acquisition decisions and planning considerations. Specifically, within Generation One, an IOC 3 and IOC 4 were added to accommodate EHF (LDR/MDR) and the start of military Ka band implementation. Generation Two was extended to FY07 to complete military Ka installations. Generation Three was changed to FY08-12 from FY06-10. This change will allow for a more comprehensive review of the Generation Three requirements to ensure alignment with Transformation Initiatives.

29 May 2007 Change

Per Vice Director for Command, Control, Communications, and Computer Systems MEMORANDUM FOR DIRECTOR, WARFARE INTEGRATION (N6F), dated 29 May 2007, the USTRANSCOM non-KPP requirement for legacy UHF-to-DSN requirement has been withdrawn from this ORD. Table 4-4 in this document has been updated to reflect this change.

OPERATIONAL REQUIREMENTS DOCUMENT (ORD)

FOR

DEPARTMENT OF DEFENSE (DOD) TELEPORT
ACAT IAM

Updated for Generation Two

1. General Description of Operational Capability

The Joint Staff, Combatant Commands, and the Services have identified, validated and approved an operational need for the rapid establishment of command, control, communications, computers and intelligence (C4I) connectivity in support of worldwide operations. As a result, the Defense Information System Network (DISN) was established to provide integrated connectivity across all levels of force structure, from Combatant Command/Service/Agency (CC/S/A) headquarters to deployed tactical elements, 24 hours per day, 365 days per year. This document articulates the Warfighter's growing requirements for emerging deployed military and commercial satellite communications (SATCOM) systems' interfaces into key DISN and legacy C4I services and incorporates the requirements identified in the Defense Planning Guidance (DPG) Fiscal Year (FY) 2004-2009 and Joint Vision 2020.

a. Mission Area

The 30 March 1995 Mission Need Statement (MNS) for the DISN established the need for a system to support the military's move into the 21st Century information age of dynamic bandwidth allocation. DISN will eventually replace individual legacy communications systems with a seamless information transport capability, which will keep pace with evolving technology and meet the changing C4I demands of the Warfighter. The DISN MNS cites the following requirements in the C4I and Space-based Systems section of the DPG FY 1996 - 2001:

- "Integrated connectivity to all theater and tactical elements through modernized, jam-resistant telecommunications network support."
- "Joint and combined interoperability in all mission areas to facilitate Joint and Combined Task Force (JTF/CTF) operations."
- "Improved systems to integrate strategic, theater, and tactical intelligence with joint/combined forces."

The DISN MNS further states; “to effectively and economically support these requirements, a common user transmission infrastructure with integrated wide area networking interfaces to local area networks, integrated networks, and system management capabilities, and value-added services transport capability must be fielded”. A vital part of the value-added transport capability is the ability to extend the DISN Long-Haul Block connectivity to the deployed forces. Although the Standardized Tactical Entry Point (STEP) partially satisfies this requirement, the throughput requirements of the Warfighter, especially by the year 2010, significantly exceed current capabilities. The shortfall is being mitigated via costly, often duplicative, interfaces between disjointed, disparate, and stovepipe networks that inefficiently consume limited bandwidth and shrinking fiscal resources.

The DISN infrastructure, as defined by the DISN Capstone Requirements Document (CRD), is divided into three major infrastructure blocks:

- “The sustaining base (i.e., base/post/camp/station) C4I infrastructure (to include legacy systems) that will interface with the long-haul network in order to support the deployed Warfighter.”
- “The long-haul telecommunications infrastructure between the fixed environment and the deployed (JTF/CTF) Warfighter.”
- “The deployed Warfighter and associated Combatant Commander telecommunications infrastructures that support the JTF/CTF.”

A goal of the Joint Staff’s DISN Deployed Strategy is for tactical deployed forces to use the DISN Long-Haul Block and its services to connect to the DISN Sustaining Base Block.

The current pre-positioned SATCOM connection into the DISN Long Haul Block is over the Defense Satellite Communications System (DSCS) Super High Frequency (SHF) system through the STEP sites. The STEP capacity is limited in the initial phases of a typical deployment to a surge capacity of about 11 megabits per second (Mbps) at a single STEP site. The Joint Chiefs of Staff (JCS) validated STEP Concept of Operations (CONOPS) dated 12 May 98 requires that a deployed JTF or Combined Task Force (CTF) be supported by a minimum of four single STEP sites for a total of about 45 Mbps. In recent years, Commercial SATCOM has provided a viable alternative to mitigate current bandwidth throughput limitations of DSCS. Beginning in 2005, DOD will begin launching Wideband Gapfiller Satellites that will provide significantly more bandwidth than DSCS. Additionally, DOD envisions removing bandwidth as a constraint with the Transformation Communications Architecture with IOC in FY09. The Integrated Communications Database (ICDB) documents all SATCOM requirements and is validated by the Joint SATCOM Panel. The ICDB is validated over a three-year horizon and can be used to accurately model tactical reachback requirements to support a variety of operational scenarios including Peace, Small Scale Contingency (SSC) and Major Combat Operations (MCO). In developing this ORD, the ICDB and Emerging Requirements Database (ERDB) were used to estimate requirements. The ERDB is approved for planning purposes by submitting Combatant

Commanders/Services/Agencies requirements for communications capability over a ten-year horizon. Modeling using the ICDB and ERDB were completed to show the throughput requirements for Peace, SSC, SSC + MCO, and two overlapping wars scenario and the results are reflected in DISA (D216) Teleport Requirements Development and Analysis study (Draft) of June 2000 based on the ERDB dated 01 April 2000. Per CJCSI 6250.01A, the ICDB and ERDB were merged and are now referred to as the Satellite Data Base (SDB) and Current Requirements (meaning present plus two years out) and Future Requirements (meaning beyond 2 years out). The SDB will be used as the primary tool for determining the objective and threshold requirements for sizing the Teleport system during Generation Two and future implementation generations. On an annual basis, the Services and Combatant Commanders will participate with the Joint Staff in the review of near term and future validated requirements so that this tool remains relevant for planning Teleport growth. Ultimately these inputs will be incorporated into future JROC validated updates of the SATCOM CRD.

As addressed in the SATCOM CRD and the Global Information Grid CRD, Warfighters need a more robust system than currently provided by STEP. Joint Task Forces (JTF) and deployed forces will rely on a variety of SATCOM capabilities – encompassing both military systems and commercial services – to satisfy tomorrow's needs. The diversity of DOD's user populations as well as their information needs demands a flexible system capable of integrating the variety of SATCOM systems used to fight or keep peace. The Unified Commands, Services, and Agencies have a wide variety of mobile Warfighter and support platforms operating in an array of environments, globally distributed as well as concentrated into theaters of operation, and they need a system that can provide integration. This system must account for many and varied needs for information types with differing volumes (usually large) and perishability (often short). At the same time, Warfighters must be relieved of the burden to carry around large amounts of communications equipment and bulky antennas to have joint communications. The DOD Teleport System is part of the solution as it will improve interoperability of joint communications systems and provide dynamic reconfiguration to meet a Commander's changing needs. As such, the DOD Teleport System is an essential combat multiplier needed in today's rapid deployment environment and is a logical progression from the original STEP design.

The Teleport system must be able to support the Joint Mission Areas (JMA) in accordance with Joint Pub 1-02, which states "JMA8 – Command and Control. The exercise of authority and direction by a properly designated commander over assigned and attached forces in the accomplishment of the mission. Command and control functions are performed through and arrangement of personnel, equipment, communications, facilities, and procedures employed by a commander in planning, directing, coordinating, and controlling forces and operations in the accomplishment of the mission."

The DOD Teleport System will provide deployed forces a point of presence into the DISN Long-Haul and legacy Tactical C4I Services & Systems (as defined in

Appendix B) with high throughput, multi-band and multi-mode telecommunications. The DOD Teleport System shall provide integration capabilities, contingency capacity, and interfaces (satellite and terrestrial) to legacy Tactical C4I Services and Systems and the DISN in a seamless, interoperable and economical manner. Using technology insertion, the DOD Teleport System can be quickly and cheaply improved to support the Warfighter as communications technology advances.

Supporting documents for the DOD Teleport System are listed in Appendix A.

b. <u>Type of System Proposed.</u>

The proposed DOD Teleport System consists of the migration of a collection of existing telecommunications capabilities and distribution points {e.g. Naval Computer and Telecommunications Area Master Station (NCTAMS), STEP sites, and non-STEP DII Gateways}. This will provide deployed forces with sufficient interfaces for multi-band and multimedia connectivity from deployed locations anywhere in the world to online DISN Service Delivery Nodes (SDN) and legacy Tactical C4I Systems. This system shall facilitate the interoperability between multiple SATCOM systems and deployed tactical networks that allows the user a seamless interface into the DISN and Legacy C4I systems.

The objective DOD Teleport facilities will integrate X-, C-, Ku-, commercial and military Ka-band satellite capabilities, Ultra High Frequency (UHF), and Extremely High Frequency (EHF) {Low Data Rate (LDR), Medium Data Rate (MDR)}, (to include applicable EHF mission planning tools) Extended Data Rate (XDR), L-band, and High Frequency (HF) capabilities to provide connectivity for deployed tactical communications systems. They shall provide worldwide, integrated communications nodes that also have the ability to modularly insert emerging systems adopted by DOD to support deployed forces and JTFs.

Where available, the commercial Internet increasingly provides warfighters a low-cost alternative transport mechanism for extending DISN services. While not providing an assured quality of service, it does provide a low-cost, easily accessible, redundant capability that should be exploited where practical and appropriate to the mission. As such, the Teleport should incorporate the ability to exploit the Internet to transport DISN services. The primary DISN services initially accessible via this method will be NIPRNET and SIPRNET but other services such as Allied and Coalition Wide Area Network access, and VTC may become feasible. Tactical forces accessing the DISN via this method need to acknowledge DISA will not be able to guarantee the same Quality of Service (QOS) as with traditional DISN connections. In addition, there is an increased risk of service interruption due to Internet Service Provider (ISP) or host government interference. These additional risks must be balanced against mission requirements and constraints.

Interfaces between the DOD Teleports and the Global Network Operations and Security Center (GNOSC), the Regional Network Operations and Security Centers (RNOSC), the Global SATCOM Support Center (GSSC), Regional SATCOM Support Centers (RSSC), Theater Communications Control Centers (TCCC), and USSTRATCOM component C2 centers will be part of the functional design. The key element is network planning, management, and control of all sub-systems for either a co-located or a distributed DOD Teleport System site.

DOD Teleport Systems shall provide standard interfaces to receive and transmit X, C, Ku, commercial and military Ka, EHF, UHF, HF, and L bands. They shall provide the capability to connect incoming voice, data, and video information with the appropriate legacy tactical C4I System and DISN SDN. They shall provide the capability to switch information from one band or medium to another band or medium with no loss of data integrity, supporting the ability to integrate multimedia, multi-band information links into standard interfaces. A DOD Teleport shall provide the capability to connect deployed tactical terminals in the theater with other deployed tactical terminals in the same theater, and will include a capability to provide seamless interoperability with the DISN backbone. DOD Teleport should have the capability to separate bundled unclassified and encrypted classified information {up to and including Top Secret/Sensitive Compartmented Information (TS/SCI)} for routing. No information above the level of SECRET will be decrypted and processed in the clear within the DOD Teleport facility. There is no requirement for High-Altitude Electromagnetic Pulse (HEMP)-protection in DOD Teleport facilities. However, DOD Teleports will not degrade the HEMP characteristics of HEMP-protected terminals installed at Teleport facilities.

DOD Teleports must be located in such a manner to allow the Warfighter at anyplace within 65°N (extended to include available polar SATCOM capabilities) and 65°S to communicate with at least two (2) DOD Teleports via Warfighter satellite system(s) across the communications satellite spectrum, provided that satellite coverage exists for that area. This capability will ensure that JTFs and deployed Warfighters are always supportable throughout the satellite frequency spectrum from any operational mission site. Analysis must be conducted to finalize the number of sites required to support Warfighter connectivity and provide reasonable redundancy. The Teleport system will be reasonably redundant in the event any one Teleport site is degraded. The DOD Teleport System, in conjunction with the DISN Long Haul, shall provide Combatant Commander and deployed forces, immediate, assured, and seamless access to DISN voice, data, and video SDNs from a wide complement of satellite, fiber or ground transmission media. Finally, DOD Teleports will require automated central management and control.

The current DOD Joint Technical Architecture (JTA) will be used to define the technical architecture for the DOD Teleport system. Information regarding the policy on governance and oversight for ensuring JTA compliance is contained in DOD Instruction 4630.8, Procedures for Interoperability and Supportability of Information Technology

(IT) and National Security Systems (NSS) and CJCSI 6212.01, Interoperability and Supportability of National Security Systems, and Information Technology Systems.

The DOD Teleport shall include time synchronization to Coordinated Universal Time (UTC) referenced to the U.S. Naval Observatory (USNO) and support legacy and planned future baseband and multiplexer communications equipment, as well as SATCOM RF equipment. It should also be able to insert emerging technology meeting Level 6 DII compliancy with the goal of achieving Level 8 as defined in the current DII Master Plan.

DOD Teleport equipment must interface with the DISN Long Haul and Deployed Block network management systems. It must also be able to interface, as appropriate, with the military and commercial satellite bandwidth management centers.

The DOD Teleport system will be the primary means to provide legacy Tactical C4I Systems and DISN services to the deployed Warfighter. Operationally, the DOD Teleport system as a whole must be able to support the deployed Warfighters' daily requirements from peacetime operations through two overlapping wars. DOD Teleports must be adequately sized to handle all the peacetime requirements plus the requirements for one SSC for the deployed Warfighter, validated in accordance with the Chairman Joint Chiefs of Staff Instruction (CJCSI) 6250.01 Satellite Data Base (SDB) process, and be able to satisfy projected 2010 MCO throughput requirements as theatre force levels increase. The DOD Teleport system should, objectively, be able to support 100 percent of the requirements associated with two overlapping wars as outlined in the National Military Strategy (NMS) and the DPG.

DOD Teleports must also be capable of providing timely support to Warfighter requirements. Access to existing legacy Tactical C4I Applications (as defined in Appendix B) and pre-positioned DISN services will be provided as identified in the implementation directive and/or termination assignment provided to the DOD Teleport facility. Access for newly defined requirements will vary based on availability of services and equipment.

The high-level functions of the DOD Teleport are shown in Figure 1-1.

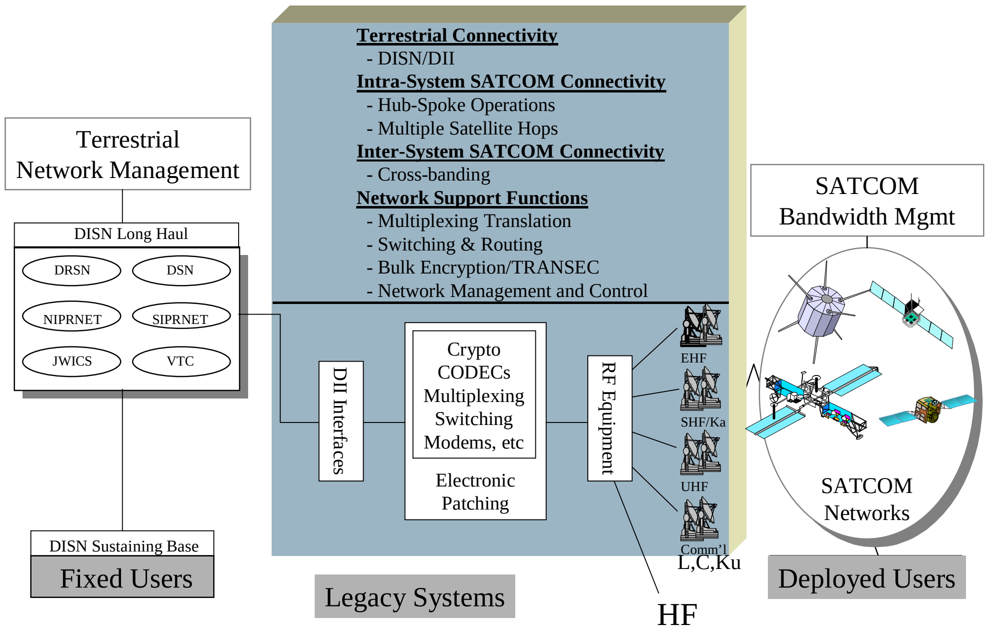


Figure 1-1 DOD Teleport Functions

Specifically, the system shall provide:

(1) DISN/GIG Enhancement Service Access. Provide on-demand, pre-positioned links to the DISN services for voice {Defense Switched Network (DSN); Defense Red Switch Network (DRSN)}; data {Unclassified-but-Sensitive IP Router Network (NIPRNET); Secret IP Router Network (SIPRNET)}; SCI voice, data and video {Joint Worldwide Intelligence Communication System (JWICS)}; and video {switched and video teleconferencing (VTC)}.

(2) Legacy Tactical C4I Services and Systems Access. Provide on-demand access to legacy Tactical C4I Systems. Refer to Appendix B (Tables B-1 & B-2).

(3) Deployed Tactical Connectivity. In a nodal configuration provide hub/spoke capability within a deployed tactical area. This connectivity provides enhanced service to deployed users, particularly rapid response units unable to transport large amounts of equipment.

(4) Relay Within the Same Band. Allow a deployed user to communicate with other users on the same satellite communication system, but on a different satellite because of satellite coverage limitations. Commonly referred to as multiple hops (M-Hop), this is an effective way to extend the range of SATCOM links that do not have a satellite cross-link capability.

(5) Cross-banding. Provide satellite communication connectivity between deployed forces using different satellite systems, e.g., SHF to UHF, EHF to SHF, MILSATCOM to Commercial systems, etc. Cross-banding shall be accomplished within the DOD Teleports at the baseband and at the digital output of the modem. Cross-banding is both an intra-theater as well as an inter-theater requirement.

(6) Multiplexer and Baseband Translation. Multiplexer translation and compatible baseband translation (e.g. Coder-Decoders (CODECs) and signaling conversion) should be implemented where practical.

(7) Switching and Routing. DOD Teleports shall support automated routing of all SATCOM accesses to the DISN Long-Haul Block or intra-theater, as required.

(8) Bulk Encryption/Transmission Security (TRANSEC). Receive and pass-through encrypted information as a normal function. It shall also separate bundled encrypted information in preparation for transmittal to and decryption by the addressed customer. All satellite links/trunks are required to be bulk encrypted, to include transmissions over commercial satellites; therefore, all military and commercial satellite trunks traversing a DOD Teleport shall be bulk encrypted.

(9) Network Management and Control. A central facility or facilities, in accordance with CJCSI 6250.01A, should be able to monitor and analyze all systems designated as part of a DOD Teleport site, to include all planning and provisioning aspects. Approved Joint Network Management systems should be able to monitor and analyze all systems designated as part of a DOD Teleport site. For definitions of control, see CJCSI 6250.01A.

(10) Intelligence, Surveillance, and Reconnaissance (ISR) Satellite Relay Services. Teleport shall have the capability to receive disseminated ISR data streams via satellite relay (such as Predator over GBS) and to receive information from forward deployed ISR ground stations that have or use commercial or military satellite relay for dissemination. ISR will not rely upon DOD Teleport for primary reachback due to Teleport's bandwidth limitations. However, ISR dissemination via DISN will permit Theater Component Commands to use Teleport for receiving ISR data. ISR data received at DOD Teleport sites will be disseminated via DISN to appropriate end-user destinations. Using commands may fund alternative connectivity (e.g. leased line services) from a DOD Teleport to end-user locations if DISN services prove unsatisfactory. Integration and procurement of any unique equipment in support of ISR terminations at a Teleport, i.e. high capacity modems and converters, will be the responsibility of the user.

(11) Support to United States Coast Guard (USCG). Provide USCG access to DOD Teleport via NCTAMS. Specific requirements are as identified in the SDB.

(12) Allied interoperability. The best established legacy Allied WAN is the NATO WAN, which supports NATO activities including exercises, operations, and contingency planning; however, other Allied and Coalition networks, including the AUSCANNZUKUS Allied WAN, are presently planned or being developed. Allied access will only be as negotiated in accordance with CJCSI 6740.01A.

(13) Homeland Security (HLS) and Homeland Defense (HLD). NORAD/USNORTHCOM mission requires interagency coordination. These agencies are appropriately equipped and approved by Combatant Commander to access Teleport sites. Current and evolving HLS and HLD missions will be documented in the SDB.

c. Operational and Support Concepts.

(1) The Teleport system will be composed of those sites that directly support tactical user requirements, (i.e. earth terminal locations and technical control facilities). The Teleport satellite communications (SATCOM) resources will be managed in accordance with CJCSI 6250.01A, Satellite Communications, dated 10 December 2001. With oversight from the Joint Staff J6 and the USSTRATCOM SATCOM Operational Manager (SOM), the USSTRATCOM Global and Regional SATCOM Support Centers (G/RSSC) manage DOD SATCOM resources authorizing access to the SATCOM RF spectrum in accordance with established priorities and procedures. Authorization for SATCOM access to the DOD Teleport System will be provided by the G/RSSC to the Teleport, the appropriate SATCOM C2 center (Wideband SATCOM Operations Centers -WSOC, NCTAMS, Milstar Space Operations Center -MSOC, etc.), Combatant Commander/TCCC, JTF/JCCC, SYSCON, and the Teleport Distant Component (as required).

The Defense Information System Agency (DISA) Global and Regional Network Operational Security Centers (G/RNOSC) manage the DISN services available via the Teleport System. Those services include DSN, DRSN, VTC, JWICS, NIPRNET and SIPRNET. JWICS is managed by the Defense Intelligence Agency (DIA) JWICS Network Operation Center (NOC). The G/RNOSC will coordinate with the JWICS NOC for access to JWICS service. The G/RNOSC will coordinate with the DISN service centers for access to each of the DISN services. Authorization to access DISN resources will be provided by the G/RNOSC to the Teleport, Combatant Commander/TCCC, JTF/JCCC, SYSCON, and the Teleport Distant Component (as required).

The Operations Department of the three NCTAMS manages the Navy's Legacy C4 requirements for their respective Area of Operations. Authorization for access to Navy Legacy C4 systems for interface with the DOD Teleport system will be provided by the NCTAMS. The Joint Fleet Telecommunications Operations Centers (JFTOC) at the NCTAMS are the single point of contact for the fleet with respect to afloat C4 systems connectivity to both Navy Legacy C4 systems and the DISN.

Authorization for access to those elements of the DCGS that interface the DOD system will be provided through that Service's headquarters or theater Combatant Commander to the Teleport, Teleport Distant Component, and the SYSCON in the form of an AUTODIN/DMS message.

Accessing and changing the status of a telecommunications service, whether RF, DISN, or Legacy C4I, requires frequent coordination between all elements of the Teleport system, i.e. the Teleport components, end-users, and the control/monitoring centers for that particular service. USSTRATCOM's SATCOM Support Center (SSC) concept provides management and control of satellite communications resources in accordance with CJCSI 6250.01A. The SSC concept provides communications planners, network managers, and users a functional structure that merges the current divested SATCOM systems operational management into a single integrated center for SATCOM resources management and allocation. The SSC concept provides the user a common day-to-day operational interface for obtaining all his SATCOM service needs. All users are assigned to either a global or regional support center, as appropriate. The Global SATCOM Support Center (GSSC) has the responsibility for maintaining a global system-of-systems SATCOM picture, coordinating the activities of the regional centers, and supporting national or global users not assigned to regional centers. The Global/Regional SATCOM Support Centers (GSSC/RSSC) are located within the major theater areas of responsibility and are generally aligned with DISA's existing end-to-end satellite and terrestrial management centers (GNOSC/RNOSC). The GSSC and/or RSSCs provide reconfiguration planning of SATCOM networks and payloads in support of the communications management responsibilities of the GNOSC and RNOSCs. Each GSSC and RSSC is provided technical assistance and engineering support by a staff of Satellite System Experts (SSE). RSSCs are expected to support the generation of terminal images in support of EHF mission requirements for deployed users accessing Teleport.

Army Space Command (ARSPACE) is responsible for managing the WSOC and GSSC/RSSC. WSOCs are responsible for the day-to-day control and operation of DSCS and WGS. The GSSC/RSSCs provide control management of SATCOM resource allocation as discussed above.

In partnership with the G/RSSCs, the SATCOM C2 Centers (SC2Cs) perform the day-to-day operational command and control of the satellite and communications payloads. The Naval Satellite Operations Center (NAVSOC), 3 and 4 Space Operations Squadrons (SOPS), WSOC, MSOC, NCTAMS, Bandwidth Management Centers (BMC), and regional Combatant Commander TCCCs will all be involved with coordinating and monitoring the full spectrum of SATCOM connectivity to the Teleport. Coordination will be via existing and/or planned coordination networks.

When accessing commercial SATCOM systems, tactical users will coordinate with the RNOSC/RSSCs, BMCs, Combatant Commander TCCCs, and/or appropriate control centers. The BMCs, through coordination with satellite operators, facilitate control of commercial transponders or circuits to ensure overall circuit quality and performance. Users will coordinate commercial SATCOM access and status changes with the BMC through commercial telephone resources or email. All MILSATCOM resources (DSCS, Milstar, UFO, etc.) will be handled through the other listed SC2Cs and TCCCs.

In the case of legacy C4I activation, the Teleport will coordinate with the distant Teleport components (i.e. earth terminal), tactical users, and the legacy C4I originating station (i.e. NCTAMS or DCGS) via NIPR/SIPRNET and/or commercial phone systems.

(2) Manning required for the control, operations, and maintenance of the DOD Teleports will consist of military, government employees, and contractor personnel. However, there shall be no net increase in Military Service end strengths to control, operate, and maintain DOD Teleports. Therefore, automation and workload reduction techniques must be employed in the DOD Teleport design to minimize manning requirements. Consolidation of existing communications nodes and network management/operations nodes is a potential “economy of scale” savings. Migration to a “lights out” remote control capability of lesser communication/network management nodes is also a goal of the DOD Teleport concept. This will help to alleviate the manpower/personnel requirements of DOD Teleports.

(3) This Teleport system must have minimal specialized field support requirements. The Military Service and DOD Agency’s existing maintenance organizations or contractor logistics support shall support the DOD Teleport. This system shall take maximum advantage of commercial products, technologies, and non-developmental items. The government may choose to retain rights regarding intellectual properties developed for the DOD Teleport. Industry proprietary components should be avoided to the maximum extent possible. This system should be designed using open systems standards to facilitate insertion of emerging technologies.

(4) Organizational level maintenance shall consist of periodic functional checks and minor maintenance, consistent with current maintenance provided by unit maintenance personnel. A supply point of line replaceable units (LRU) shall be maintained at the organizational level. When the system fails to meet acceptable performance standards, organizational maintenance personnel shall initiate Built-In-Test (BIT) troubleshooting procedures to isolate the fault to an LRU. That unit shall be replaced with an operable LRU from the supply point and the malfunctioning unit shall be shipped out for depot/contractor maintenance.

(5) Depot/Contractor Maintenance. Malfunctioning LRUs shall be repaired or replaced at a depot or through contractor support.

d. Phased Implementation Concept.

DOD Teleport is a multi-generation approach from program start in FY00 to FOC in FY12. The phasing strategy is detailed below. DOD Teleports initial implementation generation (Generation One) will establish an operational baseline capable of meeting the deployed daily operations, SSC and MCO requirements of today's Warfighter. Existing assets will be leveraged and organized to optimize throughput and add management functions that will be used as the foundation for future stages of DOD Teleport implementation. The 2012 performance objectives will be approached incrementally. Each implementation generation will be driven by validated user requirements and paced by a number of factors including: technology availability, space segment growth, terminal availability, terrestrial connectivity, and new commercial service offerings. This approach will provide just in time capability using affordable and mature technology. The implementation phase will provide information to the Services to support responsible, timely, fiscal planning.

Major terminal upgrades at the DOD Teleport sites will be implemented in one of three ways, the first being, planned installation of terminals at DOD Teleport sites, funded by the DOD Teleport program. The second approach will be to work with major SATCOM programs to coordinate implementation of their terminal equipment at the DOD Teleport to include transponder leasing, funded by the SATCOM program and integrated by the DOD Teleport program. And lastly by exercising contract options to lease entire terminals or portions of the terminal throughput funded by the DOD Teleport program.

Generation Zero (FY00 - FY02)

- Develop generation one Concept Studies/Analysis of Alternatives (AoA) based on results of Concept studies/AoA and results of the Joint Warfighter Interoperability Demonstration (JWID)
- Stand up Program Office
- Proof of Concept (JWID)

Generation One (FY02-FY06)

Generation One (Initial Operating Capability (IOC1): FY03

- Implement C, X, and Ku bands (Architecture consists of 2 core sites and 4 split sites)

Generation One (IOC2): FY04

- Implement UHF (Architecture consists of 2 core sites and 4 split sites)

Generation One (IOC3): FY06

- Implement additional C, Ku, UHF and expand to 6 core sites*
- Implement EHF (LDR/MDR) and add a secondary site in Bahrain**

Generation One (IOC4): FY06

- Implement Military Ka, and technology refreshments consistent with commercial industry standards

<u>Generation Two (FY04-FY07)</u>
- Expand to include all 2006 Military Ka requirements

<u>Generation Three (FY08-FY12)</u>

- Implement Commercial Ka, EHF (XDR), and Mobile User Objective System (MUOS)
- Implement Future Wideband Systems and Transformational Communications Architecture (TCA) into DOD Teleports
- Final Operating Capability (FOC)

Notes:
* Additional C, Ku, UHF capabilities and expansion to 6 core sites was authorized via an Acquisition Decision Memorandum of 28 October 02
** Implementation of EHF (LDR/MDR) capabilities was authorized (as Generation One, IOC3) via an Acquisition Decision Memorandum of 2 April 03

2. Threat.

a. Threat Environment.

The DOD Teleport is vulnerable to the same threats as any other military satellite and terrestrial communications systems. Locations must meet force protection requirements within the Combatant Commander current Operation Plans (OPLAN). They may be jeopardized by electromagnetic and physical threats to the communication links and/or to the system segments themselves. Electromagnetic threats include jamming, interception, exploitation, directed energy (radio-frequency weapons) high-altitude electromagnetic pulse (HEMP), and degradation of commercial and military communication links or the signal environment. Physical threats to the ground-based portion include the entire spectrum of direct and indirect fire weapons, environmental factors, chemical contamination and nuclear destruction. Further threats include the possible denial of access to commercial telecommunication systems. Compounding the threat is the fact that the General Accounting Office (GAO) has found that Federal agencies, to include DOD, account for only 10 percent of the commercial satellite market. As a result, Federal customers, to include DOD, reduce their risk by securing those system components under their control (e.g. data links); however, most components are typically the responsibility of the commercial satellite provider (e.g. the satellite itself; telemetry, tracking, and control links; and the satellite control ground stations). Presidential Decision Directive 63 (Protecting America's Critical Infrastructure) does not include the commercial satellite industry as part of the national infrastructure protection effort.

Offensive Information Operations (IO) and Information Warfare (IW) against U.S. military forces, to include electronic warfare, physical attack, and/or destruction or Computer Network Attack (CNA) and Computer Network Exploitation (CNE) has become a significant potential threat. IO now plays a significant role in the success of U.S. military operations. JTF commanders are increasingly dependent upon information and information systems to plan operations, deploy forces, and execute missions. IO multiplies the effectiveness and capabilities of military forces, but this dependence on information and information systems also creates a vulnerability to adversary IO. The Warfighter's combat effectiveness could be substantially eroded if an adversary successfully attacks an operational DOD Teleport.

Computer threat mechanisms have been grouped into four categories:

1. Compromise-of-information - when an adversary gains access to friendly information either by making an electronic copy of it or by gaining access to the hosting machine and simply reading it.
2. Data Deception or Corruption - when the data contained in a system or being transmitted over a data or sensor link is modified, whether it is intentional or unintentional.
3. Information denial or loss - when access to friendly information is disrupted. Could occur via denial of service, destruction of the bit stream, signal, or database.
4. Physical Destruction or Damage - when the original state of a system's physical components are altered or destroyed such that they no longer function according to their design.

b. <u>Threat Assessment.</u>

The MILSATCOM System Threat Assessment Report (STAR), NAIC-1574-037-03, dated July 2003, covers threats to proposed satellite communications systems to include threats to Teleport, and shall be considered part of the documented threat for this system. Threats against any associated automated information system are listed in the Information Operations Capstone Threat Capabilities Assessment, DI-15771203, published in August 2003 and shall be considered part of the documented threat for this system.

Integrating selected DOD Telecommunication sites into DOD Teleports allows for the capitalization of existing security features (physical, TEMPEST, and electromagnetic pulse (EMP) to name a few). To protect against IO/IW, the DOD Teleport software support systems must be protected through defensive IO/IW services, which include, but are not limited to, the use of Computer Network Defense (CND) measures, intrusion detection, and timely countermeasures. Consistent with DOD recommendations, the system shall exploit evolutionary development practices to improve protections to counter these threats through Pre-Planned Product Improvements

(P3I). Additional Information Warfare threats posed to joint military telecommunication systems are addressed in the Electronic Warfare Threat Environment Description (TED), NAIC-1574-0731-01, February 2001.

3. Shortcomings of Existing Systems and C4ISR Architectures.

DOD contingency operations, Operations Other Than War (OOTW), and information requirements growth within the SDB have demonstrated a critical need for increased throughput and flexibility in supporting long-haul communications connectivity. Specific shortcomings include:

a. Operational Shortcomings.

(1) Throughput requirements are growing exponentially beyond STEP and current MILSATCOM capabilities as DOD begins to develop the components necessary to ensure Joint Vision 2020 Information Superiority. STEP sites are limited to 5 Mbps of basic DISN Service Delivery Node (SDN) throughput and 11 Mbps of surge DISN SDN throughput via DSCS (X band) only. The surge DISN SDN capacity is approved by the Joint Staff and funded by the users.

(2) Access into the DISN Long-Haul Block for the Warfighter and the deployed user has major limitations. Specifically, DISN services are available to deployed forces only via DSCS (X-band) or commercially leased circuits. None of the many other frequency bands available to the Warfighter provide direct access to DISN services. This physical isolation causes DOD to create independent reach-back solutions for each deployed user each time they require the link to be activated.

(3) The STEP-only solution does not support the complete set of requirements for the Warfighter and deployed users for high throughput and multi-band communications.

(4) The time required to contract, engineer, and provide DISN access; to design, order and connect links to the nearest DISN SDN(s); and then to complete the end-to-end service(s) is unacceptable. The Warfighter and deployed user require timely access into the DISN Long Haul Block and to the DISN SDNs when they deploy. Case-by-case solutions fail to provide this timely support for fast-paced, life-and-death tactical operations.

(5) The Warfighter is faced with numerous joint and combined interoperability challenges. Three key shortcoming areas include legacy versus new technology user mix, baseband incompatibilities between forces, and multiple SATCOM RF user equipment.

(6) In the absence of a central solution, each Service and Combatant Commander has developed their solution, not allowing for synergies and the establishment of a common DOD solution.

(7) The existing Legacy Tactical C4I Systems are antiquated, stovepipe systems that provide minimum bandwidth throughput. For a list of systems, see Appendix B.

(8) Need to identify the “Long Fat Network” problem when utilizing the communications protocol “TCP” over high bandwidth/long distance circuits. There are some COTS products that have become available to help overcome this problem, however, there is no DOD standard employed and network circuits are operating at various efficiencies over satellite connections.

b. Programmatic Deficiencies.

Combatant Commanders have identified via the Integrated Priority List (IPL), the Joint Quarterly Readiness Reviews (JQRR), and the Joint Warfighting Capability Assessment (JWCA) that there are insufficient means to provide CJTFs and deployed forces with DISN requirements. There is no programmatic solution to meet Combatant Commander-identified requirements for connectivity to the Warfighter.

4. Capabilities Required.

DOD Teleports support the Warfighter by providing the capabilities of coverage, capacity, interoperability, protection, control, network management, quality of service, technology insertion, and affordability. The rationale adapted from the Advanced MILSATCOM CRD was used to develop the capabilities outlined in Table 4-1 for the Gen 1 ORD approval. The replacement SATCOM CRD crosswalk is provided in Appendix C to provide correlation of previous Advanced MILSATCOM CRD requirements with revised SATCOM CRD criteria.

Table 4-1. Rationale for Required System Capabilities

CAPABILITY	RATIONALE (WHY REQUIRED)
COVERAGE	Global national interests and threats environment. Regional conflicts/crises unpredictable in location, time, intensity and duration. Smaller US force structure; globally dispersed land, sea, air, and space operations. Time and geographically varying user population densities.
CAPACITY	Warfighter demands more military and commercial satellite bands. Connectivity cannot be a limiting factor in the application of combat power. Assured Warfighter connectivity when and where needed. DOD Teleport provides dynamic, multiple information transfer capabilities.
INTEROPERABILITY	Operations are joint in nature and execution (Land, Air, Naval, Mobility, Combat Support, And Special Operations Forces). US Forces conduct missions with Allies, Coalition Partners, and Government Agencies. Warfighters use a variety of communications to effect needed information transfers.
PROTECTION	Our C4I is a prime target and center of gravity that we expect adversaries to attack and attempt to exploit. Must deny adversaries the ability to affect our information and information systems while preserving timely, accurate, and controlled information access to authorized personnel. Must provide robust information assurance to include providing for the restoration of affected information systems. Must provide anti-jam, protection from Signals Intelligence (SIGINT), Information Security, Computer Network Defense (CND), and other information operations measures.
CONTROL & NETWORK MANAGEMENT	Warfighters need access to information and communications on demand. Warfighter must have positive control over their information domains. DOD Teleport resources must be rapidly and dynamically reconfigured to respond to changing operational situations and priorities.
QUALITY OF SERVICE	Supported warfighting and combat support systems drive high performance criteria. Information must be transferred accurately and unambiguously.
TECHNOLOGY INSERTION	Warfighter requires modern equipment and communication capability. Need to accommodate evolving doctrine, requirements, threats and technologies.
AFFORDABILITY	The program will minimize total cost to the DOD, to include the cost of procurement, manpower, O&M, and connectivity costs.

a. <u>System Performance.</u>

The DOD Teleport capabilities and the requirements needed to satisfy them are shown in Table 4-2. The specific requirements selected to satisfy the Key Performance Parameters (KPP) are restated in Table 4-3. The Deployed to DISN service throughput requirements for 2004 and 2010 are provided in the Teleport Requirements Development and Analysis study (Draft) completed by DISA in June 2000. An updated requirements analysis that accurately reflects the Generation Two, 2006 requirements is provided in Joint Staff VJ6 memo, subject “Satellite Throughput and DISN Requirements for DOD Teleport Generation Two” dated 11 Sep 03. These requirements will be revalidated by the Joint Staff J6S prior to the POM 06 budget cycle.

Table 4-2. System Requirements

COVERAGE

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
DOD Teleport will provide Warfighter coverage on a worldwide, regional, interregional and theater basis	YES	1. 365 days per year and 24 hours per day 2. In all latitudes between 65° North and 65° South (worldwide) 3. Teleports positioned to allow the Warfighter communications with at least two (2) DOD Teleports in all bands provided that satellite coverage exists for that area.	1. Teleports positioned to allow the Warfighter communications with at least three (3) DOD Teleports in all bands provided that satellite coverage exists for that area.	Gen1(IOC1) Gen1(IOC1) Gen 1(IOC1)
DOD Teleport will provide Warfighter coverage on a global, regional, interregional and theater basis	NO	1. Incorporate existing polar SATCOM capabilities. 2. Incorporate planned polar SATCOM capabilities.		Gen1 (IOC2) Gens 2 & 3

CAPACITY

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
The DOD Teleport system must have the constant capacity to provide DISN, legacy Tactical C4I services and SATCOM throughput in any given geographical region	YES	1. Provide 100% of the projected 2004 required DISN services to support day-to-day operations and one SSC.	1. Provide 100% of the projected 2004 required Legacy Tactical C4I services as identified in Appendix B.	Gen1(IOC2)
		2. Provide 100% of the projected 2004 required SATCOM throughput to support day-to-day operations and one SSC.		Gen1 (IOC2)
		3. Provide 100% of the projected 2006 required DISN services for one MCO.		Gen 2
		4. Provide 100% of the projected 2006 required SATCOM throughput for one MCO.		Gen 2
			2. Provide 100% of the projected 2010 required DISN Services to support two overlapping wars.	Gen3 (FOC)
			3. Provide 100% of the projected 2010 required SATCOM throughput to support two overlapping wars.	Gen3 (FOC)
			4. Provide 100% of the projected 2010 required Legacy Tactical C4I services as identified in Appendix B.	Gen3 (FOC)
The DOD Teleport system must have a surge capability in any given geographical region.	YES	Provide 100% of the projected 2004 required DISN and SATCOM services to support one MCO within 30 days as identified in this ORD.		Gen1 (IOC2)

INTEROPERABILITY

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DISN and Legacy Tactical C4I.	YES	Access to voice (DSN and DRSN), data (SIPRNET, NIPRNET and JWICS) and video (VTC) services. Access to DISN POPs for on demand transport and COI services.	Access to Legacy Tactical C4I services (Refer to Appendix B) Access to Legacy Tactical C4I systems (Refer to Appendix B)	Gen1 (IOC1) Gen 2
The DOD Teleport must provide common access into satellite system and be able to cross-band between them.	YES	Access to X, C, Ku bands Access to UHF Access to EHF (LDR/MDR) Access to Ka (Military) Access to Ka Commercial Bands Access to EHF (XDR)		Gen 1(IOC1) Gen 1(IOC2) Gen 1(IOC3) Gen 1(IOC4) Gen 3 Gen 3
All top level IERs will be satisfied to the standards specified in the Threshold and Objective values.	YES	100% accomplishment of the critical top-level IERs that directly interface the Teleport. 100% accomplishment of the critical top-level IERs that directly interface the Teleport.	100% accomplishment of all top-level IERs that directly interface the Teleport. 100% accomplishment of all top-level IERs that directly interface the Teleport.	Gen1 (IOC1) Gen 2
The DOD Teleport must be capable of providing hub connectivity.	YES	Access to X, C, Ku bands Access to Ka (Military) Access to Ka commercial.		Gen 1(IOC1) Gen 1(IOC4) Gen3
Teleport shall support an architecture that uses policy-based (Quality of Service) IP routing and dynamically allocated satellite bandwidth to provide a responsive network centric capability.	YES	Teleport shall provide the Warfighter access to network centric IP architecture.	Same	Gen 2
Teleports shall support the migration to IP services.	YES	Teleport shall be interoperable with IP routing requirements for tactical users in accordance with a DOD-wide standard IP routing architecture.	Teleport architecture shall conform to the Government Reference Architecture (GRA) for Transformational Communications.	Gen 3
Provide users the ability to access commercial phone services from any Teleport.	NO	Teleports shall provide capability to access Public Switched Network via DSN.		Gen 3
The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DCGS services through standard DISN networks (e.g. JWICS, SIPR, etc) or through user provided and funded alternate connectivity.	NO	Ku to DCGS connectivity Teleport shall provide the capability to receive ISR data from appropriate ISR airborne assets outfitted with military or commercial satellite relay capability (e.g. Predator, Global Hawk, F/A-22) and appropriate forward deployed ISR ground stations that have or use commercial and/or military satellite reachback (e.g. Predator GCS, Global Hawk MCE). These ISR airborne assets and/or ground stations may transmit simultaneously from any operational theater.		Gen 1(IOC2) Gen 3
The DOD Teleport must provide common access into communications systems and be	NO	Access to L band.	Add access to HF Global communications system.	Gen 2

able to cross-band between them.				
The DOD Teleport shall enable interoperability between and among COMBATANT COMMANDER, JTF Components, deployed users and DOD agencies using JTA Compliant baseband equipment.	NO	DOD Teleport must be JTA compliant.		Gen 1(IOC1)
Teleports must be interoperable with JNMS and the Integrated Network Management System (INMS).	NO	Teleport will be capable of providing alerts to INMS. JNMS will obtain any necessary information through the INMS.		When JNMS fielded.
Teleports must be interoperable with the Joint Tactical Radio System (JTRS).	NO	Must interoperate with JTRS waveforms necessary within the appropriate bands for the Teleport.		When JTRS fielded.
The DOD Teleport must provide Warfighter interconnectivity.	NO	Provide the capability to operate as a satellite terminal relay (M-hop) within the same satellite band.	Provide multiplexer and baseband translation	With implementation of access to band
The DOD Teleport must maintain system compatibility, flexibility, and interoperability with Theater Legacy Tactical Networks and systems, which are not JTA - compliant.	NO	C4I applications and systems are identified in Appendix B.		Gen 1(IOC1)
Teleports will facilitate continuity of operations for deployed users changing connections between Teleport sites.	NO	Entry into all Teleports will appear standardized to the deployed user for each transmission medium. (e.g., all X band entries will be standard. All Ku band entries will be standard. Etc.)		Gen 1(IOC1)
Teleports will be interoperable with MUOS	NO	Must interoperate with MUOS waveform when developed.		Gen 3

PROTECTION

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
The DOD Teleport must support information environment protection, attack detection, capability restoration, and attack response.	YES	1. The DOD Teleport must support information environment protection, attack detection, capability restoration, data integrity, and attack response IAW CJCSI 6510.01 Series. 2. The DOD Teleport must support bulk encryption/ TRANSEC capabilities of DISN, Legacy Tactical C4I, and SATCOM Components.	Ensure timely, accurate, and controlled information access to authorized personnel while denying adversaries the opportunity to exploit friendly information and information systems for their own purpose.	Gen 1(IOC1)
DOD Teleport must support Information Assurance.	Yes	System shall meet and maintain minimum IA Defense in Depth Standards, including certification and accreditation IAW the DITSCAP process CJCSI 6510.01C and DODI 5200.40.		Gen 2
The DOD Teleport must operate as US-controlled, US Secret facilities.	NO	All equipment, facilities and procedures must be IAW CJCSI 6510.01 Series.		Gen 1(IOC1)
The DOD Teleport must be capable of integrating HEMP-protected terminals into the Teleport facilities.	NO	Integration of HEMP-protected terminals into DOD Teleport facilities must not degrade the HEMP characteristics of these terminals as identified in MIL-STD 2169B, HEMP Environment.		Gen 1(IOC1)

CONTROL AND NETWORK MANAGEMENT

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
The DOD Teleport must be capable of resource configuration and re-configuration	YES	(A) Centralized control of all equipment with remote interfaces from one location on the floor, (B) Electronic patching controllable from central location. (C) Automated circuit redundancy where available within COTS product line.	Automated and remote	Gen 1(IOC1) or if not previously installed
The DOD Teleport must provide for system controller interfaces	NO	Interface with satellite controllers and the DISN service controllers.		Gen 1(IOC1) or if not previously installed
The DOD Teleport must provide nodal management and control for each Teleport component.	NO	Provide automated connectivity between user-to-user and user-to-DISN services with manual override.	Automated and remote.	Gen 1(IOC1)
The DOD Teleport must provide remote access to the DISN and SATCOM management and control centers (per CJCSI 6211.02A and CJCSI 6250.01A respectively)	NO	Read only.	Fully interactive.	Gen 1(IOC1)
The DOD Teleport must provide remote access to the deployed network management systems.	NO	Read only.	Fully Interactive.	Gen 1(IOC1)
DOD Teleport shall provide for configuration control and configuration management.	NO	Shall have an integrated system control of organic satellite terminals and terrestrial communications equipment.		Gen 1(IOC1)

QUALITY OF SERVICE

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
The DOD Teleport system must maintain the A_o, of existing Programs of Record.	NO	$\geq$99%	$\geq$99.9%	Gen 1(IOC1)
Individual DOD Teleport facilities must maintain the A_o, of existing Programs of Record.	NO	$\geq$95%	$\geq$99%	Gen 1(IOC1)
The DOD Teleport system must maintain circuit technical performance	NO	IAW minimum performance standards of supported systems IAW DISA Circular 300-175-9.		Gen 1(IOC1)
Maintain support of as directed mission essential deployed networks.	NO	DOD Teleport system must be capable of responding to reconfiguration direction within one hour of notification.		Gen 1(IOC1)
The system shall maintain and guarantee during transport the integrity of all information elements exchanged throughout the GIG to enable user confidence; information integrity.	NO	99.99%	99.999%	Gen 2
All transport elements (e.g. switches, routers, etc.) shall be capable of providing status changes to network management devices by means	NO	99%	99.9%	Gen 2

of an automated capability in near real time.				
The system shall have the NM capability to obtain status of networks and associated assets in near real time.	NO	99%	99.9%	Gen 2

TECHNOLOGY INSERTION

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
DOD Teleports must provide technology insertion and capacity growth capabilities to incorporate changing Warfighter requirements.	NO	Use open systems, standardized products and protocols, which are at a minimum of level 6 DII COE and JTA compliant.	Use open systems, standardized products and protocols, which are at a minimum of level 8 DII COE and JTA compliant.	Gen 1(IOC1)
DOD Teleports must provide the warfighter the capability to terminate VOIP services.	NO	The Teleport must be capable of terminating VOIP services and provide an interface into DRSN and DSN.		Gen 3

AFFORDABILITY

REQUIREMENT	KPP	THRESHOLD	OBJECTIVE	GEN
DOD Teleports must provide maximum operational capabilities to meet Warfighter requirements while optimizing acquisition, procurement, and overall life cycle costs*	NO	\$196.2M \$244.8M \$501.8M	\$158.6M \$196.7M \$366.6M	Gen 1 Gen 2 Gen 3

**In accordance with DOD Teleport Architecture Feasibility Study (Draft) June 2000, Architecture Alternative 5.*

Table 4-3 below provides a listing of only the threshold and objective KPPs for DOD Teleport.

Table 4-3. Key Performance Parameters

CAPABILITY	PARAMETER	THRESHOLD	OBJECTIVE	GEN
COVERAGE	DOD Teleport will provide Warfighter coverage on a worldwide, regional, interregional and theater basis	1. 365 days per year and 24 hours per day 2. In all latitudes between 65° North and 65° South (worldwide) 3. Teleports positioned to allow the Warfighter communications with at least two (2) DOD Teleports in all bands provided that satellite coverage exists for that area.	1. Teleports positioned to allow the Warfighter communications with at least three (3) DOD Teleports in all bands provided that satellite coverage exists for that area.	Gen 1(IOC1) Gen 1(IOC1) Gen 1(IOC1)

CAPABILITY	PARAMETER	THRESHOLD	OBJECTIVE	GEN
CAPACITY	The DOD Teleport system must have the constant capacity to provide DISN, legacy Tactical C4I services and SATCOM throughput in any given geographical region	1. Provide 100% of the projected 2004 required DISN services to support day-to-day operations and one SSC. 2. Provide 100% of the projected 2004 required SATCOM throughput to support day-to-day operations and one SSC. 3. Provide 100% of the projected 2006 required DISN services for one MCO. 4. Provide 100% of the projected 2006 required SATCOM throughput for one MCO.	1. Provide 100% of the projected 2004 required Legacy Tactical C4I services as identified in Appendix B. 2. Provide 100% of the projected 2010 required DISN Services to support two overlapping wars. 3. Provide 100% of the projected 2010 required SATCOM throughput to support two overlapping wars. 4. Provide 100% of the projected 2010 required Legacy Tactical C4I services as identified in Appendix B.	Gen 1(IOC2) Gen 1(IOC2) Gen 2 Gen 2 Gen 3(FOC) Gen 3(FOC) Gen 3(FOC)
	The DOD Teleport system must have a surge capability in any given geographical region.	Provide 100% of the projected 2004 required DISN and SATCOM services to support one MCO within 30 days as identified in this ORD.		Gen 1(IOC2)
INTEROPER-ABILITY	The DOD Teleport must provide the JTA-compliant Warfighters' suite of communications equipment access to DISN and legacy Tactical C4I.	Access to voice (DSN and DRSN), data (SIPRNET, NIPRNET and JWICS), and video (VTC) services. Access to DISN POPs for on demand transport and COI services.	Access to legacy Tactical C4I services. (See Appendix B) Access to Legacy Tactical C4I systems (Refer to Appendix B)	Gen 1(IOC1) Gen 2
	The DOD Teleport must provide common access into satellite system and be able to cross-band between them.	Access to X, C, Ku Access to UHF Access to EHF (LDR/MDR) Access to Ka (Military) Access to Ka Commercial bands Access to EHF (XDR)		Gen 1(IOC1) Gen 1(IOC2) Gen 1(IOC3) Gen 1(IOC4) Gen 3 Gen 3
	All top level IERs will be satisfied to the standards specified in the Threshold and Objective values.	100% accomplishment of the critical top-level IERs that directly interface the Teleport. 100% accomplishment of the critical top-level IERs that directly interface the Teleport.	100% accomplishment of all top-level IERs that directly interface the Teleport. 100% accomplishment of all top-level IERs that directly interface the Teleport.	Gen 1(IOC1) Gen 2
	The DOD Teleport must be capable of providing hub connectivity.	Access to X, C, Ku bands Access to Ka (Military) Access to Ka commercial		Gen 1(IOC1) Gen 1(IOC4) Gen 3
	Teleports shall support an architecture that uses policy-based (Quality of Service) IP routing and dynamically allocated satellite bandwidth to provide a responsive network centric capability	Teleport shall provide the Warfighter access to network centric IP architecture	Same	Gen 2

CAPABILITY	PARAMETER	THRESHOLD	OBJECTIVE	GEN
	Teleports shall support the migration to IP services.	Teleport shall be interoperable with IP routing requirements for tactical users in accordance with a DOD-wide standard IP routing architecture.	Teleport architecture shall conform to the Government Reference Architecture (GRA) for Transformational Communications.	Gen 3
PROTECTION	The DOD Teleport must support information environment protection, attack detection, capability restoration, and attack response.	1. The DOD Teleport must support information environment protection, attack detection, capability restoration, data integrity, and attack response IAW CJCSI 6510.01 Series. 2. The DOD Teleport must support bulk encryption/ TRANSEC capabilities of DISN, Legacy Tactical C4I, and SATCOM Components.	Ensure timely, accurate, and controlled information access to authorized personnel while denying adversaries the opportunity to exploit friendly information and information systems for their own purpose.	Gen 1(IOC1)
	DOD Teleport must support Information Assurance.	System shall meet and maintain minimum IA Defense in Depth Standards, including certification and accreditation IAW the DITSCAP process CJCSI 6510.01C and DODI 5200.40.		Gen 2
CONTROL & NETWORK MANAGEMENT	The DOD Teleport must be capable of resource configuration and re-configuration.	(A) Centralized control of all equipment with remote interfaces from one location on the floor, (B) electronic patching controllable from central location, and (C) automated circuit redundancy where available within COTS product line.	Automated and remote.	Gen 1(IOC1) or as Service implement- ed

The DOD Teleport shall be integrated into the existing DISN network in accordance with Table 4-4 below. JWICS includes both data and VTC.

Table 4-4. DISN to SATCOM Interfaces

	THRESHOLD / OBJECTIVE								
	Gen 1							Gen 2	Gen 3
	X	Ku	C	UHF	EHF (LDR)	EHF (MDR)	Ka (Mil)	L / * HF	Ka Comm / * EHF (XDR)
DSN	√	√	√		√	√	√	√	√
DRSN	√	√	√	-	-	√	√	-	√
NIPR	√	√	√	√	√	√	√	√	√
SIPR	√	√	√	√	√	√	√	√	√
JWICS	√	√	√	√	√	√	√	-	√
VTC	√	√	√	√ **	-	√	√	√ ***	√
Terr Transport	√	√	√	√	√	√	√	√	√

* ORD Objective requirements
** MUOS ORD requirement for Gen 3
*** L band only requirement

b. <u>Information Exchange Requirements.</u>

Information Exchange Requirements (IER) characterize the information exchanges to be performed by the DOD Teleport System. This set of top level IERs is used to develop interoperability KPPs. Top-level IERs identify who exchanges what information with whom, why the information is necessary, and in what manner the information exchange must occur. Top-level IERs identify warfighter information used in support of a particular mission-related task and exchanged between at least two operational systems supporting a joint or combined mission area. Table 4-5 provides the DOD Teleport Operational Information Exchange Matrix IAW CJCSI 6212.01B. To assist in understanding the IER Matrix, the specific field definitions are as follows:

1) UJTL: Is the Universal Joint Task List number which identifies it by an alpha-numeric field that documents the proposed system warfighting and Warfighter support tasks described in the current Universal Joint Task List found in (CJCSM 3500.04 series).

2) Event: A free text field that describes the event or action that triggers the need for the Information Exchange.

3) Information Characterization (Info Char): A pick list and free text field that describes the content of the information. Pick list includes, C2, Situational Awareness, Targeting, Threat, Fire Support, Logistics, Personnel, and Other specified.

4) Sending Node: Free text field that describes the node that sends the information element.

5) Receiving Node (Rec Node): Free text field that describes the node that receives the information element.

6) Critical: A logical field used to characterize the importance of the Information Exchange. An ORD critical top-level IER is one that is required to support its associated CRD critical top-level IER, or will severely and adversely impact on a warfighter mission if not accomplished.

7) Format: A pick list and free text field that describes the physical form of the information element. This is not the communications medium used to send the information. Pick list include Audio, Text, Graphics, Imagery, Video, and Data.

8) Timeliness: Numerical field measured in seconds. It represents the time between the occurrence of the event to the time it is available to the user in seconds.

9) Classification: Pick list field that describes the highest security classification that can be assigned to this information element.

Table 4-5. Top-Level Operational Information Exchange Matrix (OV-3)

<u>CONFIGURATION MANAGEMENT: RF DIRECTIVE</u>								
Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***

OP5.1.2	RF Configur-ation Manage-ment	C2 – Management Directives	G/RSSC	Teleport	Yes	Data*	Note 1	S
OP5.1.2	RF Configur-ation Manage-ment	C2 – Management Directives	G/RSSC	COMBATANT COMMANDE R/TCCC	Yes	Data*	Note 1	S
OP5.1.2	RF Configur-ation Manage-ment	C2 – Management Directives	G/RSSC	JTF/JCCC	Yes	Data*	Note 1	S
OP5.1.2	RF Configur-ation Manage-ment	C2 – Management Directives	G/RSSC	SYSCON	Yes	Data*	Note 1	S
OP5.1.2	RF Configur-ation Manage-ment	C2 – Management Directives	G/RSSC	Teleport Distant Component	Yes	Data*	Note 1	S
CONFIGURATION MANAGEMENT : DISN DIRECTIVE								
OP5.1.2	DISN Configur-ation Manage-ment	C2 – Management Directives	G/RNOSC	COMBATANT COMMANDE R/TCCC	Yes	Data*	Note 1	S
OP5.1.2	DISN Configur-ation Manage-ment	C2 – Management Directives	G/RNOSC	Teleport	Yes	Data*	Note 1	S
OP5.1.2	DISN Configur-ation Manage-ment	C2 – Management Directives	G/RNOSC	JTF/JCCC	Yes	Data*	Note 1	S
OP5.1.2	DISN Configur-ation Manage-ment	C2 – Management Directives	G/RNOSC	SYSCON	Yes	Data*	Note 1	S
OP5.1.2	DISN Configur-ation Manage-ment	C2 – Management Directives	G/RNOSC	Teleport Distant Component	Yes	Data*	Note 1	S
CONFIGURATION MANAGEMENT : LEGACY C4I MANAGEMENT								
Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.2	Legacy Configur-ation Manage-ment	C2 – Legacy C4I Management Note 2	NCTAMS	Teleport	No	Data*	Note 1	S
OP5.1.2	Legacy Configur-ation Manage-ment	C2 – Legacy C4I Management Note 2	NCTAMS	Teleport Distant Component	No	Data*	Note 1	S

OP5.1.2	Legacy Configur-ation Manage-ment	C2 – Legacy C4I Management Note 2	NCTAMS	SYSCON	No	Data*	Note 1	S
OP5.1.2	Legacy Configur-ation Manage-ment	C2 – Legacy C4I Management Note 2	DCGS	Teleport	No	Data*	Note 1	S
OP5.1.2	Legacy Configu-ration Manage-ment	C2 – Legacy C4I Management Note 2	DCGS	Teleport Distant Component	No	Data*	Note 1	S
OP5.1.2	Legacy Configur-ation Manage-ment	C2 – Legacy C4I Management Note 2	DCGS	SYSCON	No	Data*	Note 1	S
ACCESS COORDINATION: ACCESS TO RF GATEWAY								
Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Teleport	COMBATANT COMMANDE R/TCCC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	COMBATANT COMMANDER/ TCCC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Teleport	JTF/JCCC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	JTF/JCCC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Teleport	SYSCON	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	SYSCON	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Teleport	G/RSSC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	G/RSSC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	TELEPORT	BMC/OC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	BMC/OC	TELEPORT	Yes	Data*	30 Sec**	S

OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	CJTF/JCCC	BMC/OC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	BMC/OC	CJTF/JCCC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	SYSCON	BMC/OC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	BMC/OC	SYSCON	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Teleport	Distant Component	Yes	Data*	30 Sec**	S
OP5.1.1	RF Access Coordina tion	C2- Coordinate Access to RF Gateway	Distant Component	Teleport	Yes	Data*	30 Sec**	S

ACCESS COORDINATION: ACCESS TO DISN GATEWAY

Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	Teleport	COMBATANT COMMANDE R/TCCC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	COMBATANT COMMANDER/ TCCC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	Teleport	JTF/JCCC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	JTF/JCCC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	Teleport	SYSCON	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	SYSCON	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	Teleport	G/RNOSC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	G/RNOSC	Teleport	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Access Coordina tion	C2- Coordinate Access to DISN services	Teleport	Distant Components	Yes	Data*	30 Sec**	S

OP5.1.1	DISN Access Coordina tion	C2-Coordinate Access to DISN services	Distant Components	Teleport	Yes	Data*	30 Sec**	S

ACCESS COORDINATION: ACCESS TO LEGACY C4I SERVICES

Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I Services Note 2	Teleport	COMBATANT COMMANDE R/TCCC	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I Services Note 2	COMBATANT COMMANDER/ TCCC	Teleport	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I Services Note 2	Teleport	JTF/JCCC	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I Services Note 2	JTF/JCCC	Teleport	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I Services Note 2	Teleport	SYSCON	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I services Note 2	SYSCON	Teleport	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I services Note 2	Teleport	NCTAMS	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I services Note 2	NCTAMS	Teleport	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordina tion	C2-Coordinate Access to Legacy C4I services Note 2	Teleport	DCGS	No	Data*	30 Sec**	S

OP5.1.1	Legacy Access Coordination	C2-Coordinate Access to Legacy C4I services Note 2	DCGS	Teleport	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordination	C2-Coordinate Access to Legacy C4I services Note 2	Teleport	Distant Component	No	Data*	30 Sec**	S
OP5.1.1	Legacy Access Coordination	C2-Coordinate Access to Legacy C4I services Note 2	Distant Component	Teleport	No	Data*	30 Sec**	S

STATUS CHANGE: RF GATEWAY RESOURCES

Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeliness	Classi-Fication ***
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	COMBATANT COMMANDER/TCCC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	JTF/JCCC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	SYSCON	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	G/RSSC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	G/RNOSC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	BMC/OC	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Teleport	Distant Component	Yes	Data*	30 Sec**	S
OP5.1.1	RF Status Change	C2 – Report any change in status of RF Gateway	Distant Component	Teleport	Yes	Data*	30 Sec**	S

STATUS CHANGE: DISN GATEWAY RESOURCES

Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	COMBATANT COMMANDE R/TCCC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	JTF/JCCC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	SYSCON	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	G/RSSC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	G/RNOSC	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Teleport	Distant Component	Yes	Data*	30 Sec**	S
OP5.1.1	DISN Status Change	C2- Report any change in status of DISN Gateway	Distant Component	Teleport	Yes	Data*	30 Sec**	S

STATUS CHANGE: LEGACY C4I GATEWAY RESOURCES

Rationale (UJTL #)	Event	Info Char	Sending Node	Rec Node	Critical	Format	Timeli-ness	Classi-Fication ***
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	COMBATANT COMMANDE R/TCCC	No	Data*	30 Sec**	S
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	JTF/JCCC	No	Data*	30 Sec**	S
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	SYSCON	No	Data*	30 Sec**	S

OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	NCTAMS	No	Data*	30 Sec**	S
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	DCGS	No	Data*	30 Sec**	S
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Teleport	Distant Component	No	Data*	30 Sec**	S
OP5.1.1	Legacy Status Change	C2- Report any change in status of Legacy C4I Gateway Note 2	Distant Component	Teleport	No	Data*	30 Sec**	S

* As primary form, however may be audio or text form
** Or as required by operational mission/user
*** Some IER's may be unclassified due to the unavailability of a secure data and/or voice medium between sending and receiving nodes. This is predicated on the classification of the information being passed.
Note 1: Not later than 2 hours prior to activation
Note 2: As depicted in Appendix B

Figures 4-1 (a - c) provide a top-level operational concept view of the DOD Teleport System's interoperability requirements for access coordination, change in status, and configuration management within the DOD Teleport System. Figures 4-2 (a - c) provide a top-level system concept view for each of these same areas.

FIGURE 4-1(a) Top Level Operational Concept View (OV-1)

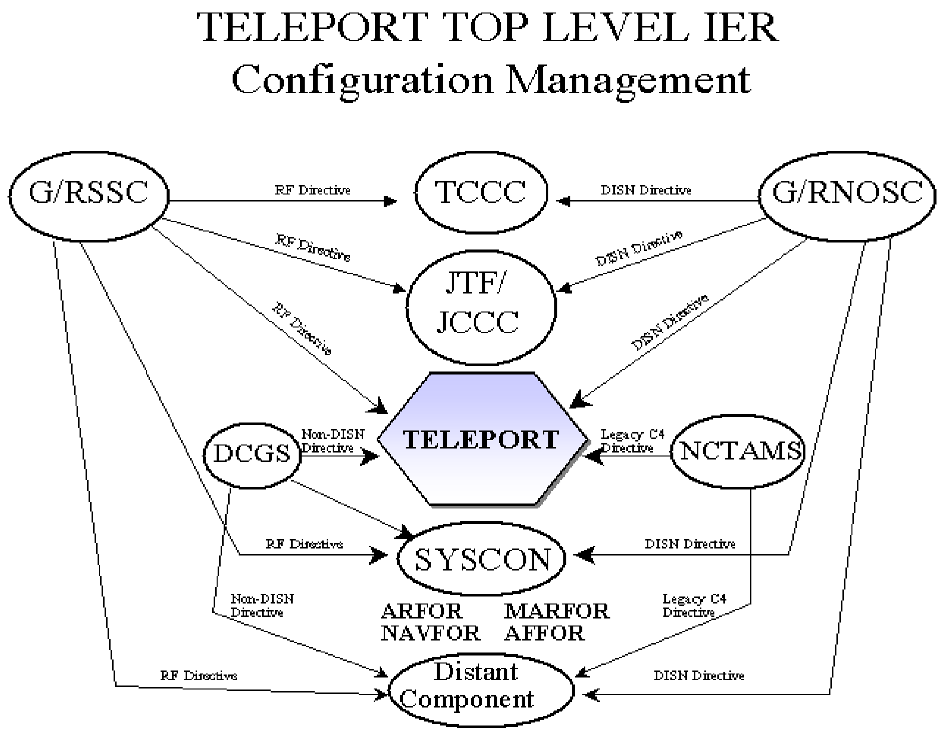


FIGURE 4-1(b) Top Level Operational Concept View (OV-1)

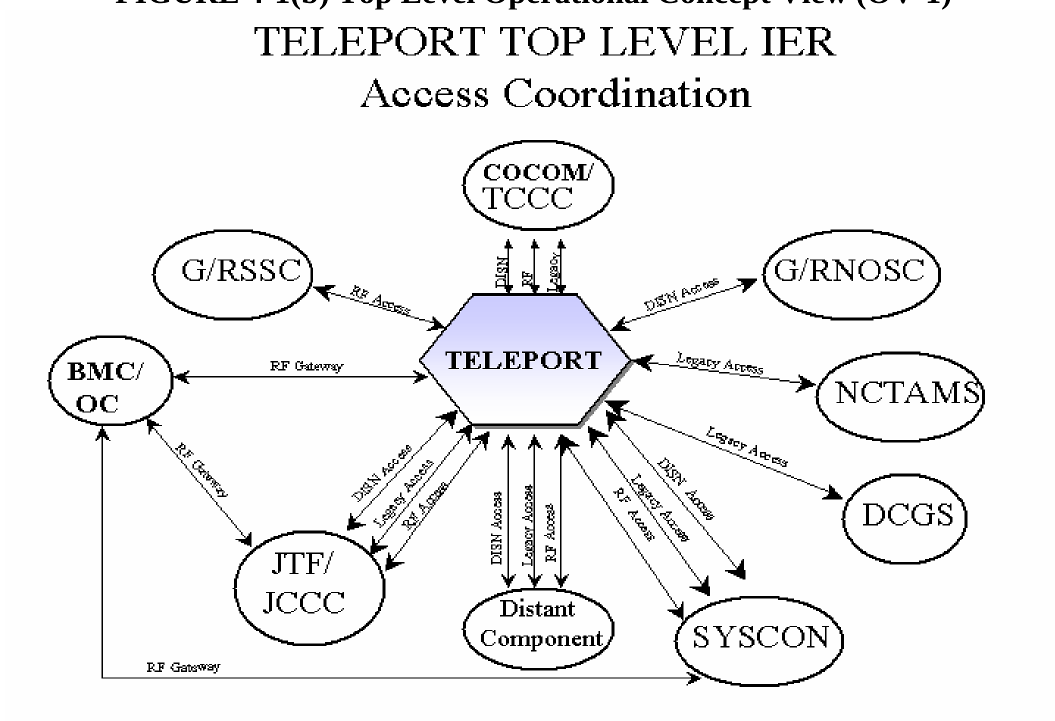


FIGURE 4-1(c) Top Level Operational Concept View (OV-1)

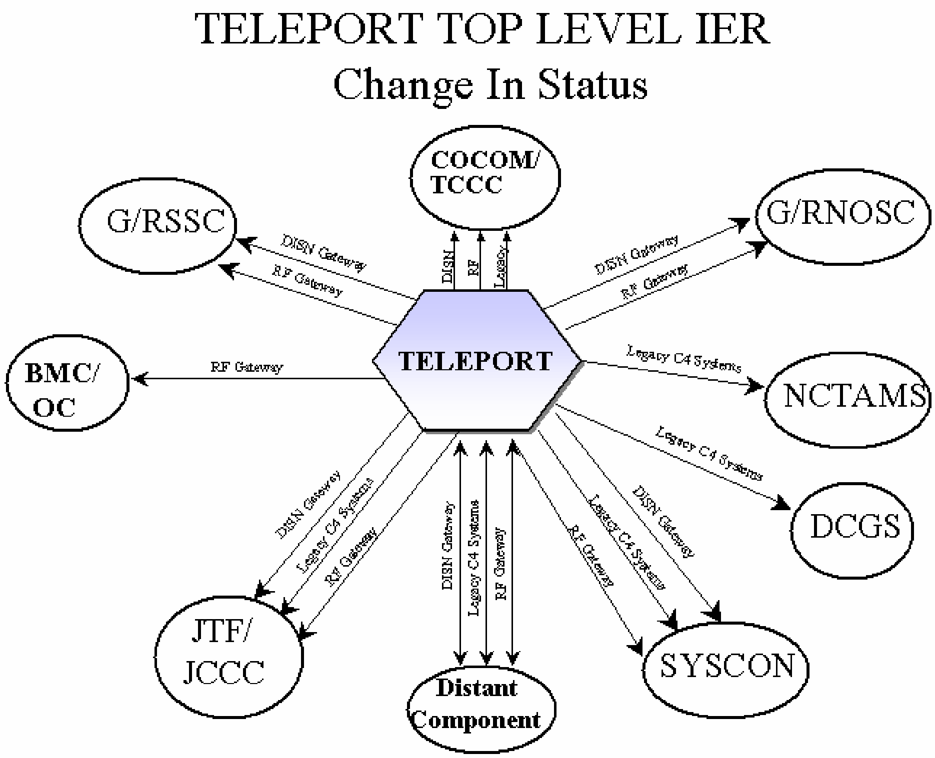


FIGURE 4-2(a) Top Level Systems Concept View (SV-1)

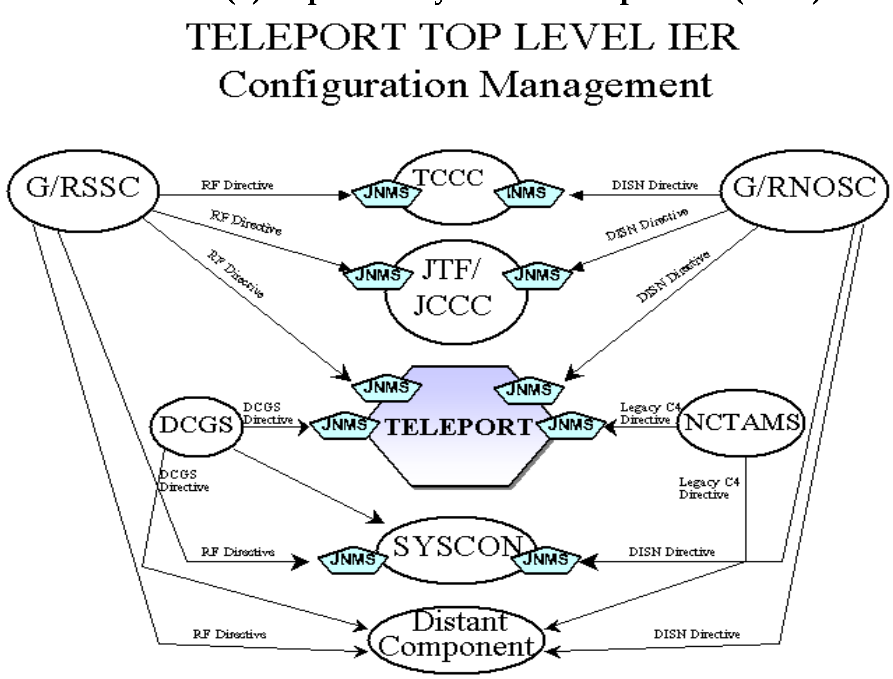


FIGURE 4-2 (b) Top Level Systems Concept View (SV-1)

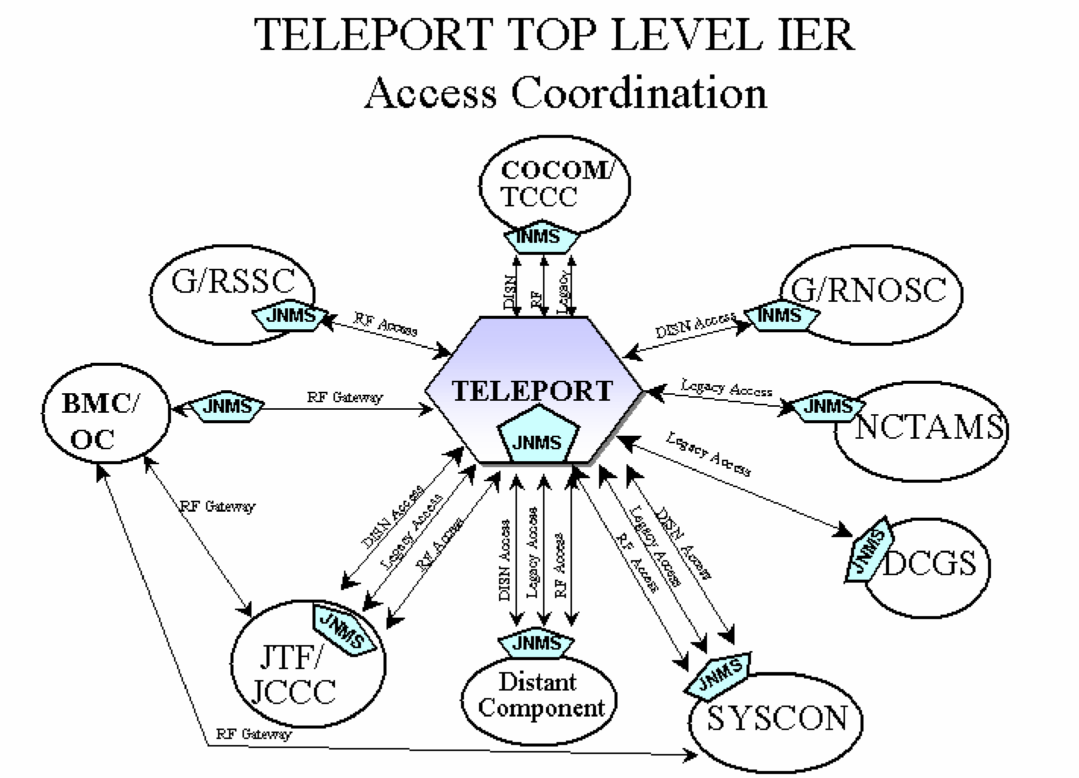
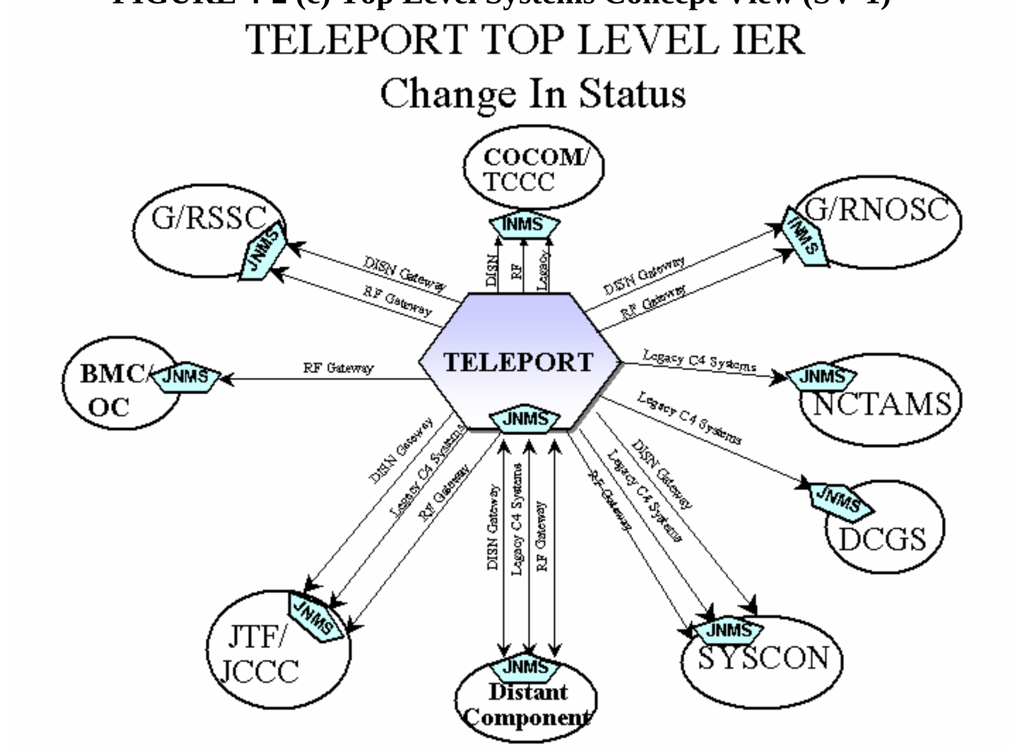


FIGURE 4-2 (c) Top Level Systems Concept View (SV-1)



c. Logistics and Readiness.

(1) Scheduled maintenance must be performed without disruption to system operations and shall not exceed one maintenance man-hour for 24 operating hours. Based on the Teleport system threshold Ao of 99%, the system threshold values are computed as follows: Mean Time Between Critical Mission Failure (MTBCMF) of 868 hours and Mean Time To Restore a Critical Mission Failure (MTTRCMF) of 2 hours, with a 6.77 hour Mean Logistics Delay Time (MLDT). Critical mission failure as applied to the Teleport System, is defined as the failure of an assigned valid communications mission that exceeds 250 milliseconds. All subsystems within DOD Teleport must operate within their validated A_o and not degrade the A_o of any other subsystem. The system shall be logistically supportable in both peacetime and wartime scenarios. Table 4-6 lists estimated Operations and Support Life Cycle costs for the Teleport system based on the preliminary Architecture Feasibility results.

Based on the Teleport facility threshold Ao of 95%, the facility threshold values are computed as follows: MTBCMF 166 hours and MTTRCMF of 2 hours, with a 6.77 hour MLDT. These values are harmonized with facility Ao and account for off-board logistic delay time and planned maintenance considerations.

The facility Ao threshold of 95% also requires maintenance of Ao for existing Programs Of Record (POR) that are integrated into the Teleport. Downtime for the facility Ao is associated with failure of any assigned valid communications mission (communications link), and is consistent with the reliability failure criteria (critical mission failure). Restoration of all communications links will re-establish availability of the facility. This facility availability shall capture suitability aspects of the site beyond, but not independent from, the individual components that are designated "existing programs of record." For example, a facility casualty to site power supplies or climate control equipment may lead to critical mission failures of the facility not specifically attributable to an integrated POR. Conversely, a failure of a POR functionality may cause a critical mission failure that affects the system Ao, as well as its own POR specific Ao. In this case, the design of the facility might allow re-configuration by redundancy to restore all valid communication links and facility Ao. The maintainability of the failed POR would then be evaluated to ensure it could be restored to full operation within its own mandated POR Ao.

(2) DOD Teleport hardware shall be supported by the Defense Logistics System. Spares must be identified and fielded. Additional logistics support requirements will be documented through the Supportability Analysis Summaries (SAS)/Logistics Management Information (LMI) process and in the Product Support Management Plan (PSMP). Where practicable, DOD Teleport components shall be common and modular in order to minimize logistic concerns and allow greater flexibility in the potential for cross matching fielded components.

Table 4-6. Operations and Support Life Cycle Costs

Cost/Parameter	**Threshold (FY2004)**	**Objective (FY2010)**
O&S Cost Drivers per Hour (C_o $) (Note 1)	$3,118	$2,835

Note (1) The Operations and Support (O&S) Life Cycle Cost for the Teleport system is based on a Total Ownership Cost (TOC) approach. The TOC approach analyzes the initial procurement cycle of the Teleport system, the long-term system operation and sustainment, and the technological refreshment of the system through the year 2010. The O&S Cost calculation includes the system operations, sustainment, and technology refreshment components of the TOC, but excludes the initial procurement cycle costs for the Teleport system. The primary costs consist of mission personnel, depot and sparing, sustaining support, technology upgrades, and indirect infrastructure costs. The operating hours for the system are based on a 24-hour daily cycle. The system is assumed to be operational at all times for this calculation. The process for measuring the DOD Teleport System, threshold and objective cost drivers for Gen 1, was captured and included in the metric table below based on architecture alternative #5 of the Architecture Feasibility Study.

<u>Operation and Support Life Cycle Affordability Metric</u>
O&S Cost per Hour Calculation
TOC O&S Costs: $223.48M
Platforms: 6
Cost per Unit: $37.25M/Unit
Operating Hours: 9 years (FY02-FY10) * 365 days/yr * 24 hrs/day
= 78,840 Total Hrs
O&S Cost per Hr (C_o) (Objective): $223.48M/78,840 Total Hours
= $2,835 per Hr
O&S Cost per Hour (C_o) (Threshold): $2835 per Hr * 110%
= $3,118 per Hr

d. <u>Other System Characteristics.</u>

The following additional design features are required by the DOD Teleport system:

(1) Cost/Performance. Cost as an Independent Variable (CAIV) techniques shall be employed in DOD Teleport design and production trade-offs.

(2) Reporting. DOD Teleport must be capable of simultaneously participating in local, theater, and worldwide Service/Agency reporting networks, as operationally required.

(3) Adaptability. Best commercial practices shall be used to ensure adaptability to improvements and evolving C4I technologies and standards.

(4) Service and Access. Subassemblies and components for all segments shall be designed to be readily accessible for removal or replacement without disturbing adjacent parts and without degrading performance.

(5) Security.

(a) DOD Teleport operators require a secure communications capability at the SECRET level.

(b) Compromising Emanations (TEMPEST). A TEMPEST site analysis must be done at each DOD Teleport location.

(c) Computer Security. Any computer or communications system interfacing with, or part of, the DOD Teleport system must comply with the appropriate security classification, Computer Network Defense directives, operating mode, and trusted systems requirements.

(d) Electronic Warfare (EW). The DOD Teleport vulnerability and susceptibility to electronic warfare threat is site/location dependent.

(e) Physical Security. Technical, physical, and procedural security will be used to control access, protect Department of Defense (DOD) information technology resources, and ensure continuous operation of the system within an accredited security posture.

(f) Security certification and accreditation. Information Technology systems that process either unclassified or classified information be registered and undergo a standardized system security certification and accreditation process IAW DODINST 5200.40 and DCID 6/3 Protecting SCI within Information Systems, 5 Jun 99.

(6) Spectrum Certification Supportability. All DOD Teleport communications – electronics equipment/subsystems including any commercial or non-developmental item (NDI) subsystems – shall comply with all DOD, national, and international spectrum management policies and regulations.

(7) Electromagnetic Environmental Effects (E3). The DOD Teleport system shall be electro-magnetically compatible within itself and with other systems in its operating environment, such that system operational performance requirements are met. Electromagnetic interference (EMI) studies must be performed at each Teleport site.

(8) Expandability. The DOD Teleport will be expanded and enhanced in generations to keep pace with technology, and infrastructure expansion, while minimizing fiscal and technical risk.

(9) HEMP Protection. High-Altitude Electromagnetic Pulse (HEMP) protection. The DOD Teleport system will not adversely impact existing HEMP protection requirements as identified in MIL-STD-188-125-1/2, "HEMP Protection for Ground Based C4I Facilities Performing Time-Urgent Missions Part I (Fixed Facilities) and Part II (Transportable Facilities)".

5. Program Support.

Analyses will be conducted and a business plan prepared to determine total ownership cost, initial buy-in costs, resource requirements for life cycle support, identify associations between capabilities and costs, and propose possible tradeoffs. Joint Potential Designator: Joint.

a. Maintenance Planning.

(1) The Teleport system will employ a two-level maintenance concept consisting of Organizational and Depot. Organizational level maintenance shall consist of periodic functional checks and minor maintenance, consistent with current maintenance provided by unit/site personnel. Critical spares (both "on-board" (hot or redundant) and shelf stock times) consisting of any LRU that would constitute a single point of failure or degrade the system operations will be maintained on site. When the system fails to meet acceptable performance standards, organizational maintenance personnel shall initiate troubleshooting procedures to isolate the problem to an LRU. The failed unit shall be replaced with an operable LRU from the supply point or requisitioned from the depot. Upon receipt of a serviceable, LRU, the defective unit will be shipped out for depot/contractor maintenance. Depot maintenance will consist of anything not identified as organizational level maintenance.

(2) Controlled Cryptographic Items (CCI) will be repaired in accordance with COMSEC maintenance procedures.

b. Support Equipment.

(1) The DOD Teleport-specific systems shall use Built-In-Test (BIT) diagnostics that will fault-isolate to a single LRU or component. Details on probability of correct fault isolation, probability of correct detection and probability of false alarm shall be incorporated in system specification documentation.

(2) The DOD Teleport-specific systems and LRUs shall be designed such that standard inventory Test, Measurement, and Diagnostic Equipment (TMDE) and off-the-shelf test equipment may be employed to the maximum extent possible. Standard

interfaces should be utilized for diagnostic ports to eliminate the need for unique or new TMDE.

c. C4I/Standardization, Interoperability, and Commonality.

The DOD Teleport system must be operationally compatible with appropriate Air Force, Army, Marine Corps, Navy, and SOF systems. All SATCOM and fiber interfaces shall comply with open system, non-proprietary network and communications protocol standards. The DOD Teleport must be certified Joint Interoperable in accordance with Department of Defense Directive (DODD) 4630.5, Compatibility, Interoperability, and Integration of C3I Systems, Department of Defense Instruction (DODI) 4630.8, Procedures for Interoperability and Supportability of Information Technology (IT) and National Security Systems (NSS), and CJCSI 6212.01B,Interoperability and Supportability of National Security Systems, and Information Technology Systems. The solution must comply with applicable Information Technology standards contained in the DOD Joint Technical Architecture (JTA) and the Defense Information Infrastructure (DII) Common Operating Environment (COE) level 6 (threshold) and level 8 (objective). Computer resources must comply with applicable information technology standards in the DOD Technical Reference Model (TRM) version 1.0.

The system shall be capable of supporting, as a minimum, the digital transmission media standards listed below as the deployed users and the DISN Long-Haul implement them:

- Fractional T1, T1, T3, DS3, OC3, OC12, E1, and E3
- Narrowband ISDN (N-ISDN)
- Broadband ISDN (B-ISDN)
- Synchronous Optical Network (SONET)

The system shall be capable of supporting, as a minimum, the digital signaling protocol standards listed below as the deployed users and the DISN implement them:

- Frame relay
- X.25 Packet Switching
- ATM
- Ethernet (including 1GBps for IOC)

And the following network layer protocols: IP.

d. Computer Resources.

Software and hardware design must emphasize simplicity, reliability, modularity, security, and Joint (multi-service) interoperability. Emphasis will be placed on utilization of open systems, non-proprietary solutions.

e. Human Systems Integration.

(1) Training. System design shall minimize training costs and time. New Equipment Training (NET) and sustainment/recurring training shall be provided by the contractor or the Services. The training solution will be determined through a Job Task Analysis, Training Task Analysis, Media Analysis and Cost Analysis. Decisions leading towards the selection of the optimal training system must result from an analysis of the job, to include the examination of the task criticality, difficulty, frequency of performance and the probability of deficient performance. All of the resulting data along with resources available, time to train, facilities available, personnel, etc. will assist in the determination of the optimal delivery method, media and instructional strategies. All new follow-on or refresher training developed must be compliant with the DOD Standard Content Object Reference Modem (SCORM).

(2) Manpower. A Human Systems Integration (HSI) evaluation plan will be implemented early in the acquisition process to identify associated workload and manpower and personnel (M&P) requirements. Tradeoffs that identify a reduction in workload or constrain resource requirements will be favored during system design and development in order to reduce life-cycle costs and maximize the return on limited Manpower, Personnel and Training (MPT) resources. Final M&P requirements will be determined based on the HSI evaluation and documented in a Training System Plan. As a threshold, the DOD Teleport System shall not increase qualitative or quantitative operators or maintainers workload. Life-cycle costs associated with changes in workload and manpower, personnel and training, shall be documented in the Product Support Management Plan (PSMP). Manning required for the control, operations, and maintenance of the DOD Teleport Systems may consist of military, government employees, and contractor personnel. However, there shall be no net increase in Military Service end strengths to control, operate, and maintain DOD Teleports.

(3) Human Engineering. All aspects of the DOD Teleport System design, implementation, support, and evaluation shall be engineered with full consideration that the operators, maintainers, and support personnel are integrated to the system. This system shall be designed to conform to applicable DOD human engineering and safety guidelines.

f. Other Logistics and Facilities Considerations.

(1) Supply Support. The existing DOD supply system and reporting procedures is anticipated to support the DOD Teleports. Other alternatives such as contractor-provided support shall be examined and evaluated to achieve maximum operational readiness. The DOD Teleport Program Management Office shall plan and coordinate the acquisition of peacetime operating stock spares and readiness spares packages to ensure the minimum essential number of spares is available at time of system delivery. In addition, sufficient spares must be in place during operational testing. Stock-listed, off-the-shelf equipment will be used whenever possible.

(2) Technical Data.

(a) Technical data describing system/LRU theory, system wiring diagrams system removal and installation; component removal, repair, and replacement; and an illustrated parts breakdown to facilitate parts ordering shall be provided to the government. All fault isolation manuals and fault reporting manuals shall be included. Validated and verified technical data and maintenance technical orders shall include instructions with sufficient detail to certify operators and maintenance technicians. The technical contractor shall provide interactive electronic technical manuals (IETM). Any technical manuals for military unique equipment shall be developed in accordance with (IAW) current IETM contract requirement documents. IETMs shall be validated, verified, fully incorporated into the baseline manuals, and delivered to the user not later than 30 days prior to production kit delivery. IETMs should be delivered in a digital format that supports a print-on-demand and view-on-demand capability. Validated, red-lined technical data shall be required for any operational tests or evaluations and associated maintenance training.

(b) Product technical data packages shall be procured as necessary for use in preparation of technical data, depot-level repair, damage and repair analysis, and modifications. For “commercial off-the-shelf/non-developmental item” (COTS/NDI) systems, unlimited rights to source drawings used to manufacture/procure or maintain components are not required. However, provisions for transfer of complete data package and rights are desired in the event a manufacturer goes out of business or drops the items from production. In addition, provisions for unlimited use of data for other than procurement purposes is desired.

(3) Facilities and Land. Existing facilities and infrastructure shall be used to the fullest extent possible. However, major node sites are at or very near their space and infrastructure limitations. For O&M commands to address Teleport facility and infrastructure-identified shortfalls with minimum impact to fielding schedules, an early identification of these concerns is necessary. The capability of the existing facilities and infrastructure to support additional electronic equipment must be verified using a holistic approach to integrate with all planned programs at the beginning of the acquisition planning process. Host Nation Approval (HNA) will need to be negotiated when establishing DOD Teleport military and commercial satellite terminals/links in foreign countries.

(4) Hazardous Materials and Ozone Depleting Chemicals (ODCs). This system shall not be constructed with or use any hazardous materials or ODCs in its development, production, operation, and maintenance. Additional requirements for hazardous waste disposal over the existing STEP system are not envisioned. The DOD Teleport system shall incorporate appropriate DOD guidance for environmental pollution prevention into all aspects of its management strategy. When disposed of in accordance with DOD standard procedures, the system shall pose no inherent environmental hazards.

(5) Supporting Command Requirements.

(a) Packaging, Handling, Storage, and Transportation. All equipment should be designed to minimize requirements for weight, volume, and complexity, and be transportable by air, land or sea.

(b) Warranty. Warranties should be pursued within the purview of applicable DOD guidance. Warranties and guarantees should cover all spares and technical data shortfalls.

g. Transportation and Basing.

The DOD Teleport system is not envisioned to be deployable. However, additional transportation and basing requirements may develop from the need to support a surge capability.

h. Geospatial Information and Services (GI&S).

National Imagery and Mapping Agency (NIMA) approved products and services will be required to determine antenna locations at the selected DOD Teleport locations.

i. Natural Environmental Support.

DOD Teleport link performance can also be degraded by tropospheric weather effects, such as heavy precipitation, icing, and high winds. Forecasts of Meteorology and Oceanography (METOC) conditions affecting SATCOM and Teleport subsystems will be provided as standard products from a Joint or Service METOC activity. These products are available through the METOC component of the GCCS.

6. Force Structure.

The DOD Teleport system will be acquired through direct acquisition or other acceptable business practices such as leasing or indirect service charges/fee-for-service basis. Every opportunity to leverage existing infrastructure, such as STEP and the NCTAMS, must be considered to lower overall program costs and to maximize efficiencies. It is estimated that as many as 19 sites will be incorporated into the global Teleport system, with six Teleport core sites planned.

Follow-on efforts will involve maintaining technological currency to ensure that DOD Teleport System is always able to support deployed forces. Fielding of new deployable systems may require increases in throughput capacity or additional capability to receive/transmit on different frequencies. The goal of follow-on efforts is to seamlessly share data with all services in support of combat and OOTW.

7. Schedule Considerations.

As previously discussed, DOD Teleport is a generational implementation, consisting of four generations, Generations 0 –3. Generation 0 consists of analysis, studies, and concept demonstrations. Generation 1 includes the system Initial Operational Capability (IOC).

a. <u>IOC (Generation One) Actions.</u>

Based on Combatant Commander and Service fielding priorities for RF media, IOC will occur in two phases. The Combatant Commanders have demanded increased SATCOM capacity particularly in C, X and Ku bands in order to support their warfighting requirements.

IOC1 will be achieved when DOD Teleports are equipped with or have access to operational terminals in the X, C, and Ku bands. This architecture consists of two core sites and four split sites. *(IOC1 was originally planned for FY 02 in the 31 Jul 00 approved ORD, however funding and acquisition decisions have delayed it until FY03)*

IOC2 will be achieved when the following actions have occurred:

(1) DOD Teleports are equipped with or have access to operational terminals in the X, C, Ku, and UHF bands. This architecture consists of two core sites and four split sites. End-to-end compatibility and linkage must exist in the communications systems and have a Joint Interoperability Certification completed by the Joint Interoperability Test Command (JITC). Also, DISN SDN circuits must be in operation to adequately support the DISN Service threshold throughput requirements for the X, C, Ku, and UHF bands.

(2) Successful completion of Initial Operational Test and Evaluation (IOT&E). DOD Teleports must also be tested and certified for use with military and commercial satellites.

(3) Operations and maintenance training programs established in all designated training avenues.

(4) Unit personnel trained on DOD Teleport System inspection, maintenance, and operation.

(5) Service organic logistics support for initial spares replenishment established.

(6) Necessary supporting documentation approved and provided.

IOC is reached when all Generation One performance threshold parameters are achieved.

IOC3 will be achieved when the following actions have occurred:

(1) DOD Teleports are equipped with additional C, Ku, and UHF terminals and expanded to six core sites.

(2) DOD Teleports are equipped with EHF (LDR/MDR) terminals. This architecture consists of six core sites and a secondary site in Bahrain.

IOC4 will be achieved when the following actions have occurred:

(1) DOD Teleports are equipped with or have access to operational terminals in the military Ka band.

(2) DOD Teleports obtain technology refreshments consistent with commercial industry standards.

b. Generation Two.

Generation Two consists of fielding the military Ka band capability in its entirety, as well as new capabilities and technology refreshments. Generation two is currently planned for FY 2004-2007. (See Appendix D).

c. Full Operational Capability (FOC) (Generation Three) Actions.

FOC will be achieved when the following actions have occurred:

(1) All DOD Teleports are equipped with operational terminals in all threshold bands and all modifications to meet threshold requirements have been completed. End-to-end compatibility and linkage must exist in the communications systems. Also, DISN SDN circuits must be in operation to adequately support all DISN Service threshold throughput requirements. Date is coincident with the end of generation three, FY 2012.

(2) Successful completion of Follow-on Operational Test and Evaluation (FOT&E). DOD Teleports must be tested and certified for use with military and commercial satellites system and have a Joint Interoperability Certification completed by JITC.

(3) Training program changed/updated to reflect lessons learned since IOC and fully implemented.

(4) Operating manual changed/updated to reflect lessons learned since IOC.

(5) Supply support expanded to accommodate increased numbers, usage, and maintenance of DOD Teleport System.

8. Program Affordability.

DOD Teleports must provide maximum operational capabilities to meet warfighter requirements while optimizing acquisition, procurement, and overall life cycle costs. Table 8-1 provides the threshold and objective cost parameters for the

Teleport system based on the Draft Architecture Feasibility study conducted by Johns Hopkins University/Applied Physics Laboratory in June 2000. A listing of the Generation One Teleport sites, and a by-site breakout of costs for each, is contained in this study.

Table 8-1 Program Costs

Cost/Parameter*	Threshold (Gen 1-3)	Objective** (Gen 1-3)
Procurement Cost (Equipment)	$177M	$128.6M
Procurement Installation	$ 63M	$ 40.9M
RDT&E	$ 53M	$ 36.5M
Operations and Maintenance (Delta)	$ 224M	$190.2M
Connectivity	$ 424M	$325.7M
Total cost	**$941M**	**$721.9M**

**Based on architecture alternative #5 estimates*

***Objective costs were determined by the Architecture Feasibility team by: removing half of the new wideband accesses, two-thirds of the additional Ka accesses, and half of the new EHF terminals; assuming a 10% per year decline in terrestrial connectivity cost; and adding one surge package capable of supporting the 2010 SSC access requirements. The result is a Teleport system that has fewer than the required accesses at each site, but can be augmented by the surge package in 30 days. Even though the accesses are decreased below the requirement, the total throughput is about that required by a 2010 SSC. The total cost decreases by 23%.*

Appendix A. List of References

(1) The DISN Mission Need Statement (MNS), JROCM 047-95, 30 March 1995.

(2) The DISN Capstone Requirements Document (CRD), JROCM 048-96, 15 April 1996.

(3) Integrated Communications Database (ICDB) and Emerging Communications Database (ERDB) 01 Apr 2000.

(4) The Satellite Communications (SATCOM) Capstone Requirements Document (CRD), US Space Command, (Draft) January 2003.

(5) DOD Requirements for Commercial SATCOM Services, Coordination Draft, DISA, September 1998.

(6) The Joint Staff DISN Deployed Strategy, J-6A 00526-96, 20 June 1996.

(7) Joint Vision 2020, June 2000.

(8) The DISN Architecture, DISA, September 1996.

(9) The Defense Planning Guidance, Fiscal Years 2004-2009, May 2002.

(10) The Defense Information Infrastructure (DII) Master Plan, Version 8.0, "Implementing the Global Information Grid", 24 May 1999.

(11) CJCSI 6250.01A, Satellite Communications, 10 December 2001.

(12) MILSATCOM System Threat Assessment Report (STAR), NAIC-1574-037-03, July 2003 (S/NF).

(13) Information Operations Capstone Threat Capabilities Assessment DI-15771203, August 2003. (S/NF).

(14) Joint Task Force Tactical Communications Architecture Functional Requirements Specifications, JIEO Report 8125, DISA, August 1996.

(15) DOD Joint Technical Architecture (JTA) Version 4.0, 17 July 2002.

(16) CJCSI 6212.01B, Interoperability and Supportability of National Security Systems and Information Technology Systems, 8 May 2000.

(17) CJCSI 6510.01C, Information Assurance and Computer Network Defense, April 2001.

(18) DOD Teleport CONOPS, Ver.3.1 dated 20 September 2002.

(19) DOD Directive 4630.5, Interoperability Supportability of Information Technology (IT) and National Security Systems (NSS), 2 May 2002.

(20) DOD Instruction 4630.8, Procedures for Interoperability and Supportability of Information Technology (IT) and National Security Systems (NSS), 2 May 2002.

(21) DOD Regulation 5000.2-R, 2 April 2002, "Mandatory Procedures for Major Defense Acquisition Programs (MDAP) and Major Automated Information System (MAIS) Acquisition Programs (Effective 30 October 2002, this document has been cancelled. The Interim Defense Acquisition Guidebook is being used in its place)."

(22) CJCSI 3170.01C, Joint Capabilities Integration Development System, 24 June 2003.

(23) MIL-STD 2169B, HEMP Environment, 17 December 1993.

(24) DOD Teleport System, Architecture Feasibility Study Results, Final Report, June 2000.

(25) CJCSI 6211.02A, Defense Information Systems Network and Connected Systems, 22 May 1996.

(26) DOD Technical Reference Manual (TRM), Version 1.0, 5 November 1999.

(27) Electronic Combat Threat Environment Description, NAIC-1574-0731-01, February 2001.

(28) DOD Instruction 5200.40, Information Technology Security Certification and Accreditation Process (DITSCAP), 7 October 1999.

(29) DOD Instruction 5000.2, "Operation of The Defense Acquisition System," 4 January 2001 (Effective 30 October 2002, this document has been cancelled and is currently under revision.)

(30) Distributed Common Ground/Surface System (DCGS) Capstone Requirements Document (CRD), 6 January 2003.

(31) DOD C4ISR Architecture Framework, Version 2.0, 18 December 1997.

(32) DOD Teleport Earth Terminal Specifications SR-110-00-0008, Nov 2001.

(33) DOD Teleport Program Support Management Plan (PSMP) Version 1.0, SL-100- 00-0001, Apr 2002.

(34) DOD Teleport Test and Evaluation Master Plan (TEMP) ST-100-00-0001, Mar 2002.

(35) DOD Teleport Acquisition Strategy SP-100-00-0001, 5 April 2002.

(36) DOD Teleport Command, Control, Communications, Computers, and Intelligence (C4I) Support Plan (C4ISP), SO-200-00-0001, 1 August 2002.

(37) Global Information Grid (GIG) Capstone Requirements Document (CRD), JROCM 134-01, 30 August 2001.

(38) NSTISSP No.12 National Information Assurance (IA) Policy for U.S. Space Systems, January 2001.

(39) Mission Information Management (MIM) Architecture, January 2001.

(40) DOD Directive 8500.1 "Information Assurance," 24 October 2002

(41) USSAN 1-69 United States Implementation of NATO Security Procedures

(42) CJCSI 6740.01A, Military Telecommunications Agreements and Arrangements Between the United States and Regional Defense Organizations or Friendly Foreign Nations, 15 May 2002.

(43) DCID 6/3 Protecting SCI within Information Systems, 5 Jun 99.

Appendix B. List of Non-DISN/Legacy Services and Systems

Table B-1. List of Non-DISN/Legacy Services

Table B-1 provides a list of warfighter services that must be supported by the Teleport System, that are not DISN services. These services are typically tactical and or Service unique and are typically transported to a processing center. The quantities reflected represent the requirement per SATCOM link for each Teleport site. The intent of the Teleport is to provide joint, interoperable communications for all warfighters; however, the warfighter still requires some legacy based systems until such time as interoperable applications can be achieved by all Services. As such, the Teleport should make an attempt to provide enough capacity to handle as many legacy circuits as deemed possible without jeopardizing the overall joint communications capability. Ultimately, the community must show an overall migration to a joint architecture with joint applications.

Navy					
TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*
BFEM					9.6 kbps x 4
HF Broadcast					9.6kbps x 4
TADIXS A		2.4 kbps	2.4 kbps (via NECC)		
IBS Simplex		2.4 kbps			
CUDIXS		2.4 kbps			
SSIXS		4.8 kbps			
MDU Direct			2.4 kbps (LDR)		
OTCIXS		2.4 kbps	2.4 kbps (via NECC)		
POTS (8kbps compression)	112 kbps			192kbps (C-band)	
POTS (9.6 kbps)				19.2 kbps (L-band)	
POTS (16 kbps)	192 kbps			192 kbps (C-band)	
RWI		2.4 kbps			
JSIPS-N	512kbps – 2048kbps			512 kbps - 2048kbps	
TACINTEL		4.8 kbps			
VIXS	128 kbps			128-384 kbps	
Tri-Tac	288-576 kbps				
ADNS	256kbps		128-384kbps	384kpbs (C-band); 32-64kpbs(L-band)	
JFN	512 – 1544 kbps		512kbps	512-1544kbps	
APTS				128 kbps (C-band); 7.2 kbps (L-band)	
NIDTS/COWAN	64kbps				
JDISS	9.6 kbps				
Serial Link 11	9.6 kbps				
Gateguard	2.4 kbps				
TESS	2.4 kbps				

Telemedicine	56kbps			56 kbps	
Teletraining with ISDN PSTN access	128kbps			128kbps	
Moving Target Enhanced System (MTES)		2.4-19.2kbps			

Air Force

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*
DCGS	X				
HFGCS					X
GBS	X				

Army

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*
TRI-TAC/MSE	1152kbps				
Trojan	1544kbps				

USMC

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*
TRI-TAC	1152kbps				
CUDIXS		2.4 kbps x 2			
TROJAN	1544kbps			1544kbps	

Glossary of services:

ADNS: The Automated Digital Network System (ADNS) is the backbone to Navy's IP migration. ADNS uses off the shelf routers and processors, and JTA compliant protocols to establish a secure, efficient, quality of service enabled, multi-path, adhoc networking architecture. The ADNS architecture provides advanced IP routing and security features including: a single shore autonomous system to simplify shifts of IP services, multiple path IP routing, bandwidth reservation for different security levels, dynamic reallocation of unused bandwidth, IP quality of service, IP packet filtering and access controls.

BFEM: Battle Force E-Mail utilizes HF spectrum to pass battle group email via ship to ship or shore to ship.

CUDIXS: The Common User Digital Information Exchange Subsystem is an automated GENSER communications message processing network that supports surface ships and other mobile users over half duplex UHF and EHF LDR SATCOM networks.

DCGS: Distributed Common Ground/Surface System (DCGS) Direct Downlink. Provides the Intelligence, Surveillance and Reconnaissance (ISR) platforms with additional downlink locations to augment the DCGS. DCGS currently uses Tri-band (X, C, & Ku) SATCOM for its ISR reach back architecture. SATCOM terminals are migrating to support Ka-band in the future. The DCGS receives information from various ISR platforms and sends that information over its own WAN to provide for distributed processing and analysis.

HFGCS: The HFGCS system is a DOD Global Network, providing Automatic Link Establishment (ALE) for HF radio. As aircraft and other communications systems are replaced with Joint Tactical Radio System (JTRS) software compliant architecture radios, the need for cross banding radio systems will increase.

IBS Simplex: Integrated Broadcast Service Simplex, formerly TDDS, is a multi-source, global satellite broadcast dissemination system that supports the rapid and continuous delivery of vital, time-critical intelligence data to operational users deployed world-wide at the secret level over a UHF satellite broadcast. Simplex is a hub-based architecture, with the network management center (NMC) receiving all intelligence over the terrestrial communications network (TCN). The NMC will prioritize the information in accordance with each Combatant Command's guidance and transfer the intelligence to the simplex uplink sites for broadcast.

OTCIXS: The system manages computer-to-computer communications for the receipt and transmission of target and other tactical contact data over half duplex UHF and EHF LDR SATCOM networks. They are particularly important for Over-the-Horizon Targeting (OTH-T) where accurate real time data is crucial in generating a fire control solution.

Mission Data Update (MDU) Direct: Tomahawk missile mission data transferred over and EHF LDR point-to-point circuit or "call" from one of two shore-based cruise missile support activities.

RWI: Radio Wireline Interface: A shore based device providing the capability for a shore operator to connect a tactical user of a half duplex secure voice network (i.e. HF, UHF SATCOM or EHF LDR) to a STU-III user ashore over a DSN or Public voice network.

SI SSIXS: The SI Submarine Satellite Information Exchange Subsystem is an automated SI/SCI communications message-processing network specifically for submarine communications over UHF SATCOM.

SSIXS: The Submarine Satellite Information Exchange Subsystem is an automated communications message-processing network specifically for submarine communications over UHF SATCOM.

TACINTEL: The Tactical Information Exchange Subsystem is an automated SI/SCI communications message-processing network specifically for surface ship communications over UHF SATCOM.

TADIXS A: Tactical Data Information Exchange Subsystem is an automated, one-way, shore-to-ship communications processing network for the broadcast of tactically relevant over-the-horizon data to the battle force commanders over UHF or EHF LDR SATCOM.

TRI-TAC: Tri-Service Tactical Communications System.

VIXS: The Video Information Exchange System (VIXS) is a secret and unclassified level Video Teleconferencing (VTC) network established to meet the CNO levied requirements for a VTC system that supports dedicated, highly responsive, real-time combat and combat support collaboration for CNO, Fleet Commanders, Numbered Fleet Commanders, Battle Force Staffs and individual ships.

The intent of Table B-2 is to provide a list of Service communications systems that will be used to access a Teleport site. Each Teleport site will therefore be capable of matching each of these system's capabilities to ensure compatibility with each Service system and user.

Table B-2. List of Service Communications Systems

Navy

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*	*Mil-Ka*
TRI-TAC	288-576 kbps					
AN/WSC-6(V)5 and 7	X					
AN/WSC-6(V)9	X			X C-band		
AN/WSC-3(V)		X				
AN/WSC-8(V)1 and 2				C-band		
AN/USC-38(V)1 and 2			X LDR only			
AN/USC-38(V)4,5 and 9			X LDR&MDR			
AN/USC-38(V)8,11 &12	X		X LDR&MDR			
INMARSAT B-Nera Saturn B w/128kbps modem upgrade				X L-band		
Timeplex Link 2+	X			X C-band		
ON-143(V)6&14		X Data circuits				
Navy EHF Communications Controller			X Data Circuits LDR only			

Air Force

TITLE	*SHF*	*UHF*	*EHF*	*COML (C/Ku)*	*HF*	*Mil-Ka*
TCS-152 LMST	X			C, Ku		X
TSC-65(V) LMST	X			C, Ku		
USC-60A FTSAT	X			C, Ku		
TSC-94A	X					
TSC-100A	X					
TSC-85B	X					
TSC-93B	X					
PSC-5		X				
A/N PSC-11 (SCAMP)			X			
A/N TSR-4,7 (GBS)	X			X		
USC-60A	X			C/Ku		
TRI-TAC	X					
AN/TSC-154 (SMART-T)			LDR/MDR			

LST-5D		X				
NSM 8448 TSSP	X			C, Ku		
TD-1337 TSSP	X			C, Ku		
SMX-1000	X			C, Ku		
MD-1026	X					
SB-3865	X					
TTC-39	X					
TTC-42	X					
IGX*C voice switch	X			C, Ku		
FCC-100	X			C, Ku		
PROMINA 400	X			C, Ku		
PROMINA 800	X			C, Ku		
AN/TRC-170 (Tropo)	X			C, Ku		
INMARSAT A,B,C, M, Mini-M, M-4, Aero-H, Aero-I, with 128kbps modem upgrade				L Band		
DCGS requirements as identified in SDB.						

Army

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*	*Mil-Ka*
TRI-TAC	1152kbps					
AN/PSC-5		X				
AN/TSC-86A/B/C	X					
AN/TSC-85B/C	X					
AN/TSC-93B/C	X					
AN/TSQ-19O(V) Trojan Spirit II				Coml Ku		
TFT-C						
AN/TSC-154 (SMART-T)			X LDR/MDR			
Large Fixed Ka						
AN/ GSC-52	X					
AN/GSC-39	X					
AN/FSC-78	X					
Phoenix	X			C and Ku		
AN/PSC-11 SCAMP			X LDR			
INMARSAT B, M, Mini-M, M-4, M4-128 kbps				L Band		
AN/USC-60A	X			X		
AN/TRC-143	X,C,Ku-band					

USMC

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*	*Mil-Ka*
TRI-TAC	Consists of Terminals listed below, with					

	noted data rates.					
AN/TSC-85C	1152kbps					
AN/TSC-93C	1152kbps					
LMST AN/USC-65	2048kbps			1024 kbps		
SMART-T AN/TSC-154			1.544kbps			
TROJAN	1544kbps			1544kbps		
AN/TSQ-190	2048kbps			2048kbps		
TROJAN LITE	1544kbps			1544kbps		
OTHER						
AN/TSC-96A		2.4kbps				
AN/PSC-5		16kbps				
INMARSAT Various				128kbps *Earth Station termination not required		
HF Radio					2.4 kbps	

Special Operations Forces (SOF)

TITLE	*SHF*	*UHF*	*EHF*	*COML*	*HF*	*Mil-Ka*
AN/USC-160	X			C/Ku		
AN/USC-65	X			C/Ku		
AN/PSC-11(SCAMP)			X			
Inmarsat B,M,M-4, Mini-M				L-Band		
LMST	X			C/Ku		
GBS				Ku		
AN/TSC-85C	X					
AN/TSC-93C w/TSSP	X					
AN/TSQ-190 Trojan Spirit	X			Ku		
JBS		X			X	
AN/PSC-5C		X				
AN/PSC-5D(MBMMR)		X				
AN/PRC-117 (HSW)		X				
AN/PRC-150					X	
AN/TRQ-43					X	
AN/PRC-137F(SMRS)					X	
IBS		X				

Appendix C.
SATCOM/DISN CRD KPP Crosswalk

C.1 Introduction

The operational requirements for the DOD Teleport are derived from the SATCOM, Defense Information Systems Network (DISN), and Global Information Grid (GIG) Capstone Requirements Documents (CRD). The SATCOM CRD's Capstone KPPs and other requirements provide the formal broad framework, boundary conditions, performance goals, and essential SATCOM considerations addressed by the Teleport ORD. The DISN CRD provides the critical baseline capabilities required for system or C4I specific ORDs that require DISN connectivity to support the warfighter. Although it does not include Capstone KPPs, the DISN CRD does provide critical system characteristics necessary for any DISN related ORDs. The GIG CRD was formalized 30 Aug 01. A Teleport system crosswalk is provided in Appendix D, Part 2 and applies to Generations Two and Three. While the DCGS CRD is referenced, a crosswalk is not applicable. The Teleport System will not directly interface DCGS and is not included in the DCGS Family of Systems.

C.2 CRD-ORD Requirements Linkage

C.2.1 The Satellite Communications (SATCOM) Capstone Requirements Document (CRD)

The DOD SATCOM Architecture is comprised of a number of individual SATCOM systems, each supporting specific user requirements. The Senior Warfighter Forum (SWarF) developed a SATCOM course of action that assigned the CRD requirements against specific existing and future systems. Currently, DOD Wideband SATCOM services are supported by DSCS and UFO-GBS. These systems will be augmented by the Wideband Gapfiller System (WGS) until they are eventually replaced by the future wideband systems. DOD Narrowband SATCOM services are currently supported by FLTSATCOM and UFO, with the Mobile User Objective System (MUOS) planned for future support. DOD Protected SATCOM services are supported by Milstar, UFO-EHF and the EHF Polar Package today with the Advanced EHF program scheduled to replace the Milstar satellites. DOD will also continue to supplement these MILSATCOM systems with commercial SATCOM services. Beginning in FY09, DOD will implement the Transformation Communications Architecture, which will significantly increase MILSATCOM capabilities. The SATCOM elements of the time phased Teleport requirements were derived from this course of action. Table C-1 was extracted from the SATCOM CRD (Table 4-1) and summarizes each of the CRD's Capstone KPPs. The following paragraphs describe the linkage between the SATCOM CRD KPPs and Teleport System requirements.

Table C-1 - SATCOM CRD Key Performance Parameters

KPP	*Threshold*	*Objective*
COVERAGE	Coverage of users operating anywhere in the worldwide and North Polar areas is required.	Coverage of users operating anywhere in the global coverage area is required.
CONNECT-IVITY	The SATCOM FoS must provide SATCOM and data relay capacity to support the full range of projected DOD, IC, and NASA operations.	Threshold plus each individual SATCOM system or service within the SATCOM FoS should have the reserve capacity to accommodate surge loading and support multiple operations.
	Provide protected capacity to users and networks deemed most at risk to disruption of their SATCOM links with the protective features (e.g., AJ, AS, nuclear effects resistant, HEMP-hardened, LPI/LPD/LPE, and/or US Control) sufficient for them to operate as intended in their postulated threat environments.	Provide protected capacity to all users and networks with the protective features (e.g., AJ, AS, nuclear effects resistant, HEMP-hardened, LPI/LPD/LPE, and/or US Control) sufficient for them to operate as intended in their postulated threat environments.
INFORMATION ASSURANCE	SATCOM systems and services must have the ability to avoid, prevent, negate, or mitigate the degradation, disruption, denial, unauthorized access or monitoring, and/or exploitation of sensitive and classified information that originates in, is conveyed by, or provided to them.	SATCOM systems and services must have the ability to avoid, prevent, and/or negate the degradation, disruption, denial, unauthorized access or monitoring, and/or exploitation of sensitive and classified information that originates in, is conveyed by, or provided to them.
OPERATIONAL MANAGEMEN T	The operational management systems supporting the overall SATCOM FoS/SoS and its constituent parts must provide platform, payload, and network management, control, and configuration capabilities to plan and perform spacecraft launch and early orbit (L&EO) activities, on-orbit operations, network support, and satellite disposal.	Objective equals threshold.
INTER-OPERABILITY	The SATCOM FoS and its constituent systems must satisfy all of the top-level IERs designated critical in Appendix E.	The SATCOM FoS and its constituent systems should satisfy all of the top-level IERs identified in Appendix E.
	Interoperability must exist between and among the SATCOM networks of all operational elements (ground, air, SOF, maritime, intelligence, and support forces and related activities) with which they will form military or inter-agency mission task forces or otherwise be conducting operations.	Objective equals threshold.
OPERATIONAL SUITABILITY	The various constituents of the SATCOM FoS must comply with the minimum performance standards of their intended user communities' information systems.	The various constituents of the SATCOM FoS must comply with objective performance standards of their intended user communities' information systems.

C.2.1.1 Connectivity.

At the threshold level, the CRD Connectivity KPP requires the ability to provide world-wide SATCOM coverage. "World-wide coverage" is generally defined as the surface of the earth between 65° S and 65° N latitude and at all longitudes. North polar coverage is defined as the area above 65° N latitude and at all longitudes.

The Teleport Coverage KPP builds on the CRD requirement, calling for "world-wide coverage" and Warfighter access to a minimum of two (2) DOD Teleports provided that satellite coverage exists for that area. The interoperability and capacity requirements are time-phased in accordance with the current SATCOM Course of Action (COA) and the anticipated satellite constellations. The additional detail in interoperability and capacity ensure that space segment limitations do not make Teleport requirements unattainable.

The SATCOM CRD Connectivity KPP calls for providing the requisite amounts of the various types of wideband and narrowband capacities to support warfighters and their supporting infrastructure. As a threshold, the KPP calls for enough wideband capability to meet the essential needs of the deployed warfighting forces and the OCONUS requirements of the DOD's supporting infrastructure (e.g., DISN, NSA, and DTS). The CRD defines capacity in terms of the kinds and amounts of throughput available to the targeted user population as measured against specific employment scenarios.

The Teleport Capacity KPP calls for constant capacity to DISN and Legacy C4I Services via wideband, narrowband, protected, and commercial SATCOM systems. The common thread for all capacity requirements is the focus on deployed forces and OCONUS warrior support activities.

C.2.1.2 Information Assurance.

The SATCOM CRD Protection KPP calls for SATCOM system capabilities to avoid, prevent, negate, and/or mitigate the degradation, disruption, denial, unauthorized access, or exploitation of services or resources. Protection requirements including survivability against nuclear effects, protection against jamming, limiting the probability of signal detection or interception, control over the SATCOM resources and the prevention of unauthorized access to, or disclosure of, information are levied against specific systems and programs.

The Teleport, although not intended to support all protection requirements (e.g. nuclear survivable protection requirements), includes Information Assurance and necessary control requirements in the protection KPP. Through the SATCOM systems included in the Teleport interoperability KPP, varying levels of protection are specified for access to DISN and Legacy C4I Services.

C.2.1.3 Operational Management.

The SATCOM CRD Operational Management KPP addresses the ability of Unified Combatant Commands and JTF Commanders to dictate resource utilization of their apportioned resources and the ability to rapidly and dynamically configure/reconfigure resources. At the threshold level this includes the capabilities needed to plan, allocate, and schedule apportioned resources and the capability to rapidly and dynamically configure SATCOM resources.

The Teleport Control and Network Management KPP supports the SATCOM Operational Management requirement through timely, automated control measures at the Teleport. Teleport control functions will ensure that internal Teleport control functions are in step with any changes to SATCOM loadings or configurations.

C.2.1.4 Interoperability.

At the threshold level the SATCOM CRD Interoperability KPP requires interoperability between and among the satellite communications capabilities of Unified Command and JTF components and between the DOD's satellite communications and the rest of the Global Information Grid (GIG).

Through access to DISN services and SATCOM systems, the Teleport Interoperability KPP supports the CRD KPP for integration with the GIG. Additionally, cross banding requirements provide interoperability between Joint and/or Allied and Coalition forces operating in different bands.

C.2.1.5 Operational Suitability.

The SATCOM CRD Operational Suitability KPP requires compliance with the minimum performance standards of their intended user communities' information systems.

The Teleport Technology Insertion requirement calls for use of open systems, standardized products and protocols that are at a minimum of level 6 DII COE and JTA compliant.

C.2.2 Defense Information Systems Network (DISN) Capstone Requirements Document (CRD) *(Applies to Generation One; Generation Two GIG CRD crosswalk is incorporated in Appendix D)*

As described and defined in section 1.a., the DISN consists of three infrastructure blocks; sustaining base, long-haul and deployed. Each of the blocks is a "system of systems" that provides the necessary interfaces and services to garrison and deployed forces. The following paragraphs describe the CRD critical characteristics and the corresponding ORD KPPs.

C.2.2.1 Global Coverage and Connectivity

"Coverage and connectivity to provide worldwide access to the network. The operating forces require high-quality, specialized support anywhere in the world to any size or type of force. In addition, joint and combined task force operations require full horizontal and vertical communications support. Users must be provided access to voice, data, and video services without regard to geographical locations. Modeling and simulation of high-capacity networks in joint- deployment scenarios should be used extensively. Small, light, and very reliable telecommunications and processing equipment, such as commercial mobile satellites services using low, medium, and high-earth orbiting satellites, must be available. To be cost effective, commercial telecommunications products must be adapted to support DOD requirements. These adaptations must include the military features such as surge capacity, security, priority services, reliability, and survivability."

The Teleport KPPs for coverage, capacity, and interoperability support this characteristic providing worldwide access to critical DISN services from at least two different Teleport facilities. Integrating both DOD and commercial SATCOM systems into the DISN, the Capacity KPP ensures adequate day to day and surge capabilities are in place.

C.2.2.2 Interoperability

Interoperability is required across all infrastructure blocks and within technology groups without regard to service or organizational affiliation. This includes connectivity among US entities, between supporting and deployed elements, between the DISN and the commercial infrastructure, and between the DISN and the communications infrastructure of allied nations. NATO and other allied forces must continue to have the same level, or better, of interoperability as experienced with the DCS. Interoperability includes the requirement to integrate satellite, airborne, sea-based and terrestrial based (wire and wireless) transmission and switching systems (strategic and tactical). It must also provide seamless interfaces to commercial networks as required to support increased traffic during surge and contingency conditions. All DISN management activities must be guided by common standards and procedures for integrated management of network and systems resources.

By providing the requisite SATCOM and DISN integration necessary to support operations from garrison, through transition, deployment and redeployment, the Teleport KPP for interoperability supports all aspects of the DISN CRD critical characteristic.

C.2.2.3 Responsiveness To Provide Assured Service

Assured connectivity is the ability of the network to provide end-to-end service based upon mission requirements. This capability must allow the deployed warfighter to prioritize service to various mission elements, based upon DISN resources available. The DISN must be flexible and responsive so warfighters receive continuous C4I connectivity as they transition from the sustaining base to their deployed areas of operation. Once deployed, the network must be dynamic enough to handle rapid expansion of connectivity and bandwidth (dynamic reallocation) to support JTF/CTF operations.

The Teleport KPP for Access and Control provides for automation of control processes at the Teleport facility. Coupled with the planning and management requirements and capabilities of the existing DISN and SATCOM components, Teleport will support assured connectivity.

C.2.2.4 Security

"Security measures the ability of the DISN to process and protect the confidentiality and security of unclassified, classified, and compartmentalized data throughout the system at any level, and to ensure against unauthorized access to such data. The decision process to select and implement safeguards must balance the reduction of risk against the cost of protection. Using available information security techniques and tools, the network must support information transfer at all classification levels unclassified but sensitive to Top Secret Sensitive Compartment Information, in accordance with governing security regulations, to include DCID 6/3 Protecting SCI within Information Systems. Security must also provide personal, physical, and electronic protection against unauthorized access to information. Each segment of the DISN must incorporate appropriate safeguards commensurate with the existing or projected threat, and as required by governing policies. Additionally, operational security and facilities protection must be instituted to enforce physical protection standards at DISN facilities and to ensure access to information is consistent with mission assignments and clearance levels. The DISN must provide a number of privacy and security features, including the separation of user traffic operating at different security levels or communities of interest, and protection of user data against threats from outside the network. These security capabilities must be provided through techniques such as access control procedures, link encryption, Multi Security Level (MSL) and end-to-end encryption."

Providing for all classification levels of traffic, necessary TRANSEC bulk encryption, as well as physical and virtual network protection, the Teleport protection KPP directly supports the security aspects described in the CRD. Non-KPP requirements regarding facility physical security augment the KPP as well.

C.2.2.5 Network Management

"DISN must provide network management and control services via management centers. The management centers must be cost-effectively automated and have the capability to perform network management functions associated with fault detection, isolation and correction, performance monitoring, configuration management, security management, and accounting."

The G/RNOSC and G/RSSC management centers are a part of the Teleport hierarchy. The KPP for Access and Control calls out the necessary reporting and control exchanges to capitalize on the existing network management capabilities for both the SATCOM and DISN components.

C.2.2.6 Technology Insertion

"DISN must allow for the rapid insertion of new, commercially available technology and any key/critical leading-edge services. Any DISN service/system contracts must be structured to encourage the rapid insertion of new technology. The insertion of new technology must be coordinated across the DISN infrastructure blocks to ensure interoperability and the efficient delivery of services to the end user meets operational requirements. The current telecommunication industry's continuous and fast-paced evolution is encouraging new demands while providing better performance at lower prices. Throughout the life of the DISN, satisfaction of emerging validated requirements in light of emerging technological opportunities and the assessment of their merits for inclusion into the DISN architecture and design must proceed on a continual basis. Unique regional differences and requirements in the different theaters must dictate when and how new DISN services can be introduced or enhanced. The timing will be based on the level of technology available in the region."

There are only non-KPP requirements for technology insertion included in the Teleport ORD. However, the nature of the time phased requirements included in the Teleport "Generations" create an implied requirement for technology insertion, and drive the standard and coordinated fielding of new Teleport capabilities.

Appendix D
Generation Two (FY04-07)
Part One

D.1 General Description

The focus of this Appendix is to identify the additions and or changes to the DOD Teleport System brought on by the implementation of Generation Two requirements. Generation Two will add RF capabilities for the following bands: military Ka-band, L-band, and HF Global Communications System. Additional capabilities for Generation Two include access to terrestrial transport services. The capacity requirements identified for the Generation One Teleport System, will be revised accordingly to reflect the most recent Satellite Database (SDB) requirements as applied to current Defense Planning Guidance (DPG) scenarios for the FY 2006 timeframe.

As the linchpin between legacy systems and the future Transformational Communications Architecture, it is critical that the Teleport system evolve to support this future network centric environment. This evolution will occur primarily in Generation Three, however initial efforts may begin in the later stages of Generation Two.

D.2 Mission Area

The mission area of the DOD Teleport System will not change from the Generation One description in that it will provide deployed forces high capacity, multi-band, and multi-mode telecommunications and access to terrestrial networks. Generation One was the first phase of implementing additional RF and DISN service capabilities to those STEP sites that will evolve to Teleport sites. Joint Task Forces (JTFs) and deployed forces will continue to rely on a variety of SATCOM capabilities – encompassing both military systems and commercial services – to satisfy their needs.

As addressed in the SATCOM CRD and GIG CRD, Warfighters need a robust system that will support Network Centric Warfare concepts. The Teleport System for Generation Two must consider the following areas:

- The Joint Technical Architecture (JTA)
- The GIG Architecture
- The Mission Information Management (MIM) Architecture
- Network Centric Enterprise

D.2.1. Type of System Proposed for Generation Two.

The Generation One DOD Teleport system integrates X-, C-, Ku-, UHF and EHF (LDR/MDR) satellite capabilities at seven existing STEP sites (Wahiawa, Northwest, Lago Patria, Roberts, Buckner, Landstuhl, Ramstein). Generation Two is expected to incorporate military Ka-band, L-band, and HF Global Communications System capabilities to provide increased RF access for deployed tactical communications systems. These additional capabilities will provide worldwide, integrated communications nodes that also have the ability to modularly insert emerging systems adopted by DOD to support deployed forces and JTFs. Implementation of these commercial and military RF systems throughout the generations shall provide the capability to interconnect incoming voice, data, and video systems with the appropriate legacy tactical C4I Systems, DISN Service Delivery Nodes (SDN), and DISN/GIG Transport systems. Additionally, the Teleport system will support interoperability with NATO WAN and other allied and coalition networks.

As in Generation One, the current DOD JTA will be used to define the technical architecture for the Generation Two DOD Teleport system. The system will be developed, tested, and deployed in a manner that is compliant with all appropriate treaties and international agreements. The DOD Teleport system must be capable of interfacing the DISN/GIG network management systems. It must also be capable of interfacing, as appropriate, the military and commercial satellite Bandwidth Management Centers. Additionally, periodic analysis should be conducted to determine whether the number of Teleport sites determined in the Generation One AoA is still adequate to support Warfighter connectivity and provide reasonable redundancy. A refinement of the requirements in Table 4-2 will be provided under a separate cover letter based on analysis of the July 2002 SDB, CMCO scenario 6.1, which will be coordinated through Combatant Command and Services prior to publication by the Joint Staff J6. Finally, the requirements in this document are at the overall Teleport system level. The Teleport Program Office will conduct appropriate analysis to determine specific Teleport site requirements. The aggregate of Teleport site capabilities will be used to meet the overall system requirements.

D.3. Threat.

The Teleport's Generation Two threat environment is not expected to be materially different from the Generation One environment. The current MILSATCOM Systems Threat Assessment Report (STAR), AIC-1574-037-03, dated July 2003 covers threats to proposed satellite communications systems, to include threats to Teleport, and shall be considered part of the documented threat for this system. Threats against any associated automated information system are listed in the Information Operations Capstone Threat Capabilities Assessment, DII-15771203, dated August 2003 and shall be considered part of the documented threat for this system. Additional Information Warfare threats posed to joint military telecommunication systems are addressed in the Electronic Warfare Threat Environment Description (TED), NAIC-1574-0731-01, February 2001.

D.4. Operational Shortcomings of Existing Systems and C4ISR Architectures.

The DOD warfighter must continually support diverse and ever changing missions. As such, the C4I infrastructure deployed to support these missions must be diverse and capable of supporting ever-changing requirements. The following are considered shortcomings of the existing Teleport system:

a. Access to on-demand transport services are not available in the current system design. This access is essential in providing tactical users transport capability not currently provided through one of the six DISN services being offered in Generation One.
b. The time required to contract, engineer, and provide access to GIG Transport and Commercial SATCOM is unacceptable. The deployed user requires timely access into the systems managed by DISA.
c. Access to DISN and Legacy C4I services via HF Global Communications System is not available in the current system design.
d. Access to Teleport voice services is not available to users of the Voice Over Internet Protocol (VOIP).
e. An Unmanned Aerial Vehicle (UAV) access gateway is not available in the current system design. This capability would allow for UAV aircraft to relay ISR data through the Teleport gateway in order to gain access to the Distributed Common Ground/Surface System (DCGS) via GIG Transport Service.
f. Access to Public Key Infrastructure (PKI) is required to ensure information security over all voice, video, and data transmission IAW CJCSI 3170.01B.

D.5. Program Support.

The Teleport Program will build on the supportability structure established during Generation One and will incorporate the supportability structure developed by the Wideband Gapfiller Program Manager. Any concerns identified with the supportability structure will continue to be addressed by the Teleport JILSMT and resolved at the direction of the Teleport Program Manager.

a. Maintenance. Maintenance planning will be based on two levels: On-site/organizational and Off-site/depot. On-site/organizational maintenance shall consist of periodic functional checks and minor periodic and corrective maintenance consistent with procedures provided by Service life cycle support organizations and Original Equipment Manufacturers (OEMs). A stock of on-site spares must be provided at the time of fielding in order to maintain 99% Ao. When the system fails to meet acceptable performance standards, on-site maintenance personnel shall initiate troubleshooting procedures to isolate the problem to an LRU. The defective LRU shall be replaced with an operable LRU from the on-site spares and the malfunctioning LRU shall be shipped out for depot/contractor repair.

Off-site/depot maintenance may be provided by either a government-operated depot, or through the commercial vendor for COTS type items. No new government depot requirements will be added for the Teleport Program as a system. Individual equipment segments may result in new depot requirements being established if a business case analysis determines that it would be best to bring the work into a government depot. For COTS items, the planning assumption is that the commercial source will be used.

b. Supply Support. The DOD Teleport system will be composed of a mixture of Government Off the Shelf (GOTS) and COTS items. The GOTS items will be supported under the logistics system established by the original developing or acquiring activity. COTS items will generally be supported by the commercial vendor under contractor logistics support (CLS) unless a business case analysis is performed by the acquiring activity and this analysis determines that it would be cost beneficial and operationally effective to bring the COTS spares into the Defense supply system. In making this analysis, consideration must be given to the implementation of the Supply Chain Management Initiatives.

c. Training. Training on commercial equipment segments will be performed by the vendor unless such training is already available through the Service-training infrastructure. This training can be provided either to site personnel during installation, or as I&KP training given to a training cadre to establish a new equipment training capability in a lead Military Service. Sustainment and follow-on training shall be provided by computer based training (CBT) or distance learning to the maximum extent possible. Training Programs of Instruction or training curriculum should address system level as well as segment level training.

d. Manpower and Personnel. The program implementation of Generation One introduced significant increases in manpower requirements. The Human Systems Integration (HSI) Study initiated during Generation One will be completed to identify associated workload and manpower and personnel requirements for those items being built out from Generation One. For items being developed by another Program for incorporation into the Teleport site (E.g., WGS), the HSI data from that program office will be extrapolated into the requirements for the Teleport Program. Regardless of the increased manpower requirements, there will be no increase in military end strengths as a result of program implementation.

e. Program Affordability. Generation One funding was based on feasibility study analysis conducted by Johns Hopkins University in 2000. Current Generation Two funding levels may be sufficient; however an additional Analysis of Alternatives (AOA) should be performed due to program refinements and technology enhancements.

Appendix D
Part Two - GIG CRD KPP Crosswalk

GIG CRD COMPLIANCE CHECKLIST

ORDs that fall under a CRD must comply with any of the particular CRD's KPPs and other capability requirements as appropriate and applicable. Close examination of the ORD operational concept and information exchange requirements will help determine which KPPs/capability requirements apply. Determination of which GIG KPPs and other capability requirements are applicable to an ORD must be based on the principle that the GIG operates as a globally interconnected, end-to-end, interoperable system of systems, wherein all systems that comprise the GIG shall be GIG-enabled so as to allow "plug and play" interoperability among systems. A system shall be considered GIG-enabled if it includes the capabilities described in this CRD from the seven GIG functions in the GIG CRD necessary to fulfill the system's operational mission(s). In order to establish whether a system is a part of the GIG, and thus subject to GIG CRD compliance, or not, the GIG definition should be considered very carefully (see Chapter I.A.3.a). If a system is judged to be a part of the GIG, then the next step is to determine where within the system is the origination/termination point for information to be exchanged externally with another system or systems across the enterprise (i.e. two or more systems connected or networked together end-to-end across any portion of the GIG) for the purpose of satisfying one or more information exchange requirements. It should be noted that "information" as used in the GIG information flow/exchange context discussed in this section and elsewhere within this CRD, does NOT include any unprocessed raw data that is not yet recognizable or usable as information. In the final analysis, any system that has external information exchange requirements (or which supports and facilitates system-to-system, end-to-end flow and exchange of information, such as a transport system, network management system, etc.) must be GIG-enabled with whatever information capabilities are necessary to ensure that information exchanges can be carried out successfully and in an interoperable manner. CJCSI 3170.01B does not specify a particular format to be used for the mandatory ORD to CRD crosswalk; the following GIG CRD Compliance Checklist lists the areas that are applicable to the DOD Teleport System.

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
CHAPTER I: GENERAL DESCRIPTION OF OPERATIONAL CAPABILITY						
GIG Reference	I.B.3	Does the GIG CRD appear in the Related Documents section?	APP A	REF (36)		YES
Operational Concept	I.D.3	If the OV-1 depicts information exchange relationships, are the producer, user, and command node entities identifiable?		4-1 (A).(B)(C)	Var	YES
		Does the operational concept include external information exchange?		4-1 (A -C)	Var	YES
GIG Implementation Guidelines	I.E	Have each of the following GIG implementation guidelines been considered and applied in the ORD as appropriate?				YES
		GIG implementation done in accordance with the standards included in the most current version of the *DOD JTA*?	6	1.B.		YES
		All new Command, Control, Communications, Computers and Intelligence (C4I) emerging systems and upgrades to be fielded as level 6 DII COE compliant with the goal of achieving level 8 compliance?		5.C.		YES
		System is either standards-based or employs commercial-off-the-shelf (COTS) technologies to: - Facilitate joint, allied, and coalition interoperability? - Simplify integration? - Reduce both long and short-term costs?		1.B.		YES
		System is to be scalable, affordable, sustainable and extensible with respect to its functionality?		1.B.		YES
		System is designed to accommodate change and facilitate the integration of future systems and technologies as they evolve?		4.D. (8)		YES
		Additional manpower requirements are minimized?		1.C. (2)		YES
		Reliability, availability, survivability, and maintainability features of the system are designed to support all functions necessary to meet the requirements documented in Chapter IV, including the ability to recover from critical failures?		4.C.		YES
		Emphasis is placed on reducing the complexity, time, and cost of training?		5.E.		YES
		Software design is aimed at enhancing interoperability and commonality among GIG-enabled systems?		5.D.		YES

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
GIG Implementation Guidelines	I.E	System designed using an open systems approach and adhering to applicable standards within the JTA?		5.D.		YES
		Bandwidth and throughput requirements along with implications to strategic, fixed, theater, and tactical architectures are considered?	6	1.B.		YES
		System will be developed, tested, and deployed in a manner that is compliant with all appropriate treaties and international agreements?	APP D	D.2.1		YES
		System will be tested and certified for interoperability IAW Joint Interoperability Test Command (JITC) procedures?		7.A.		YES
		System mitigates security risks and meets all current security provisions articulated in appropriate DOD and IC policies, procedures, and instructions including DODD 8500.aa?	APP D			YES
		System uses standards-based rather than system-unique security mechanisms?		5.C.		YES
		ORD considers ongoing developments and evolving specifications in the following areas (as applicable): - Joint Operational Architecture (JOA)? - Nuclear C2 Systems Technical Performance Criteria (NTPC)? - GIG Architecture? - Mission Information Management (MIM) Architecture?	APP D	D.2.		 YES N/A YES N/A
		Time-phased requirements developed in ORD, with associated objectives and thresholds, IAW DODI 5000.2?		TBL 4-2 & 4-3		YES
CHAPTER II: THREAT						
Threat to be Countered	II.	If information exchange is fundamental to the ORD, does Chapter II mention Information Operations, Computer Network Attack, Computer Network Exploitation, Electronic Warfare, and Electromagnetic Pulse?		2.B.		YES

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
CHAPTER III: SHORTCOMINGS IN MISSION AREA AND EXISTING SYSTEMS						
Shortcomings	III	Does the ORD describe shortcomings or absence of existing capabilities and systems to fulfill the needs of the GIG functions described in Chapter I? As applicable, are GIG shortcomings addressed such as: lack of interoperable applications; limited ability to rapidly catalog, search, and retrieve required information; limited ability to effectively and efficiently use existing RF spectrum; limited ability to move digital information seamlessly; lack of asset visibility resulting in limited ability to effectively manage a common user network; limited means to prioritize information and establish profiles; limited ability to support multilevel security operations?		3.A.		YES
CHAPTER IV: CAPABILITIES REQUIRED – PROCESS FUNCTION						
						N/A
CHAPTER IV: CAPABILITIES REQUIRED – STORE FUNCTION						
						N/A
CHAPTER IV: CAPABILITIES REQUIRED – TRANSPORT FUNCTION						
Switching/Routing/ Transmission	IV.B.4b	System providing switching, routing, and transmission control capabilities/mechanisms shall be fully interoperable and work seamlessly across the entire GIG in accordance with *DOD JTA* (**Threshold**)?		5.C.		YES
Spectrum Supportability/ Electromagnetic Environmental Effects	IV.B.4c	System shall optimize use of the available electromagnetic spectrum through efficient frequency reuse and advanced modulation, compression and filtering techniques, and shall comply with DOD, National and International spectrum management policies as applicable (**Threshold**)? System shall be mutually compatible with other systems, including allied and coalition systems, in the operational environment and shall not be degraded by electromagnetic environmental effects (**Objective**)?		4.D. (6) (7)		YES

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
Quality of Service	IV.B.4d	Transport system shall provide QoS capabilities that ensure that information identified as priority is delivered ahead of regular traffic 99% of the time (**Threshold, KPP**) and 99.9% of the time (**Objective, KPP**)**?** Required QoS factors include: • Prioritization. End users shall be able to assign priority to information targeted for transport (**Threshold**)? • Response Time. All transport capabilities shall be designed to meet or exceed customer stated response times (**Threshold**)? • Precedence. Data shall receive expedited handling during transport in accordance with the commander's policy and user assigned priority (**Threshold**)? • Reliability. Delivery of information shall be guaranteed in accordance with its assigned broadcast level (**Threshold**)? • Latency. It shall be possible to deliver information in real and/or near real time as required (**Threshold**)?		Table 4-2 (Inter-operability)		YES
Information Integrity	IV.B.4e	System shall maintain and guarantee during transport the integrity of all information elements exchanged throughout the GIG to enable user confidence; information integrity shall be 99.99% (**Threshold, KPP**) and 99.999% (**Objective, KPP**).		Tbl 4-2 (QOS)		YES
Standards	IV.B.4f	To ensure system interoperability across the GIG and to support uninterrupted service, all GIG transport capabilities shall be standards-based using *DOD JTA* and DOD CIO prescribed standards, as applicable (**Threshold**)?		Tbl 4-2 (Interop)		YES
Connectivity	IV.B.4g	Transport system shall provide connectivity on demand to all fixed and deployed locations/users (**Threshold**)? Transport systems shall have the ability to maintain network connectivity on-the-move to meet Service/JTF requirements in all warfighting environments (afloat, sub-surface, airborne, in space, on the ground) (**Objective**)?		Tbl 4-2 (Interop)		YES

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
Capacity	IV.B.4h	With minimal exceptions, GIG transport capacity shall be viewed as an open system that is available to transport information from all domains utilizing unicast, multicast, and broadcast techniques to provide information on demand to the warfighter/decision maker (**Threshold**)? Transport system shall have the reserve capacity to accommodate surge loading and support multiple military operations as described in Defense Planning Guidance (**Objective**)?		Tbl 4-2 (Capacity)		YES
Technology Insertion	IV.B.4i	To effectively keep pace with advances in technology that have the potential to render existing systems obsolete shortly following acquisition, the GIG shall enable and support the seamless and efficient insertion and incorporation of emerging (future) technologies into the transport domain (**Threshold**)?		Tbl 4-2 (Tech Insertion)		YES
Security	IV.B.4j	System shall provide link and transmission security based on the level of risk acceptable to the user, and the GIG security architecture shall support use of clear headers if and when necessary (**Threshold**)?		Tbl 4-2 (Protection)		YES
Robustness	IV.B.4k	To avoid any single point of failure, the GIG shall use multiple connectivity paths (not susceptible to the same threat) and media (**Threshold**)?		Tbl 4-2 (Coverage)		YES
Scalability	IV.B.4l	Transport capability shall be scalable and adaptable to meet dynamic needs of users (**Threshold**)?		Tbl 4-2 (Capacity & Interop)		YES
Survivability	IV.B.4m	Transport system shall be protected against all potential threats commensurate with the operating environment and the criticality of the information being transported, and shall also ensure connectivity through the total threat environment (i.e. conventional and nuclear) (**Threshold**)?		Tbl 4-2 (Protection)		YES
Availability/ Reliability	IV.B.4n	Transport capabilities shall be available to provide reliable information exchange services to the warfighter/decision maker on demand and shall be responsive to the criticality of the information to be exchanged (**Threshold**)?		Tbl 4-2 (Interop)		YES
Transport Element Status	IV.B.4p	All transport elements (e.g., switches, routers, etc.) shall be capable of providing status changes to network management devices by means of an automated capability in near real time 99% (**Threshold, KPP**) and 99.9% (**Objective KPP**) of the time?		Tbl 4-2 (QOS)		YES
CHAPTER IV: CAPABILITIES REQUIRED – HUMAN-GIG INTERACTION (HGI) FUNCTION						

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
						N/A
CHAPTER IV: CAPABILITIES REQUIRED – NETWORK MANAGEMENT (NM) FUNCTION						
Situational Gig End to End Awareness	IV.B.6.a.(2)	To accomplish GIG end-to-end situational awareness, system shall have the NM capability of automatically generating and providing an integrated/correlated presentation of network and all associated network assets (**Threshold**)?		Tbl 4-2 (Control & Network Mgmt)		YES
Dynamic, Predictive Planning	IV.B.6.a. (3)	System shall have the NM capability to perform dynamic, predictive planning by gathering, storing and using knowledge about GIG assets/resources, so as to optimize their utilization (**Threshold**)? System shall have the NM capability to create/modify/distribute network plans and orders IAW user requirements (**Threshold**)?		Tbl 4-2 (Control & Network Mgmt)		YES
Distributed and Partitioned Network Control	IV.B.6.a. (4)	System shall have the NM capability to rapidly transfer control of one or more objects or groups of varying size, and reestablish control when relinquished without hindering end-to-end visibility by the senior network manager, while maintaining continuous control (**Threshold**)?		Tbl 4-2 (Control & Network Mgmt)		YES
Remote Object and Network, Control and Configuration	IV.B.6.a. (5)	System shall have a NM capability that leverages existing and evolving technologies and has the ability to perform remote network device configuration/reconfiguration of objects that have existing DOD JTA management capabilities (**Threshold**)?		Tbl 4-2 (Control & Network Mgmt)		YES
Network Status	IV.B.6.a. (6)	System shall have an automated NM capability to obtain the status of networks and associated assets in near real time 99% (**Threshold, KPP**) and 99.9% (**Objective, KPP**) of the time.		Tbl 4-2 (Control & Network Mgmt)		YES
Automated Fault Management	IV.B.6.a. (7)	Systems shall have the NM capability to perform automated fault management of the network, to include problem detection, fault correction, fault isolation and diagnosis, problem tracking until corrective actions are completed, and historical archiving (**Threshold**)?		Tbl 4-2 (Control & Network Mgmt)		YES
CHAPTER IV: CAPABILITIES REQUIRED – INFORMATION DISSEMINATION MANAGEMENT (IDM) FUNCTION						
						N/A

GIG CRD COMPLIANCE CHECKLIST						
CRD Section Heading	**CRD Para #**	**CROSSWALK ITEMS**	**ORD Page #**	**ORD Para #**	**ORD Line #**	**YES, NO, N/A**
CHAPTER IV: CAPABILITIES REQUIRED – INFORMATION ASSURANCE (IA) FUNCTION						
Information Integrity and Availability	IV.B.6.c. (2)	System shall be robust, survivable and capable of rapid restoration, to support IA across the GIG (**Threshold**)? System shall have an IA capability to define, control, and defend enclave boundaries (**Threshold**)? System shall have an IA capability to provide timely, reliable access to processes and data even in the event of a denial of service attack (**Threshold**)? System shall have an IA capability to ensure information and process integrity throughout the system (during storage, processing, transmission and presentation) to prevent unauthorized or unintended changes, in accordance with mission specific criteria (**Threshold**)?		Tbl 4-2 (Protection)		YES
Prevent Opportunity to Attack	IV.B.6.c. (3)	System shall be developed in accordance with IA Defense in Depth standards (CJCSI 6510.01C) to prevent or at least minimize the opportunity for attack; and shall have, in the event of an attack, the IA capability to immediately define, detect and respond appropriately to anomalies/attacks/disruptions from external threats, internal threats and natural causes (**Threshold**)?		Tbl 4-2 (Protection)		YES
Detection and Responses	IV.B.6.c. (5)	System shall incorporate a detection, reporting and response IA infrastructure that enables rapid detection of and reaction to anomalous events, and enables operational situation awareness and responses (**Threshold**)?		Tbl 4-2 (Protection)		YES
Authentication/ Confidentiality/Non-repudiation	IV.B.6.c. (7)	System shall meet and maintain minimum IA Defense in Depth standards, including certification and accreditation IAW DITSCAP process (e.g., *CJCSI 6510.01C*, *DODI 5200.40*) (**Threshold/Objective, KPP**)? System shall utilize/interoperate with the security management infrastructure (e.g., key management and DOD public key infrastructure) (**Threshold**)? System shall provide proof of information origin and receipt as required (**Threshold**)?		4d. (5) (f)		YES
CHAPTER IV: CAPABILITIES REQUIRED – INTEROPERABILITY						
Interoperability	IV.C	System shall satisfy all critical IER attributes to the threshold level (**Threshold, KPP**) and satisfy all IER attributes to the objective level (**Objective, KPP**)?		Tbl 4-5 (IER Tables)		YES

PART 1 - ABBREVIATIONS AND ACRONYMS

A_o	Operational Availability
AFFOR	Air Force Forces
AIS	Automated Information System
AoA	Analysis of Alternatives
AOR	Area of Responsibility
ARFOR	Army Forces
ARG	Amphibious Ready Group
ASD	Assistant Secretary of Defense
ATE	Automatic Test Equipment
ATM	Asynchronous Transfer Mode
AUTODIN	Automated Digital Network
BER	Bit Error Rate
B-ISDN	Broadband Integrated Services Digital Network
BIT	Built-in-Test
BMC	Bandwidth Management Center
C_o	Operation Support Costs
C3I	Command, Control, Communications, and Intelligence
C4I	Command, Control, Communications, Computers, and Intelligence
C^4ISR	Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance
CAIV	Cost as an Independent Variable
CALS	Continuous Acquisition and Life-Cycle Support
CBT	Computer Based Training
CCI	Controlled Cryptographic Item
CEC	Cooperative Engagement Capability
CJCS	Chairman of the Joint Chiefs of Staff
CJCSI	Chairman of the Joint Chiefs of Staff Instruction
CJTF	Commander, Joint Task Force
CNA	Computer Network Attack
CND	Computer Network Defense
COA	Course of Action
CODEC	Coder-Decoder
COE	Common Operating Environment
COI	Community of Interest
COMNAVFOR	Commander, Naval Forces
COMSEC	Communications Security
CONOPS	Concept of Operations
CONUS	Continental United States
COTS	Commercial-Off-The-Shelf
COWAN	Coalition Wide Area Network

CRD	Capstone Requirements Document
C-C/S/A	Combatant Commander/Service/Agency
CTF	Combined Task Force
CUDIXS	Common User Digital Information Exchange Subsystem
CVBG	Carrier Battle Group
DAMA	Demand Assigned Multiple Access
DCGS	Distributed Common Ground/Surface System
DCS	Defense Communications System
DIA	Defense Intelligence Agency
DII	Defense Information Infrastructure
DISA	Defense Information Systems Agency
DISN	Defense Information System Network
DMS	Defense Message System
DOD	Department of Defense
DODD	Department of Defense Directive
DODI	Department of Defense Instruction
DPG	Defense Planning Guidance
DRSN	Defense Red Switched Network
DSCS	Defense Satellite Communications System
DSN	Defense Switched Network
DTS	Diplomatic Telecommunications Service
E3	Electromagnetic Environment Effect
EHF	Extremely High Frequency
EMI	Electro-Magnetic Interference
EMP	Electromagnetic Pulse
ERDB	Emerging Requirements Database (replaced by SDB)
EW	Electronic Warfare
FOC	Full Operational Capability
FoS	Family of Systems
FOT&E	Final Operational Test and Evaluation
FY	Fiscal Year
GBS	Global Broadcast Service
GEO	Geosynchronous Earth orbit
GI&S	Geospatial Information and Services
GIG	Global Information Grid
GNOSC	Global Network Operations and Security Center
GOTS	Government-Off-The-Shelf
GRA	Government Reference Architecture
GSSC	Global SATCOM Support Center
GUI	Graphical User Interface

HEMP	High-Altitude Electromagnetic Pulse
HLD	Homeland Defense
HLS	Homeland Support
HNA	Host Nation Approval
HQ	Headquarters
IA	Information Assurance
IAW	In Accordance With
ICDB	Integrated Communications Database (replaced by SDB)
IER	Information Exchange Requirement
IETM	Interactive Electronic Technical Manual
IF	Intermediate Frequency
ILS	Integrated Logistics Support
INMS	Integrated Network Management System
IO	Information Operations
IOC	Initial Operational Capability
IOT&E	Initial Operational Test and Evaluation
IPL	Integrated Priority List
ISDN	Integrated Services Digital Network
ISP	Internet Service Providers
ISR	Intelligence, Surveillance and Reconnaissance
IW	Information Warfare
JCCC	Joint Communications Control Center
JCS	Joint Chiefs of Staff
JFN	Joint Fires Network
JFTOC	Joint Fleet Telecommunications Operations Center
JIEO (DISA)	Joint Interoperability and Engineering Organization
JITC	Joint Interoperability Test Command
JKMS	Joint Key Management System
JNMS	Joint Network Management System
JROC	Joint Requirements Oversight Council
JROCM	Joint Requirements Oversight Council Memorandum
JSOTF	Joint Special Operations Task Force
JTA	Joint Technical Architecture
JTF	Joint Task Force
JTRS	Joint Tactical Radio System
JQRR	Joint Quarterly Readiness Review
JWCA	Joint Warfighting Capabilities Assessment
JWICS	Joint Worldwide Intelligence Communications System
JWID	Joint Warfighter Interoperability Demonstration

KPP	Key Performance Parameter
LDR	Low Data Rate
LEO	Low Earth Orbit
LMI	Logistics Management Information
LOS	Line-of-Sight
LRU	Line Replaceable Unit
LSA	Logistics Support Analysis
MARFOR	Marine Forces
Mbps	Megabits per second
MDR	Medium Data Rate
MEO	Medium Earth Orbit
M-hop	Multiple Hop
MILSATCOM	Military Satellite Communications
MIL-STD	Military Standard
MLS	Multi-Level Security
MLDT	Mean Logistics Delay Time
MNS	Mission Need Statement
MPT	Manpower, Personnel, and Training
MSOC	Milstar Space Operations Center
MSS	Mobile Satellite Services
MTBCMF	Mean Time Between Critical Mission Failure
MTBF	Mean Time Between Failures
MTBOMF	Mean Time Between Operational Mission Failures
MTTRCMF	Mean Time To Restore a Critical Mission Failure
MCO	Major Combat Operations
MUOS	Mobile User Objective System
NAVFOR	Naval Forces
NAVSOC	Naval Satellite Operations Center
NCTAMS	Naval Computer and Telecommunications Area Master Station
NDI	Non-Developmental Item
NDS	Non-Developmental Software
NET	New Equipment Training
NIMA	National Imagery and Mapping Agency
NIPRNET	Non-secure Internet Protocol (IP) Router Network
N-ISDN	Narrowband Integrated Services Digital Network
NMS	National Military Strategy
NSA	National Security Agency
OC	Operations Center
ODC	Ozone Depleting Chemicals

OEM	Original Equipment Manufacturers
O&M	Operations and Maintenance
ONI	Office of Naval Intelligence
OOTW	Operations Other than War
OPLAN	Operation Plan
ORD	Operational Requirements Document
OSA	Open System Architecture
OTCIXS	Officer-In-Tactical Command Information Exchange Subsystem
P3I	Pre-Planned Product Improvements
PDC	Project Designator Code
PKI	Public Key Infrastructure
POP	Point of Presence
POTS	Plain Old Telephone System
PSMP	Product Support Management Plan
PSTN	Public Switched Telephone Network
RF	Radio Frequency
RFI	Radio Frequency Interference
RNOSC	Regional Network Operations and Security Center
RSSC	Regional SATCOM Support Center
SA	System Administrator
SAE	Service Acquisition Executive
SAS	Supportability Analysis Summary
SATCOM	Satellite Communications
SC2C	SATCOM C2 Center
SDB	Satellite Data Base
SDN	Service Delivery Node
SHF	Super High Frequency
SIGINT	Signals Intelligence
SIPRNET	Secret Internet Protocol (IP) Router Network
SOM	SATCOM Operational Manager
SONET	Synchronous Optical Network
SOPS	Space Operations Squadron
SoS	System of Systems
SSC	Small-Scale Contingency
SSIXS	Submarine Satellite Information Exchange Subsystem
STAR	System Threat Assessment Report
STEP	Standardized Tactical Entry Point
SYSCON	Systems Control
TADIL	Tactical Digital Information Link
TADIXS	Tactical Data Information Exchange

	Subsystem
TAFIM	Technical Architecture Framework for Information Management
TBD	To Be Determined
TBP	To Be Provided
TCA	Transformational Communications Architecture
TCCC	Theater Communications Control Center
TED	Threat Environment Description
TMDE	Test, Measurement, and Diagnostic Equipment
TRANSEC	Transmission Security
TRM	Technical Reference Model
TS/SCI	Top Secret/ Sensitive Compartmented Information
UAV	Unmanned Aerial Vehicle
UFO	UHF Follow-On
UHF	Ultra High Frequency
UJTL	Universal Joint Task List
USCG	United States Coast Guard
USSTRATCOM	U.S. Strategic Command
VOIP	Voice Over Internet Protocol
VTC	Video Teleconference
WAN	Wide Area Network
WGS	Wideband Gapfiller System
WSOC	Wideband Satellite Operations Center
XDR	Extended Data Rate

PART II Glossary of Terms

Analysis of Alternatives (AOA). The evaluation of the operational effectiveness and estimated costs of alternative material systems to meet a mission need. The analysis assesses the advantages and disadvantages of alternatives being considered to satisfy requirements, to include the sensitivity of each alternative to possible changes in key assumptions or variables. The AOA assists decision makes in selecting the most cost-effective material alternative to satisfy a mission need.

Access. An individual one-way transmission of a signal carrying user information through a communications satellite. As an example, a full duplex transmission between two individual sites would be two accesses. A simplex broadcast from a central location to multiple receive locations would be a single access. Also, The right to enter a SATCOM network and make use of communications payload resources.

Adjudication. Adjudication refers to the apportionment decision made between two or more users contending for the same resources.

Allocation. The operational real-time assignment of SATCOM communications payload resources to an approved user for use in activating a communications link or network.

Automated Connectivity. Automatic DISN-to-user and user-to-user (for cross-banding, hubbing, and M-Hops) circuit configuration, reconfiguration, and redundancy for equipment within the Teleport.

Availability. Availability is the measure of the degree to which a system is operable and capable of initiating a mission at an unknown (random) time. Availability defines the percentage of time that a system or item of equipment is operational in accordance with a minimum set of prescribed operational or functional specifications or criteria. Space segment availability reflects the space segment's ability to meet the threshold set of communications requirements as a function of the connectivity key parameter.

Bandwidth Management Center. Perform the actual command and control of the satellite and communications payloads.

Broadcast. One-way transmission from a single, uplink source to an area or earth coverage downlink listening area.

Capacity. The communications throughput and/or number of accesses provided by a SATCOM system or service as measured against specific user scenarios. The scenarios specify user parameters.

Capstone Requirements Document (CRD). A document that contains capabilities-based requirements that facilitates the development of individual ORDs by providing a common framework and operational concept to guide their development. It is an oversight tool for overarching requirements for a system-of-systems.

C band. The portion of the radio frequency spectrum between 4 and 8 gigahertz (GHz)

Combatant Command. One of the unified or specified commands established by the President. (Joint Pub 1-02)

Commercial Satellite Communications. The satellite communications resources provided by commercial entities using commercial frequencies.

Compatibility. Capability of two or more items or components of equipment or material to exist or function in the same system or environment without mutual interference.

Connectivity. The ability to provide the requisite magnitude of the demanded types of protected and unprotected throughput communications services to the target user terminal populations as dispersed and/or concentrated within the deployed geographical areas. Connectivity encompasses coverage in terms of the physical geometry between the satellite, the earth, and the user terminal population; and capacity in terms of the relative data throughput.

Coverage. The portion of the earth's surface over which SATCOM services are provided.

Cross-band. An activity that occurs at a Teleport Site when the incoming signal from a satellite in one frequency band (e.g., SHF) is re-transmitted to a terminal in another frequency band (e.g., Ka). Cross-banding should not be confused with the normal frequency translation performed by communications satellite transponders.

DAMA. Demand Assigned Multiple Access is a method of automated channel sharing (multiplexing). User terminals automatically interact with a control station, which allocates a portion of a satellite channel for that user's communications. Allocation of the channel is on the basis of current needs and priorities, i.e., not pre-assigned. Any unused DAMA resource is available on a priority basis and is available for all. The ad hoc nature of DAMA limits the waste caused by "long term" pre-assigned satellite resources.

Defense Information Infrastructure (DII). Resources identified by the Defense Information Systems Agency (DISA) as critical for the flow of information within the Department of Defense. Interoperability and multi-path technologies are being applied to the DII to make it as flexible as possible.

Defense Information System Network (DISN). A network of networks that support information transfer within the DOD. Networks that comprise the DISN include DSN, DRSN, NIPRNET, SIPRNET, DVSG, and DATMS.

Distant Component. A component (i.e. earth terminal) of the Teleport System separated geographically from the Teleport core site.

Distributed Common Ground/Surface System (DCGS). An intelligence, surveillance, and reconnaissance (ISR) system of systems that connects geographically separated fixed and deployable ISR ground stations. The DCGS will simultaneously task, receive, process, exploit, and disseminate data from national, theater, tactical, and commercial collection assets in a distributed environment. The DCGS will support national, joint, combined, and Service operations.

EHF. Extremely High Frequency. The portion of the radio frequency spectrum between 30 and 300 gigahertz (GHz).

Emerging Requirements Database. ERDB was replaced by the Satellite Data Base. A comprehensive database of emerging network connectivity requirements maintained by the Defense Information Systems Agency. Adapted from the SDB, it is intended to capture mid- and far-term information needs of emerging warfighter and support systems and platforms that will require SATCOM.

End-to-End System. System connectivity that begins at the input to the baseband instrument of the transmitting user, includes all hardware, software, and processes to the input port of a transmit terminal through the receive terminal and all baseband equipment to the instrument at the receiving user.

Full period. Communications link in which the capability to transfer information is continuously provided. Once activated, the link is not released until the end of the requested period. Capacity is typically reserved whether or not traffic is flowing.

Gateway. A ground station that acts as a relay between satellites in a system.

Global Coverage. Coverage of all latitudes and longitudes.

Global SATCOM Support Center. The integrated SATCOM support center responsible for system level global SATCOM resource management and constellation configuration management.

Global Broadcast System. GBS is an information technology, national security system providing mission essential network-centric warfare communications. As an element of the GIG, GBS provides shared access to worldwide high capacity, one-way transmission of video, data, imagery and other information as required, supporting forces in garrison, in transit and in theater. GBS segments consist of broadcast management, space, and terminals. Modularity of and between each of the segments allows for the use of multiple space assets and permits flexibility in selection of transmit and receive components. Information is organized, controlled and fed to the satellite uplink by a Satellite Broadcast Manager (SBM). The broadcast can be complemented with receive suite reachback using asymmetrical networked connectivity.

HF. High Frequency. The portion of the radio frequency spectrum between 3 and 30 megahertz (MHz).

High Data Rate (HDR). Rates greater than 2.048 Mbps.

Hub/Spoke. An arrangement where disadvantaged terminals ("spokes") communicate to a usually larger "hub" terminal, which then relays their traffic to other spoke terminals. Disadvantaged terminals can be those terminals (shipboard or ashore) that have limited look angles due to mast blockage or terrain limitations.

Information Exchange Requirements. The requirement for information to be passed between and among forces, organizations, or administrative structures concerning ongoing activities. Information exchange requirements identify who exchanges what information with whom, as well as why the information is necessary and how that information will be used. The quality (i.e. frequency, timeliness, security) and quantity (i.e., volume, speed, and type of information such as data, voice, and video) are attributes of the information exchange included in the information exchange requirement.

Information Operations. Actions taken to affect adversary information and information systems while defending one's own information and information systems

Information Superiority. The capability to collect, process, and disseminate an uninterrupted flow of information, while exploiting or denying an adversary's ability to do the same.

Integrated Communications Database (ICDB). ICDB has been replaced by the Satellite Data Base. A DISA-maintained database of Combatant Commander/ Service/Defense Agency-provided, Joint Staff approved, SATCOM requirements. The database includes satellite requirements data and information on all DOD communications. The SATCOM portion of the database contains information necessary to support peacetime operational management of all SATCOM systems; wartime deliberate planning; and future needs for architecture preparation and system development."

Integrated Broadcast Service (IBS) Simplex. Formerly TDDS, IBS Simplex is a multi-source, global satellite broadcast dissemination system that supports the rapid and continuous delivery of vital, time-critical intelligence data to operational users deployed world-wide at the secret level over a UHF satellite broadcast. Simplex is a hub-based architecture, with the network management center (NMC) receiving all intelligence over the terrestrial communications network (TCN). The NMC will prioritize the information in accordance with each Combatant Command's guidance and transfer the intelligence to the simplex uplink sites for broadcast

Interoperability. The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users. The degree of interoperability should be defined when referring to specific cases.

Joint Fleet Telecommunications Operations Center (JFTOC). The JFTOC is located at each NCTAMS and is the single point of contact in executing the Navy Combatant Commander communications requirements for the fleet with respect to the operational and network management of C4I systems.

Joint Network Management System (JNMS). JNMS is a Combatant Commander and Commander, Joint Forces (CJF), Joint Communications Planning and Management System.

Joint Tactical Radio System: A family of software defined radios capable of covering 2Mhz to 2Ghz frequency range.

Joint Technical Architecture. The JTA provides DOD systems with the basis for the needed seamless interoperability. The JTA defines the service areas, interfaces, and standards (JTA elements) applicable to all DOD systems, and its adoption is mandated for the management, development, and acquisition of new or improved systems throughout DOD. The JTA consists of two main parts: the JTA core, and the JTA Annexes. The JTA core contains the minimum set of JTA elements applicable to all DOD systems to support interoperability

Ka band. The portion of the radio frequency spectrum between 27 and 40 gigahertz (GHz).

Key Performance Parameters (KPP). Those capabilities or characteristics considered most essential for successful mission accomplishment. Failure to meet an ORD KPP threshold can be cause for the concept or system selection to be reevaluated or the program to be reassessed or terminated. Failure to meet a CRD KPP threshold can be cause for the family-of-systems or system-of-systems concept to be reassessed. KPPs are validated by the JROC.

Ku band. The portion of the radio frequency spectrum between 12 and 18 gigahertz (GHz).

L band. The portion of the radio frequency spectrum between 1 and 2 gigahertz (GHz).

Legacy Systems. Systems used by a single Service that are not interoperable. These systems are being replaced by common user functional systems.

Line Replaceable Unit (LRU). Essential support item which is removed and replaced at Organizational Maintenance level to restore the end item to an operationally ready condition.

Link. A general term used to indicate the existence of communications (facilities) between two points.

Long Fat Network. An Internet path with high bandwidth (>56Kbps) and long round-trip delays (common on SATCOM and long-haul terrestrial circuits) is known as a "long, fat pipe", and a network containing this path is known as a “long fat network”. The significant parameter is the product of bandwidth (bits per second) and round-trip delay (RTT in seconds) which affects the performance of the Transmission Control Protocol (TCP).

Low Data Rate (LDR). Data rates falling between 75 bps and 2.400 bps.

Maintainability. Reflects the ability of a system to be retained in or restored to a specified operational condition.

Medium Data Rate (MDR). Rates greater than 4.800bps up to and including 1.544 Mbps (T1) including voice grade circuits.

Military Satellite Communications. The satellite communications resources that are owned and operated by DOD primarily in the government frequency bands.

Mission Need Statement. A formatted non-system-specific statement containing operational capability needs and written in broad operational terms. It describes required operational capabilities and constraints to be studied during the Concept Exploration and Definition Phase.

Narrowband. Encompasses data rates less than 64 kilobits per second.

NATO Wide Area Network. IP network used by NATO in support of exercises, operations and contingency planning.

Network Control. Network control is the ability to plan and effectively manage user accesses to satellite communications capabilities. It encompasses the techniques and procedures over networks, terminals, satellites, and up/downlinks to effectively plan, monitor, control, and configure/reconfigure those assets in dynamic response to user needs while optimizing the overall throughput of the satellite.

Network Manager. A combatant command, component, or other organization that uses or manages a SATCOM apportionment and allocation. The network manager has operational control over the communications payload as defined by the CJCS-provided apportionment.

Network. A communications service of terminals connecting users (or a set of users) in order to provide a particular communication function.

Objective Requirement. The objective value is that desired by the user and which the program manager is attempting to attain.

Off-board logistic delay time. Is the logistics delay associated with obtaining a spare part from the depot level and or commercial vendor since part is not carried as an onboard repair part.

On-call. Links in which the capability to transfer information is not continuously provided throughout the day. Rather, telecommunications sessions supporting information transfer sessions are activated and released on-demand as needed throughout the day.

Other System Characteristics. Special categories of characteristics, which tend to be design, cost, and risk drivers to the program.

Point of Presence: A POP refers to a long distance company's equipment that is connected to the local telephone company's central office. The POP is the point at which telephone and data calls are handed off between local telephone companies and long distance telephone companies.

Point-to-Multipoint. Service from a hub or other location to multiple remote sites, which select and acknowledge data.

Point-to-Point. Uni-directional or bi-directional transmission from one point to another.

Priority. A user category defined by CJCSI 6250.01A that ranks SATCOM services based on mission and support of national survival. There are currently seven levels of priority.

Protected SATCOM. The protection provided by a SATCOM system is defined as the system's ability to avoid, prevent, negate, or mitigate the degradation, disruption, denial, unauthorized access, or exploitation of communications services by adversaries or the environment. There are various methods of providing levels of protection to SATCOM systems, both on the spacecraft and on the ground. The level of protection varies with the data rate with highly protected communications at the lower data rates (75 – 2400 bps) and less protection at the higher data rates (up to 8 Mbps in future systems).

Protection. Measures taken to ensure continuous access to satellite communications. This includes defensive Information Warfare, Anti-Jam, covertness, nuclear survivability, resistance to physical destruction, and US control of SATCOM access.

Reach Back. A satellite link from a deployed location to a STEP site or a link to the home station via the DII.

Regional SATCOM Support Center. The regional SATCOM support centers that provide the peacetime operational management of SATCOM resources in support of designated Combatant Commands, Services and Defense agencies and other users.

Relay. The retransmission of a received signal in the same information format in which it was received.

Robustness. A measure of signal quality used with SATCOM systems. A more robust signal is more protected during jamming, scintillation, or adverse weather conditions. Increasing a signal's robustness allows the receive terminal to operate with a lower Eb/No (margin) to close the communications link/service.

SATCOM Command and Control (C2) Centers. The operations centers responsible for satellite control and payload control execution.

SATCOM Operational Manager. The organization responsible for peacetime operations and resource management of SATCOM systems. Primary responsibility is maximizing system efficiency to support user requirements.

SATCOM System Expert. The component or designated organization responsible for providing the technical planning and functions in support of the operational management of a specific SATCOM constellation.

Satellite Communications (SATCOM). The term SATCOM includes military satellite communications, and DOD use of commercial, allied and civil satellite communications.

Satellite Data Base (SDB): Per CJCSI 6250.01A, the ICDB and ERDB were merged and are now referred to as the Satellite Data Base (SDB) with " Current Requirements" formerly ICDB meaning (present plus two years out) and " Future Requirements" formerly ERDB meaning (beyond 2 years out).

Secure Interoperability. Interoperable information exchange between and amongst entities such that information can be obtained and shared, based on a need-to-know basis, with authenticity, confidentiality, and integrity -- across systems and networks at various levels of classification -- while it is protected in storage, during processing, and in transit.

Service. The term "service" is used in two ways: If the word is capitalized, as in "Service", it indicates a Military Service (e.g., Army, Navy, Air Force, or Marine Corps). If the word is displayed un-capitalized, as in "service", it denotes a type of communications connectivity (e.g., two-way point-to-point, broadcast, etc.).

Simplex. One-way transmission through the satellite. Consists of the uplink signal, actions of the satellite on the signal (e.g., frequency translation or filter-switch-and-route) and the downlink.

Split-Based Operations. Operations in which a minimum of logistical and combat support functions and activities are deployed into the forward area of operations with the bulk of the support provided as and when needed from overseas or CONUS bases. A split-based concept relies on assured communications and readily available transportation between the forward-deployed warfighting elements and the supporting bases. Assured communications are necessary for the rearward elements to maintain situation awareness of the theater of operations and to receive and act on requests for support. Readily available transportation is necessary to deploy the requested support into theater immediately upon request.

Surge Capability: The ability to provide additional capacity and throughput from any Teleport site.

Telemedicine. The real time/near real time exchange of medical or dental knowledge, experience, and information between different geographical locations via telecommunications means using various applications: digitized live video and audio, images, data, text, graphics, and digitized vital signs. Generally refers to actions taken in consultation or in support of ongoing medical/dental procedures.

TEMPEST. Unclassified term for compromising emanations.

Terminal. Refers to X, C, Ku, Ka, UHF, EHF-band ground SATCOM terminals operating over various satellites, includes all system and subsystem components including modems and antennas.

Threshold. The threshold value is the minimum acceptable value that, in the user's judgment, is necessary to satisfy the need. If threshold values are not achieved, program performance is seriously degraded, the program may be too costly, or the program may no longer be timely.

UHF. The portion of the radio frequency spectrum between 0.3 and 3 gigahertz (GHz). Also a portion of the frequency spectrum between 225 and 399.95 megahertz (MHz) assigned to the U.S. military.

User to DISN Services (IOC1 and beyond). An individual satellite link (user) terminating at a Teleport and being routed to the DISN Service Delivery Node (SDN) or DISN Point of Presence (POP).

User to User (IOC1 and beyond). An individual satellite link (user) terminating at a Teleport and being re-transmitted by either cross-banding or an M-hop to another satellite link (user).

U. S. Control. The ability to deny adversarial entities (or others) the opportunity to deliberately or accidentally interfere with, interrupt, disrupt, deny, or exploit communication services that are not otherwise protected by anti-jam or LPI/LPD/LPE features. “U.S. Control” usually means the system is under the direct operational control of a corporate, private, or government activity that is subordinate or immediately responsive to the national legal direction of U.S. authority.

Wideband. Encompasses data rates greater than 64 kilobits per second.

Worldwide Coverage. Coverage between 65 degrees North and 65 degrees South latitude and at all longitudes.

X band. The portion of the radio frequency spectrum between 8 and 12 gigahertz (GHz).

ANNEX A

to the

DEPARTMENT OF DEFENSE (DOD) TELEPORT

GENERATION TWO UPDATE

Net Ready Key Performance Parameter (KPP) Update

5 January 2005

(INTENTIONALLY BLANK)

TABLE OF CONTENTS

LIST OF TABLES

LIST OF FIGURES

ANNEX A to the

DEPARTMENT OF DEFENSE (DOD) TELEPORT GENERATION TWO UPDATE ACAT IAM

Prepared for Generation Two Net Ready Update and Milestone C

1. Purpose

This document updates the DOD Teleport Generation Two Operational Requirements Document (ORD) Interoperability Key Performance Parameters (I KPP) of 9 July 2004 to include all applicable established components of the Net Ready Key Performance Parameter (NR KPP) in accordance with the CJCSI 6212.01C.

2. Net Ready Key Performance Parameter (NR KPP)

a. The NR-KPP assesses net-readiness; information assurance requirements; and both the technical exchange of information and the end-to-end operational effectiveness of that exchange. The NR-KPP consists of measurable and testable characteristics and/or performance metrics required for the timely, accurate, and complete exchange and use of information to satisfy information needs for a given capability.

b. The NR-KPP consists of the following elements: Information Assurance; Compliance with the Net-Centric Operations and Warfare Reference Model (NCOW RM); Compliance with applicable GIG Key Interface Profiles (KIPs); and Supporting Integrated Architecture Products. The following defines the interoperability requirements for the DOD Teleport System in accordance with the CJCSI 6212.01C:

(1) Information Assurance. The DOD Teleport System fully complies with DOD Directive 8500.1 and DOD Instruction 8500.2, and with Phase 3 Validation of the DITSCAP (DOD Instruction 8500.40), and has made the required Information Assurance documentation available to the Joint Staff J-6 for review. The DOD Teleport System Generation One IOC 1 System Security Authorization Agreement (SSAA) was approved April 2004.

In accordance with NSTISSP#11, GAO-04-678, DODD 8500.1, DODI 8500.2, and NSTISSAM COMPUSEC/1/99, the DOD Teleport System will use properly evaluated Commercial-off-the-shelf (COTS) IA equipment and IA-enabled software and NSA evaluated Government-off-the-shelf (GOTS) IA capabilities. Independent verification and validation of non-controlled software source code and hardware will be performed in compliance with GAO-04-678 at a level of thoroughness based on acceptance standards, such as a penetration test. Cross-domain solutions are a high security risk and should not be used unless there is a significant impact to Teleport missions. Teleport data integrity controls are in compliance with DODI 8200.2, DCID 6/1, and DCID 6/3. The Teleport system will comply with DODI 8551.1 for high risk protocols identified in the Teleport TV-1.

(2) Net-Centric Operations and Warfare Reference Model (NCOW RM). The NCOW RM is a description of enterprise level activities, services, technologies, and concepts (e.g. data strategy) that enable a net-centric environment for crossbanding, business, and management operations. The DOD Teleport System is compliant with the NCOW RM v1.0 and will be updated as refinements and more architecture products are added to the reference model. This supplement provides additional detail for the DOD Teleport System architectural views and descriptions not included in the Generation Two ORD.

(3) GIG Key Interface Profiles (KIPs). The DOD Teleport System was identified as one of the eight communication KIPs and was selected as a "pilot" KIP for development of an Interface Control Document that will serve as the basis for forming the Teleport System standards profile for the GIG architecture. The Teleport KIP Generation One (IOC1) of 1 October 2002 consists of the Interface Control Document and System View (SV) to Technical View (TV) Bridge and TV-1 and TV-2.

The DOD Teleport KIP defines the interfaces, standards, and protocols for both terrestrial and space interfaces and therefore is the only applicable KIP. However, as defined by Reference (s), other potentially applicable KIPs are shown in Table A-1. The DOD Teleport System and Generation Three will be coordinated with the development of these KIPs to ensure compliance.

KIP	DESCRIPTION	APPLICABLE	NOT APPLICABLE
1	Logical Networks to DISN Transport Backbone		√
2	Space to Terrestrial Interface	√	
3	JTF to Coalition		√
4	JTF Components to JTF Headquarters		√
5	STEP and Teleport	√	
6	Joint Interconnection Service		√
7	DISN Service Delivery Point	√	
8	Secure Enclave Service Point	√	
9	Applications to Database Server		√
10	Client/Server		√
11	Applications to COE/CCP		√
12	End System to PKI		√
13	Management Systems to (Integrated) Management Systems	√	
14	Management Systems to Managed Systems		√
15	IDM to Distributed Infrastructure		√
16	Information Servers to IDM Infrastructure		√
17	Applications to Shared Data		√

Table A-1: Applicable KIPs

(4) Integrated Architectural Products. The CJSCI 6212.01C requires, at a minimum, the following integrated architectural products be incorporated into the NR KPP and used to assess information exchange and use for a given capability. Appendix A of this document provides all of these products.

Framework Products	Framework Product Name
AV-1	Overview and Summary Information
OV-2	Operational Node Connectivity Description
OV-4	Organizational Relationship Chart
OV-5	Operational Activity Model
OV-6c	Operational Event-Trace Description
SV-4	System Functionality Description
SV-5	Operational Activity to Systems Functional Traceability Matrix
SV-6	Systems Data Exchange Matrix
TV-1	Technical Standards Profile

Table A-2: Principal Integrated Architecture Products

3. Net Ready KPP Statement

The DOD Teleport NR KPP statement is provided in Table A-3.

KPP	Threshold	Objective
All activity interfaces, services, policy-enforcement controls, and data-sharing of the NCOW-RM and GIG-KIPs will be satisfied to the requirements of the specific Joint integrated architecture products (including data correctness, data availability, and data processing), and information assurance accreditation, specified in the Threshold (T) and Objective (O) values.	100% of interfaces; services; policy-enforcement controls; and data correctness, availability and processing requirements designated as enterprise-level or critical in the Joint integrated architecture.	100% of interfaces; services; policy-enforcement controls; and data correctness, availability and processing requirements in the Joint integrated architecture.

Table A-3: NR KPP Statement

(INTENTIONALLY BLANK)

APPENDIX A: ARCHITECTURE FRAMEWORK

A.1 Overview and Summary Information (AV-1)

The DOD Teleport Generation Two Architecture provides an overarching picture of the functions and the information exchanges required to accomplish assigned Teleport missions. This architecture is a living representation of the DOD Teleport system, allowing for the efficient integration of new missions, functions, and technology.

A.1.1 Architecture Project Identification

Name:	**Department of Defense (DOD) Teleport Generation Two Architecture v1.0**
Short Name:	DOD Teleport, Generation Two Architecture v1.0
Architect:	Defense Information System Agency (DISA) Teleport Program Office (TPO) (GE441)
Coordinating Architects:	Navy (N71C1)
Organization Developing Architecture:	DISA TPO (GE 441)
Assumptions and Constraints:	None
Approval Authority:	Joint Requirement Oversight Council (JROC)
Milestone Decision Authority:	Office of the Secretary of Defense (OSD) Network and Information Integration (NII)
Date:	5 January 2005
Architecture Classification:	UNCLASSIFIED

A.1.2 Scope: Architecture Views and Products Identification

The DOD Teleport architecture, described by this effort, is an environment that provides deployed forces with sufficient interfaces for multi-band and multi-media connectivity from worldwide locations to Defense Information System Network (DISN) Service Delivery Nodes (SDN) and tactical command, control, communications, computers, and intelligence (C4I) systems. This system will facilitate the interoperability between multiple Satellite Communications (SATCOM) systems and deployed tactical networks, thus providing the user a seamless interface into the DISN and C4I systems.

The scope of this architecture defines the organizations, systems, information exchanges, and activities required to provision RF, DISN, and C4I system accesses from the deployed Warfighter to the DISN point of presence or backbone via the space segment and the Teleport site.

A.1.3 Architecture Views and Products Developed

This update is composed of the following DOD Architectural Framework (version 1.0):

1) Overview and Summary Description (AV-1)
2) High Level Operational Concept Graphic (OV-1)
3) Operational Node Connectivity Description (OV-2)

4) Organizational Relationship Chart (OV-4)
5) Operational Activity Model (OV-5)
6) Operational Event Trace Description (OV-6c)
7) System Functionality Description (SV-4)
8) Operational Activity to Systems Functional Traceability Matrix (SV-5)
9) Systems Data Exchange Matrix (SV-6)
10) Technical Standards Profile (TV-1)
11) Level of Information System Interoperability (LISI) Profile

A.1.4 Architecture Timeframe

This architecture product update addresses the Teleport Generation Two increment (FY04-07) and was developed IAW with the CJCSI 6212.01C for Net Ready Compliance. This architecture will be updated for Generation Three.

A.1.5 Organizations Involved

- Defense Information Systems Agency (DISA)
- OSD (NII)
- Navy (N61C1, SPAWAR)
- Army (PM DCATS)
- USJFCOM (J621)
- USSTRATCOM (CL183)

A.1.6 Purpose and Viewpoint

The requirement for Teleport is traced from the Joint Requirements Oversight Council (JROC) approved DOD Teleport Operational Requirements Document (ORD), dated 31 July 2000 (S/N 567-06-00) and the DOD Teleport Operational Requirements Document (ORD) Generation Two Update, dated 9 July 2004. These documents are the foundation of Teleport System and define its operational capability through Generation Two. The need for the Teleport System also flows from the Satellite Communications (SATCOM) System, Capstone Requirements Document (CRD), dated 25 May 2003; the Defense Information System Network (DISN), Capstone Requirement Document (CRD), dated 30 March 1995; and the Global Information Grid (GIG), Capstone Requirements Document (CRD), dated 30 August 2001.

This document further defines that architecture and incorporates the Net Ready Key Performance Parameter in accordance with the CJCSI 6212.01C. This document focuses on the Teleport Generation Two architecture.

Teleport is a global system but is considered a theater asset under the control of Joint Force Commander (JFC). Because of this, the architecture focal point is on theater assets subject to theater Operational Control (OPCON)/Tactical Control (TACON). This version is developed from a Joint perspective and is aligned with Joint Doctrine, JP 6-02, *Joint Doctrine for Employment of Operational/Tactical Command, Control, Communications, and Computer (C4) Systems Support*, 1 October 1996.

A.1.7 Intended Use

The intended uses for this architecture are:

- Further development and refinement of the Teleport architecture
- Support requirements development and milestone decision reviews
- Support CONOPS maturation, evolution, and configuration management
- Validate the Generation Two Teleport
- Provide the foundation for future additional architecture development
- Promote standardization format to compare and collaboratively build architectures

A.1.8 Anticipated Decisions

This architecture product update is intended to support Generation Two Milestone C and the NR KPP compliance.

A.1.9 Context

The operational environment depicted in the high level operational concept (OV-1) is based on Net-Centricity and represents a piece of the overall Network-Centric Operations and Warfare Reference Model (NCOW RM). Net-Centricity is the realization of a robust, globally interconnected, network environment (including infrastructure, systems, processes, people) in which data is shared timely and seamlessly among users, applications and platforms. By securely interconnecting people and systems, independent of time or location, net-centricity enables substantially improved military situational awareness and significantly shortened decision making cycles. Users are empowered to better protect assets; more effectively exploit information; more efficiently use resources; and unify our forces by supporting extended, collaborative communities to focus on the mission. The Teleport system represents a portion of that end-to-end infrastructure, systems, and processes enabling global connectivity.

A.1.10 Scenario

This architecture addresses the operational procedures for SATCOM mission planning, both peacetime and wartime, as it pertains to Teleport support, mission planning, users, operators, and managers. As the DoD transitions to a network-centric environment, this document provides a focus on the need for integration, synchronization, coordination, and reliable operations-related data flow among Teleport Operations Officers, SATCOM Support Centers (SSCs), Military and Commercial SATCOM operations centers and customers using Government-owned or leased transponder services.

A.1.11 Authoritative Development Sources

- DOD Architecture Framework, Version 1.0 – Final Draft, 30 August 2003
- Network Centric Operations and Warfare Reference Model, Version 1.0

- CJCSM 3170.01 Series, "Operation of the Joint Capabilities Integration and Development System."
- CJCSI 3170.01 Series, "Joint Capabilities Integration and Development System."
- CJCSI 6212.01 Series, "Interoperability and Supportability of Information Technology and National Security Systems (IT and NSS)."

A.1.12 Tools and File Formats

The following tools were used to build these architecture products:

- Microsoft Word, Power Point, and Excel
- DISROnline
- InspeQtor Tool and the Level of Information System Interoperability (LISI) Survey

File Formats

- Microsoft Office (Word, Power Point, Excel)
- Hypertext Markup Language (HTML)

A.2 High Level Operational Concept Graphic (OV-1)

a. Product Purpose. The purpose of this high-level OV-1 is to present the high level interfaces and operational concepts of the DOD Teleport.

b. Product Description. The DOD Teleport System is a telecommunications collection, transport, and distribution point providing deployed forces with multi-ban and multimedia worldwide reach-back capabilities to the DISN. At the highest level, the DOD Teleport OV-1 interfaces between the fixed DISN and the deployed DISN as shown in Figure A-1. This system will provide interoperability among multiple military satellite communications (MILSATCOM) systems, commercial SATCOM systems, and fixed DISN. The Operational Management of these SATCOM systems and its implementation, as part of Teleport, will be defined within the USSTRATCOM Consolidated Systems Control and Operations Concept (SCOC).

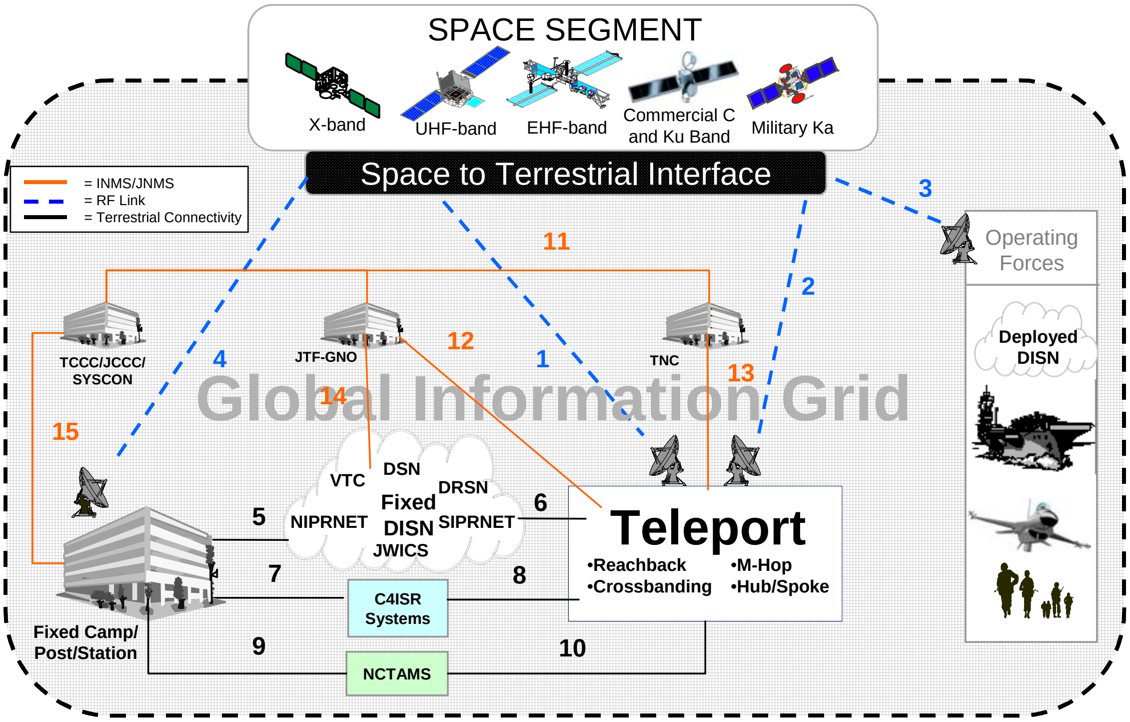


Figure A-A-1: DOD Teleport OV-1

The DISN is a sub-element of the GIG and is the DOD's consolidated worldwide enterprise level telecommunications infrastructure that provides end-to-end information transfer network for supporting military operations. The DOD Teleport, as this interface, provides the deployed Warfighter access to the following DISN network services:

(1) Defense Switched Network (DSN) for nonsecure voice
(2) Defense Red Switched Network (DRSN) for secure voice
(3) Secret Internet Protocol Router Network (SIPRNET) for secure Internet Protocol (IP) service
(4) Nonclassified Internet Protocol Router Network (NIPRNET) for nonsecure IP service
(5) Joint Worldwide Intelligence Communications System (JWICS)
(6) Video Teleconference Center (VTC) services

c. Data Elements for High Level Operational Concept (OV-1). The DOD Teleport High Level OV-1 data definitions are provided in Table A-A-1.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Asset Icon		
	Name	Space Segment
	Representation Type	Space Satellite Communication Assets
	Description	The space segment consists of the different satellites and mediums that interface with DOD Teleport System interfaces. For Generation Two, the mediums are X-band, UHF-band, EHF-band, Commercial C/Ku-band, and Ka-band.
	Generic Location	Space

Data Elements	Attributes	Example Values/Explanation
	Name	Operating Forces
	Representation Type	Deployed Operating Forces
	Description	The Operating Forces consist of those deployed Joint or Coalition Warfighter assets, platforms, weapons, and forces and interface with the deployed DISN.
	Generic Location	Worldwide
	Name	Teleport
	Representation Type	Enterprise Satellite Communication Station
	Description	The DOD Teleport supports the Combatant Commanders with extended multi-band satellite communication capability and seamless access to terrestrial components of the DISN.
	Generic Location	Six primary land based locations: Camp Roberts, Northwest, Ramstein/Landstuhl, Lago Patria, Fort Buckner, and Wahiawa. One secondary site at Bahrain.
	Name	Fixed Camp/Post/Station
	Representation Type	Land-based installation
	Description	The fixed Camp/Post/Station consists of all fixed land-based military locations that interface with the fixed DISN and C4I Systems.
	Generic Location	Worldwide
	Name	TNC
	Representation Type	Theater Network Center
	Description	The TNC will develop, monitor, and maintain a timely and accurate event-based GIG situational awareness view for the theater, enabling the Combatant Commander to direct, command, control, and coordinate NetOps resources that impact operations in the AOR. For Teleport missions, the TNC receives communication plans from the CONEX and provides operational direction for the activation and troubleshooting of Teleport DISN services, among them DSN, DRSN, VTC, JWICS, NIPRNET, and SIPRNET. Teleport O&M personnel will coordinate with the TNC, DISN Service Centers, Teleport Earth Terminals, and tactical users through NIPRNET/SIPRNET, commercial phone systems, or orderwire systems between the elements.
	Generic Location	Worldwide
	Name	TCCC/JCCC/SYSCON
	Representation Type	Theater Communications Control Center/Joint Communications Control Center/Systems Control Center
	Description	The Combatant Commander or JTF/J6 directorate, or the equivalent office, is responsible for joint communications management, which combines centralized control with decentralized execution. The SYSCON manages and controls the communications networks for an individual service component and usually reports to the JTF/JCCC in a joint environment.
	Generic Location	Worldwide
	Name	JTF-GNO
	Representation Type	Joint Task Force – Global Network Operations
	Description	With oversight from the Joint Staff (JS) J6 and the U.S. Strategic Command (USSTRATCOM) SATCOM Operational Manager (SOM), the JTF-GNO, in conjunction with the Unified Combatant Commander (UCC) and resource managers, will

Data Elements	Attributes	Example Values/Explanation
		lead and direct continuous Enterprise Service Management (ESM), Information Assurance and Computer Network Defense (CND) and Content Staging/Information Dissemination Management (IDM) throughout the GIG, maintaining near real-time situational awareness, end-to-end management and dynamic DOD network defense, assuring Global Decision Superiority. The JTF-GNO will operate the GIG through the Global NetOps Center (GNC) and the Global NetOps Support Center (GNSC). The GNC is responsible for the daily operation and defense of the GIG. The GNSC is responsible for the day-to-day technical operation, control and management of GIG assets that are not assigned to a COCOM.
	Generic Location	Peterson AFB (GSSC), Macdill AFB (RSSC-CONUS), Wahiawa (RSSC-Pacific), and Patch Barracks (RSSC-Europe)
	Name	NCTAMS
	Representation Type	Naval Computer and Telecommunications Area Master Station
	Description	The Operations Departments of the three NCTAMS manage the Navy's command, control, communications, and computers (C4) requirements for their areas of operations. The numbered Fleet Commander and the Unified Combatant Commander (UCC) will validate access requests and the TNC will authorize X-band. The numbered Fleet Commander will validate and the DSCS Manager will authorize Navy only C-band, NCTAMS JFTOC will make satellite and frequency assignment. NCTAMS will coordinate baseband (DISN) requirements with the deployed user.
	Generic Location	Wahiawa, Naples, and Norfolk
Graphical Arrow Types		
Line		
	Identifier	1 and 2
	Descriptive Name	Communications Link
	Description	Radio Frequency (RF) communication link in either X, UHF, EHF, C, Ku, L, and/or Ka-band
	"From"	Teleport Site; Space Segment
	"To"	Space Segment; Teleport
	Identifier	3
	Descriptive Name	Communication Link
	Description	Radio Frequency (RF) communication link in either X, UHF, EHF, C, Ku, L, and/or Ka-band
	"From"	Space Segment; Operating Forces
	"To"	Operating Forces; Space Segment
	Identifier	4
	Descriptive Name	Communication Link
	Description	Radio Frequency (RF) communication link in either X, UHF, EHF, C, Ku, L, and/or Ka-band
	"From"	Space Segment; Fixed Camp/Post/Station
	"To"	Fixed Camp/Post/Station; Space Segment
	Identifier	5
	Descriptive Name	Communication Link
	Description	Series of fixed network connections
	"From"	Fixed Camp/Post/Station; DISN
	"To"	DISN; Fixed Camp/Post/Station
	Identifier	6

Data Elements	Attributes	Example Values/Explanation
	Descriptive Name	Communication Link
	Description	Series of fixed network connections
	“From”	Teleport; DISN
	“To”	DISN; Teleport
	Identifier	7
	Descriptive Name	Communications Link
	Description	Series of fixed network connections
	“From”	Fixed Camp/Post/Station; C4I Systems
	“To”	C4I Systems; Fixed Camp/Post/Station
	Identifier	8
	Descriptive Name	Communication Link
	Description	Series of fixed network connections
	“From”	Teleport; C4I Systems
	“To”	C4I Systems; Teleport
	Identifier	9
	Descriptive Name	Communications Link
	Description	Series of fixed network connections
	“From”	NCTAMS; Fixed Camp/Post/Station
	“To”	Fixed Camp/Post/Station; NCTAMS
	Identifier	10
	Descriptive Name	Communications Link
	Description	Series of fixed network connections
	“From”	NCTAMS’ Teleport
	“To”	Teleport; NCTAMS
	Identifier	11
	Descriptive Name	Needline
	Description	RF Access; RF Gateway; RF Directive
	“From”	GNC; G/RNSOC; TCCC/JCCC/SYSCON
	“To”	TCCC/JCCC/SYSCON; GNC; TNC
	Identifier	12
	Descriptive Name	Needline
	Description	RF Access; RF Gateway; RF Directive
	“From”	TNC; Teleport
	“To”	Teleport; TNC
	Identifier	13
	Descriptive Name	Needline
	Description	RF Access; RF Gateway; RF Directive
	“From”	Teleport; GNC
	“To”	GNC; Teleport
	Identifier	14
	Descriptive Name	Needline
	Description	RF Access; RF Gateway; RF Directive
	“From”	TNC; Fixed DISN
	“To”	Fixed DISN; TNC
	Identifier	15
	Descriptive Name	Needline
	Description	RF Access; RF Gateway; RF Directive
	“From”	TCCC/JCCC/SYSCON; Fixed Camp/Post/Station
	“To”	Fixed Camp/Post/Station; TCCC/JCCC/SYSCON

Table A-A-1: Teleport High Level Operational View (OV-1) Data Elements

A.3 Operational Employment Concept

a. The capabilities or service offerings of the DOD Teleport System can be grouped into the following operational mission areas:

1) Provision of connectivity to the DISN services (pre-positioned via the Teleport sites) and C4I services for deployed tactical users

2) Provision of hub/spoke functionality for deployed tactical users

3) Provision of digital and frequency crossbanding functionality between deployed tactical forces using different satellite systems (e.g., super high frequency [SHF]-to-UHF, MILSATCOM-to-commercial systems)

4) Provision of relay or multiple-hop functionality for deployed tactical forces using different satellites or satellite antenna beams.

b. Reference (b) specified that mission support of the Teleport System fell into three categories or operations: Configuration Management, Access Coordination, and Change in Status. In conformance with the NCOW RM, these mission support operations have been updated to reflect activities performed in a Net-centric environment and are the foundation for OV-2 development.

DOD TELEPORT ACTIVITY	NCOW ACTIVITY	NCOW ACTIVITY #	RATIONALE
Configure System and Network Controls	Configure System and Network Controls	A523	The DOD Teleport M&C Subystem establishes, maintains, and coordinates the software and hardware operating parameters used by the system and network control mechanisms within the information environment. The Teleport System provides system and network configuration for RF and DISN Directive Management and C4I Management.
Provide Communication Resources	Provide Communication Resources	A42	The DOD Teleport provides access to satellite communication, C4I systems and services (e.g., Reference (b), Appendix B), and DISN network services (i.e., DSN, DRSN, NIPRNET, SIPRNET, JWICS, VTC)
Perform Day-to-Day Control of Electromagnetic Spectrum in Theater	Perform Day-to-Day Control of Electromagnetic Spectrum in Theater	A5234	The DOD Teleport Management and Control System coordinates change in status to the RF Gateway, DISN Gateway, and C4I services. Control of the Electromagnetic Spectrum is governed by Reference (e).

A.4 Reachback and Hub/Spoke (OV-2)

A.4.1 Configure Network and Systems Controls (OV-2)

a. Product Purpose. The purpose of the Configure Network and Systems Controls OV-2 is to define the connectivity and information exchanges between the DOD Teleport and its operational nodes for network configuration to the RF and DISN Directive Management and C4I Management.

b. Product Description. The DOD Teleport Monitoring and Control (M&C) Subystem establishes, maintains, and coordinates the software and hardware operating parameters used by the system and network control mechanisms within the information environment. The Teleport System provides system and network configuration and re-configuration for RF and DISN Directive Management and C4I Management. The Teleport M&C System configures access to the RF Gateway, the DISN gateway, and C4I systems and services as shown in Figure A-A-2a. These operational concepts are further defined in paragraph 1.c of Reference (b).

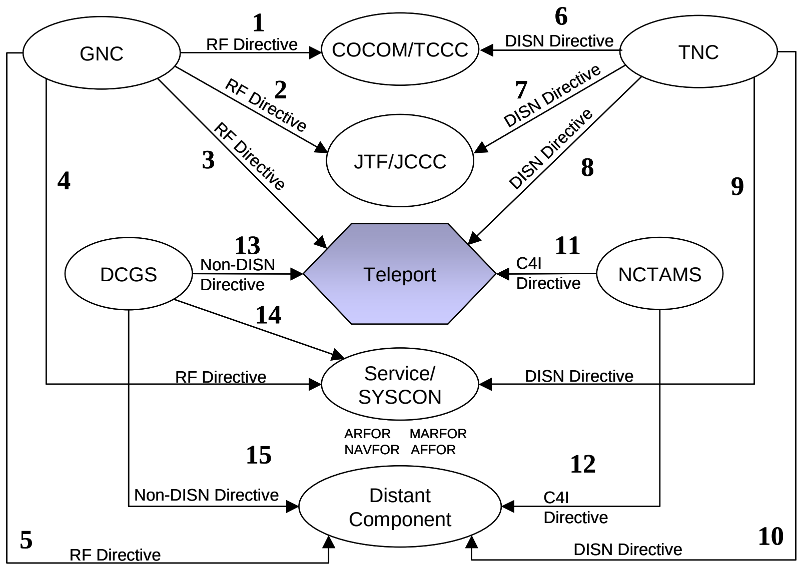


Figure A-A-2a: Reachback and Hub/Spoke Configure Network and Systems Control (OV-2)

c. Data Elements for Configure Network and Systems Control (OV-2). The DOD Teleport Configure Network and Systems Control (OV-2) data elements are provided in Table A-A-2a.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Operational Node		
	Name	Teleport
	Representation Type	Enterprise Satellite Communication Station
	Description	The DOD Teleport Station is a fixed location consisting of the following elements: satellite terminal, SATCOM modems, TRANSEC, Multiplexers, bandwidth resource managers, DISN interfaces, miscellaneous equipment, and management and control (M&C) functions. The satellite terminal includes antenna, amplifiers, and other RF/IF components for each frequency band.
	Generic Location	Six primary locations: Camp Roberts, Northwest, Ramstein/Landstuhl, Lago Patria, Fort Buckner, and Wahiawa. One secondary site is Bahrain.
	Name	JTF-GNO
	Representation Type	Joint Task Force – Global Network Operations
	Description	With oversight from the Joint Staff (JS) J6 and the U.S. Strategic Command (USSTRATCOM) SATCOM Operational Manager (SOM), the JTF-

Data Elements	Attributes	Example Values/Explanation
		GNO will lead and direct continuous Enterprise Service Management (ESM), Information Assurance and Computer Network Defense (CND) and Content Staging/Information Dissemination Management (IDM) throughout the GIG, maintaining near real-time situational awareness, end-to-end management and dynamic DOD network defense, assuring Global Decision Superiority. . The JTF-GNO will operate the GIG through the Global NetOps Center (GNC) and the Global NetOps Support Center (GNSC). The GNC is responsible for the daily operation and defense of the GIG. The GNSC is responsible for the day-to-day technical operation, control and management of GIG assets that are not assigned to a COCOM.
	Name	TNC
	Representation Type	Theater NetOps Center
	Description	The TNC will develop, monitor, and maintain a timely and accurate event-based GIG situational awareness view for the theater, enabling the Combatant Commander to direct, command, control, and coordinate NetOps resources that impact operations in the AOR. For Teleport missions, the TNC receives communication plans from the CONEX and provides operational direction for the activation and troubleshooting of Teleport DISN services, among them DSN, DRSN, VTC, JWICS, NIPRNET, and SIPRNET. Teleport O&M personnel will coordinate with the TNC, DISN Service Centers, Teleport Earth Terminals, and tactical users through NIPRNET/SIPRNET, commercial phone systems, or orderwire systems between the elements.
	Generic Location	Worldwide
	Name	Combatant Commander/TCCC
	Representative Type	Combatant Commander/Theater Communications Control Center
	Description	The Combatant Commander or JTF/J6 directorate, or the equivalent office, is responsible for joint communications management, which combines centralized control with decentralized execution.
	Generic Location	Worldwide
	Name	JTF/JCCC
	Representative Type	Joint Task Force/Joint Communications Control Center
	Description	The Combatant Commander or JTF/J6 directorate, or the equivalent office, is responsible for joint communications management, which combines centralized control with decentralized execution.
	Generic Location	Worldwide
	Name	DCGS
	Representative Type	Distributed Common Ground System
	Description	Teleport will provide connectivity to DCGS through standard DISN networks (e.g., JWICS, SIPRNET) or through user provided and funded alternative connectivity. DCGS receives information from various Intelligence, Surveillance and Reconnaissance (ISR) platforms and sends that information over its own WAN to provide for distributed processing and analysis.
	Generic Location	Worldwide

Data Elements	Attributes	Example Values/Explanation
	Name	NCTAMS
	Representative Type	Naval Computer and Telecommunications Area Master Station
	Description	The Operations Departments of the three NCTAMS manage the Navy's command, control, communications, and computers (C4) requirements for their areas of operations. The numbered Fleet Commander and the Unified Combatant Commander (UCC) will validate access requests and the TNC will authorize X-band. The numbered Fleet Commander will validate and the DSCS Manager will authorize Navy only C-band, NCTAMS JFTOC will make satellite and frequency assignment. NCTAMS will coordinate baseband (DISN) requirements with the deployed user.
	Generic Location	Wahiawa, Naples, Norfolk, and Bahrain (NCTS)
	Name	Service/SYSCON
	Representative Type	Military Services Systems Control Center
	Description	The SYSCON manages and controls the communications networks for an individual service component and usually reports to the JTF/JCCC in a joint environment.
	Generic Location	Worldwide
	Name	Distant Component
	Representative Type	System
	Description	Teleport Component (i.e., earth terminal) geographically separated from Teleport core site
	Generic Location	Worldwide
Graphical Arrow Types		
Needline		
	Identifier	1
	Descriptive Name	RF Directive
	Description	Transmit Satellite Access Authorization (SAA)
	"From Operational Node"	GNC
	"To Operational Node"	TCCC
	Identifier	2
	Descriptive Name	RF Directive
	Description	Transmit SAA
	"From Operational Node"	GNC
	"To Operational Node"	JCCC
	Identifier	3
	Descriptive Name	RF Directive
	Description	Transmit SAA
	"From Operational Node"	GNC
	"To Operational Node"	DOD Teleport
	Identifier	4
	Descriptive Name	RF Directive
	Description	Transmit SAA
	"From Operational Node"	GNC
	"To Operational Node"	SYSCON
	Identifier	5
	Descriptive Name	RF Directive
	Description	Transmit SAA
	"From Operational Node"	GNC
	"To Operational Node"	Distant Component
	Identifier	6
	Descriptive Name	DISN Directive

Data Elements	Attributes	Example Values/Explanation
	Description	Transmit Gateway Access Request (GAR)
	"From Operational Node"	TNC
	"To Operational Node"	TCCC
	Identifier	7
	Descriptive Name	DISN Directive
	Description	Transmit GAA
	"From Operational Node"	TNC
	"To Operational Node"	JCCC
	Identifier	8
	Descriptive Name	DISN Directive
	Description	Transmit GAA
	"From Operational Node"	TNC
	"To Operational Node"	DOD Teleport
	Identifier	9
	Descriptive Name	DISN Directive
	Description	Transmit GAA
	"From Operational Node"	TNC
	"To Operational Node"	SYSCON
	Identifier	10
	Descriptive Name	DISN Directive
	Description	Transmit GAA
	"From Operational Node"	TNC
	"To Operational Node"	Distant Component
	Identifier	11
	Descriptive Name	C4I Directive
	Description	Coordinate Distant Component and backhaul connections via Teleport
	"From Operational Node"	NCTAMS
	"To Operational Node"	DOD Teleport
	Identifier	12
	Descriptive Name	C4I Directive
	Description	Coordinate Navy connectivity to Teleport
	"From Operational Node"	NCTAMS
	"To Operational Node"	Distant Component
	Identifier	13
	Descriptive Name	Non-DISN Directive
	Description	Coordinate DCGS connectivity to Teleport
	"From Operational Node"	DCGS
	"To Operational Node"	DOD Teleport
	Identifier	14
	Descriptive Name	Non-DISN Directive
	Description	Coordinate DCGS connectivity to SYSCON
	"From Operational Node"	DCGS
	"To Operational Node"	SYSCON
	Identifier	15
	Descriptive Name	Non-DISN Directive
	Description	Coordinate DCGS connectivity to Deployed Warfighter
	"From Operational Node"	DCGS
	"To Operational Node"	Distant Component

Table A-A-2a: Configure Network and System Controls (OV-2) Data Elements

A.4.2 Provide Communication Resources (OV-2)

a. Product Purpose. The purpose of the Provide Communication Resources OV-2 is to define the connectivity and information exchanges between the DOD Teleport System and its operational nodes for access coordination to the RF and DISN Gateway and C4I systems and services.

b. Product Description. The DOD Teleport provides access to satellite communication, C4I systems and services (e.g., NCTAMS, Reference (b), Appendix B), and DISN network services (i.e., DSN, DRSN, NIPRNET, SIPRNET, JWICS, VTC). The Teleport communication resources (e.g., bandwidth, frequency) include access coordination to the RF Gateway, DISN Gateway, and C4I systems and services. These operational concepts are further defined in paragraph 1.c of Reference (b).

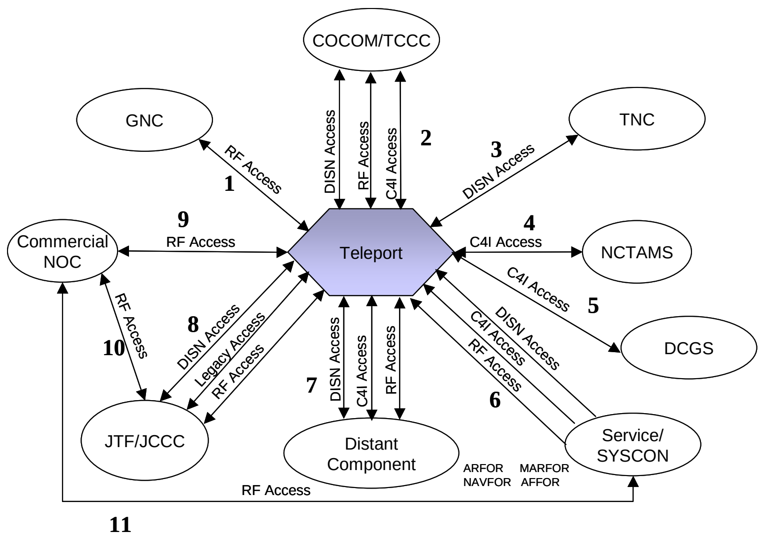


Figure A-A-2b: Reachback and Hub/Spoke Provide Communication Resources (OV-2)

c. Data Elements for Provide Communication Resources (OV-2). The DOD Teleport Provide Communication Resources (OV-2) data elements are provided in Table A-A-2b.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Operational Node		
	Name	Teleport
	Description	As previously defined.
	Name	GNC
	Description	As previously defined.
	Name	TNC
	Description	As previously defined.
	Name	Combatant Commander/TCCC
	Description	As previously defined.
	Name	JTF/JCCC
	Description	As previously defined.
	Name	DCGS
	Description	As previously defined.

Data Elements	Attributes	Example Values/Explanation
	Name	NCTAMS
	Description	As previously defined.
	Name	Service/SYSCON
	Description	As previously defined.
	Name	Distant Component
	Description	As previously defined.
	Name	Commercial NOC
	Representative Type	Commercial Network Operations Center
	Description	Operational management of commercial transponders and circuits is performed by the commercial network operations center associated with each commercial satellite access. Users will coordinate commercial SATCOM access and status changes via commercial telephone resources or e-mail.
Graphical Arrow Types		
Needline		
	Identifier	1
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	"From Operational Node"	GNC; DOD Teleport
	"To Operational Node"	DOD Teleport; GNC
	Identifier	2
	Descriptive Name	RF, DISN, C4I Access
	Description	Coordinate RF, DISN, or C4I connectivity between TCCC and Teleport
	"From Operational Node"	TCCC;DOD Teleport
	"To Operational Node"	DOD Teleport;TCCC
	Identifier	3
	Descriptive Name	DISN Access
	Description	Coordinate DISN resources
	"From Operational Node"	TNC; DOD Teleport
	"To Operational Node"	DOD Teleport; TNC
	Identifier	4
	Descriptive Name	C4I Access
	Description	Coordinate backhaul connectivity between NCTAMS and Teleport
	"From Operational Node"	NCTAMS; DOD Teleport
	"To Operational Node"	DOD Teleport; NCTAMS
	Identifier	5
	Descriptive Name	C4I Access
	Description	Coordinate connectivity between DCGS and Teleport
	"From Operational Node"	DCGS;DOD Teleport
	"To Operational Node"	DOD Teleport; DCGS
	Identifier	6
	Descriptive Name	RF, DISN, C4I Access
	Description	Coordinate RF, DISN, and C4I connectivity between Teleport and SYSCON
	"From Operational Node"	SYSCON
	"To Operational Node"	DOD Teleport
	Identifier	7
	Descriptive Name	RF, DISN, C4I Access
	Description	Coordinate RF, DISN, or C4I connectivity between Distant Component and Teleport
	"From Operational Node"	Distant Component; DOD Teleport

Data Elements	Attributes	Example Values/Explanation
	"To Operational Node"	DOD Teleport; Distant Component
	Identifier	8
	Descriptive Name	RF, DISN, C4I Access
	Description	Coordinate RF, DISN or C4I connectivity between JCCC and Teleport
	"From Operational Node"	JCCC; DOD Teleport
	"To Operational Node"	DOD Teleport; JCCC
	Identifier	9
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	"From Operational Node"	Commercial NOC; DOD Teleport
	"To Operational Node"	DOD Teleport; Commercial NOC
	Identifier	10
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	"From Operational Node"	Commercial NOC; JCCC
	"To Operational Node"	JCCC; Commercial NOC
	Identifier	11
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	"From Operational Node"	Commercial NOC; SYSCON
	"To Operational Node"	SYSCON; Commercial NOC

Table A-A-2b: Provide Communication Resources (OV-2) Data Elements

A.4.3 Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2)

a. Product Purpose. The purpose of the Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2) is to define connectivity and information exchanges between the DOD Teleport System and its operational nodes for changes in status of the RF and DISN Gateway and C4I systems and services.

b. Product Description. The DOD Teleport Management and Control System coordinates change in status to the RF Gateway, DISN Gateway, and C4I services. Control of the Electromagnetic Spectrum is governed by Reference (e). The Teleport System coordinates change in status to the RF Gateway, DISN Gateway, and C4I systems and services. These operational concepts are further defined in paragraph 1.c of Reference (b).

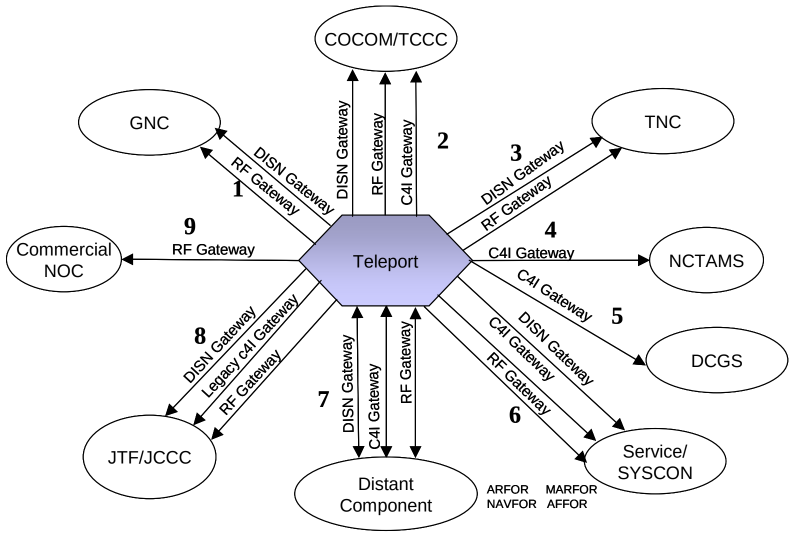


Figure A-A-2c: Reachback and Hub/Spoke Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2)

c. Data Elements for Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2). The DOD Teleport Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2) data elements are provided in Table A-A-2c.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Operational Node		
	Name	Teleport
	Description	As previously defined.
	Name	GNC
	Description	As previously defined.
	Name	TNC
	Description	As previously defined.
	Name	Combatant Commander/TCCC
	Description	As previously defined.
	Name	JTF/JCCC
	Description	As previously defined.
	Name	DCGS
	Description	As previously defined.
	Name	NCTAMS
	Description	As previously defined.
	Name	Service/SYSCON
	Description	As previously defined.
	Name	Distant Component
	Description	As previously defined.
	Name	Commercial NOC
	Description	As previously defined.
Graphical Arrow Types		
Needline		

Data Elements	Attributes	Example Values/Explanation
	Identifier	1
	Descriptive Name	RF or DISN Gateway
	Description	Transmit SATCOM connectivity status message
	"From Operational Node"	DOD Teleport
	"To Operational Node"	GNC
	Identifier	2
	Descriptive Name	RF, DISN, or C4I Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	"From Operational Node"	DOD Teleport
	"To Operational Node"	TCCC
	Identifier	3
	Descriptive Name	RF or DISN Gateway
	Description	Transmit network connectivity status message
	"From Operational Node"	DOD Teleport
	"To Operational Node"	TNC
	Identifier	4
	Descriptive Name	C4I Gateway
	Description	Manage backhaul and Navy distant component access
	"From Operational Node"	DOD Teleport
	"To Operational Node"	NCTAMS
	Identifier	5
	Descriptive Name	C4I Gateway
	Description	Transmit link connectivity status message
	"From Operational Node"	DOD Teleport
	"To Operational Node"	DCGS
	Identifier	6
	Descriptive Name	RF, DISN, or C4I Gateway
	Description	Transmit system status message
	"From Operational Node"	DOD Teleport
	"To Operational Node"	SYSCON
	Identifier	7
	Descriptive Name	RF, DISN, or C4I Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	"From Operational Node"	DOD Teleport; Distant Component
	"To Operational Node"	Distant Component; DOD Teleport
	Identifier	9
	Descriptive Name	RF, DISN, or C4I Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	"From Operational Node"	DOD Teleport
	"To Operational Node"	JCCC
	Identifier	9
	Descriptive Name	RF Gateway
	Description	Manage SATCOM resources
	"From Operational Node"	DOD Teleport
	"To Operational Node"	Commercial NOC

Table A-A-2c: Perform Day-to-Day Control of Electromagnetic (OV-2) Spectrum Data Elements

A.5 Crossbanding and Multiple Hop (OV-2)
A.5.1 Configure Network and Systems Controls (OV-2)

a. Product Purpose. The purpose of the Crossbanding and Multiple Hop Configure Network and Systems Controls OV-2 is to define the connectivity and information exchanges between the DOD Teleport and its operational nodes for network configuration to the RF and DISN Directive Management and C4I Management.

b. Product Description. The DOD Teleport Monitoring and Control (M&C) Subystem establishes, maintains, and coordinates the software and hardware operating parameters used by the system and network control mechanisms within the information environment. The Teleport System provides system and network configuration for RF and DISN Directive Management and C4I Management. The Teleport System configures access to the RF Gateway, the DISN gateway, and C4I systems and services as shown in Figure A-A-3a. These operational concepts are further defined in paragraph 1.c of Reference (b).

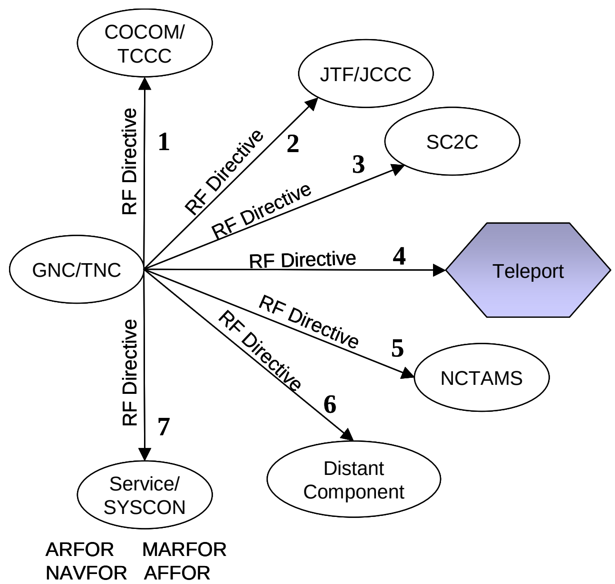


Figure A-A-3a: Crossbanding & Multiple Hop Configure Network & System Controls (OV-2)

c. Data Elements for Configure Network and Systems Control (OV-2). The DOD Teleport Configure Network and Systems Control (OV-2) data elements are provided in Table A-A-3a.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Asset Icon		
	Name	Teleport
	Description	As previously defined
	Name	GNC/TNC
	Description	As previously defined
	Name	Combatant Commander/TCCC
	Description	As previously defined
	Name	JTF/JCCC

Data Elements	Attributes	Example Values/Explanation
	Description	As previously defined
	Name	NCTAMS
	Description	As previously defined
	Name	Service/SYSCON
	Description	As previously defined
	Name	Distant Component
	Description	As previously defined
	Name	SC2C
	Representative Type	Satellite Command and Control Center
	Description	Satellite Command and Control Centers (SC2C) perform commanding functions for SATCOM and payload control. SC2Cs include 3SOPS, 4SOPS, NAVSOC, and WSOC.
	Generic Location	Worldwide
Graphical Arrow Types		
Line		
	Identifier	1
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	TCCC
	Identifier	2
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	JCCC
	Identifier	3
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	SC2C
	Identifier	4
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	DOD Teleport
	Identifier	5
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	NCTAMS
	Identifier	6
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	Distant Component
	Identifier	7
	Descriptive Name	RF Directive
	Description	Transmit SAA
	“From”	GNC/TNC
	“To”	Service; SYSCON

Table A-A-3a: Configure Network and System Controls (OV-2) Data Elements

A.5.2 Provide Communication Resources (OV-2)

a. Product Purpose. The purpose of the Crossbanding and Multiple Hop Provide Communication Resources OV-2 is to define the connectivity and information exchanges between the DOD Teleport and its operational nodes for access coordination to the RF and DISN Gateway and C4I systems and services.

b. Product Description. The DOD Teleport provides access to satellite communication, legacy C4I, (e.g., NCTAMS, Appendix E), and DISN network services (i.e., DSN, DRSN, NIPRNET, SIPRNET, JWICS, VTC). The Teleport System configures access to the RF Gateway, the DISN gateway, and C4I systems and services as shown in Figure A-A-3b. These operational concepts are further defined in paragraph 1.c of Reference (b).

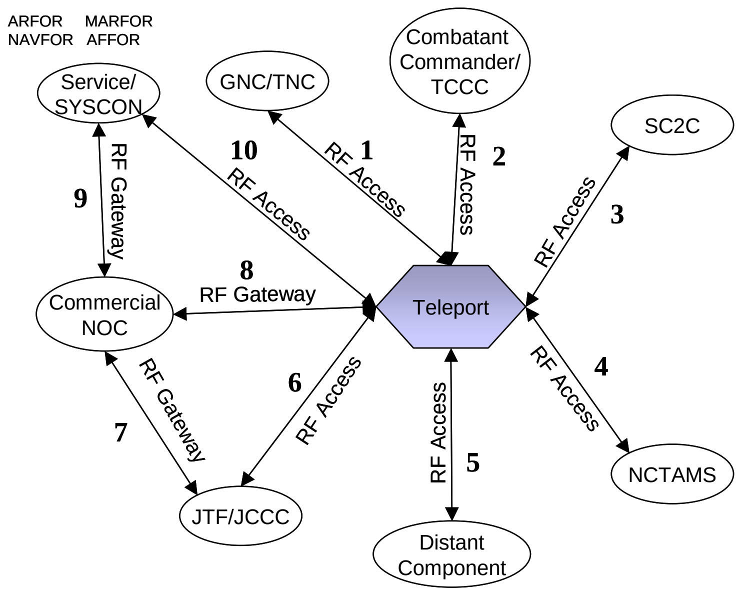


Figure A-3b: Crossbanding and Multiple Hop Provide Communication Resources (OV-2)

c. Data Elements for Provide Communication Resources (OV-2). The DOD Teleport Provide Communication Resources (OV-2) data elements are provided in Table A-A-3b.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Asset Icon		
	Name	Teleport
	Description	As previously defined
	Name	GNC/TNC
	Description	As previously defined
	Name	Combatant Commander/TCCC
	Description	As previously defined
	Name	JTF/JCCC
	Description	As previously defined
	Name	NCTAMS
	Description	As previously defined

Data Elements	Attributes	Example Values/Explanation
	Name	Service/SYSCON
	Description	As previously defined
	Name	Distant Component
	Description	As previously defined
	Name	Commercial NOC
	Description	As previously defined
	Name	SC2C
	Description	As previously defined
Graphical Arrow Types		
Line		
	Identifier	1
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	“From”	GNC/TNC; DOD Teleport
	“To”	DOD Teleport; GNC/TNC
	Identifier	2
	Descriptive Name	RF Access
	Description	Coordinate RF Access
	“From”	TCCC; DOD Teleport
	“To”	DOD Teleport; TCCC
	Identifier	3
	Descriptive Name	RF Access
	Description	Coordinate UHF SATCOM resources
	“From”	SC2C; DOD Teleport
	“To”	DOD Teleport; SC2C
	Identifier	4
	Descriptive Name	RF Access
	Description	Coordinate RF connectivity
	“From”	NCTAMS; DOD Teleport
	“To”	DOD Teleport; NCTAMS
	Identifier	5
	Descriptive Name	RF Access
	Description	Coordinate RF Access
	“From”	Distant Component; DOD Teleport
	“To”	DOD Teleport; Distant Component
	Identifier	6
	Descriptive Name	RF Access
	Description	Coordinate RF connectivity
	“From”	JCCC; DOD Teleport
	“To”	DOD Teleport; JCCC
	Identifier	7
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	“From”	JCCC; Commercial NOC
	“To”	Commercial NOC; JCCC
	Identifier	8
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources
	“From”	Commercial NOC; DOD Teleport
	“To”	DOD Teleport; Commercial NOC
	Identifier	9
	Descriptive Name	RF Access
	Description	Coordinate SATCOM resources

Data Elements	Attributes	Example Values/Explanation
	"From"	Commercial NOC; SYSCON
	"To"	SYSCON; Commercial NOC
	Identifier	10
	Descriptive Name	RF Access
	Description	Coordinate RF connectivity
	"From"	SYSCON; DOD Teleport
	"To"	DOD Teleport; SYSCON

Table A-A-3b: Provide Communication Resources (OV-2) Data Elements

A.5.3 Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2)

a. Product Purpose. The purpose of the Crossbanding and Multiple Hop Provide Communication Resources OV-2 is to define the connectivity and information exchanges between the DOD Teleport and its operational nodes for changes in status of the RF and DISN Gateway and C4I systems.

b. Product Description. The Teleport System configures access to the RF Gateway and manages Crossbanding and Multiple Hop missions as shown in Figure A-A-3c. These operational concepts are further defined in paragraph 1.c of Reference (b).

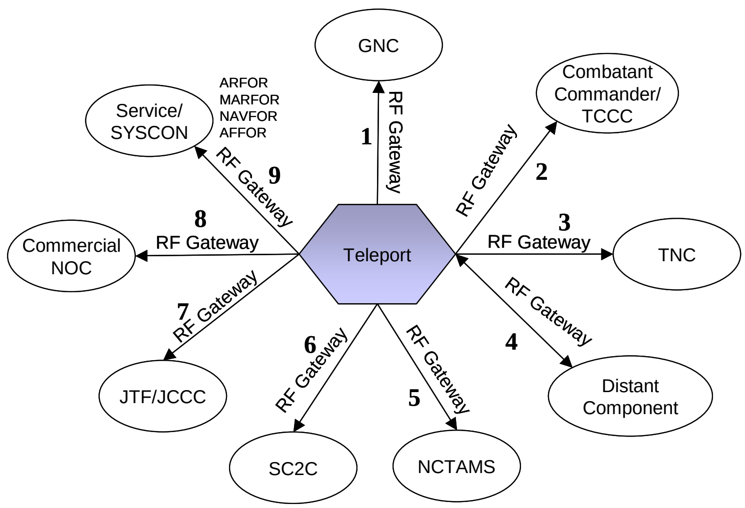


Figure A-A-3c: Crossbanding and Multiple Hop Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2)

c. Data Elements for Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2). The DOD Teleport Perform Day-to-Day Electromagnetic Spectrum (OV-2) data elements are provided in Table A-A-3c.

Data Elements	Attributes	Example Values/Explanation
Graphical Box Types		
Asset Icon		
	Name	Teleport
	Description	As previously defined

Data Elements	Attributes	Example Values/Explanation
	Name	GNC
	Description	As previously defined
	Name	TNC
	Description	As previously defined
	Name	Combatant Commander/TCCC
	Description	As previously defined
	Name	JTF/JCCC
	Description	As previously defined
	Name	NCTAMS
	Description	As previously defined
	Name	Service/SYSCON
	Description	As previously defined
	Name	Distant Component
	Description	As previously defined
	Name	Commercial NOC
	Description	As previously defined
	Name	SC2C
	Description	As previously defined
Graphical Arrow Types		
Line		
	Identifier	1
	Descriptive Name	RF Gateway
	Description	Transmit SATCOM connectivity status message
	“From”	DOD Teleport
	“To”	GNC
	Identifier	2
	Descriptive Name	RF Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	“From”	DOD Teleport
	“To”	TCCC
	Identifier	3
	Descriptive Name	RF Gateway
	Description	Transit network connectivity status message as required
	“From”	DOD Teleport
	“To”	TNC
	Identifier	4
	Descriptive Name	RF Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	“From”	DOD Teleport; Distant Component
	“To”	Distant Component; DOD Teleport
	Identifier	5
	Descriptive Name	RF Gateway
	Description	Transmit system status message
	“From”	DOD Teleport
	“To”	NCTAMS
	Identifier	6
	Descriptive Name	RF Gateway
	Description	Transmit UHF connectivity status message

Data Elements	Attributes	Example Values/Explanation
	"From"	DOD Teleport
	"To"	SC2C
	Identifier	7
	Descriptive Name	RF Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	"From"	DOD Teleport
	"To"	JCCC
	Identifier	8
	Descriptive Name	RF Gateway
	Description	Transmit SATCOM connectivity status message
	"From"	DOD Teleport
	"To"	Commercial NOC
	Identifier	9
	Descriptive Name	RF Gateway
	Description	Coordinate connectivity and monitor link performance to ensure mission requirements are met
	"From"	DOD Teleport
	"To"	SYSCON

Table A-A-3c: Perform Day-to-Day Control of Electromagnetic Spectrum (OV-2) Data Elements

A.6 Organizational Relationship Chart (OV-4)

a. Product Purpose. The Teleport Organizational Relationship Chart (OV-4) illustrates the command structure or relationships among operational nodes or organizations that are key players within the Teleport architecture.

b. Product Description. The Teleport OV-4 defines the key command relationships and organizations that control and apportion the electromagnetic spectrum, DISN, and gateway access. Per references (e), USSTRATCOM is the SATCOM Operational Manager (SOM) and directs the day-to-day operation of DOD-owned and leased SATCOM resources to provide authorized users with global SATCOM support as operations and evolving requirements dictate. Reference (u) establishes the Joint Task Force – Global Network Operations (JTF-GNO) and the Theater NetOps Center (TNC) and the operational construct to use and defend the GIG. The JTF-GNO will be established through the merger of the JTF-Computer Network Operations (CNO), the DISA GNOSC, the DOD Computer Emergency Response Team (DOD-CERT) and the Global SATCOM Support Center (GSSC). The TNC will be established through the merger of the DISA RNOSC, the DISA RCERT, and the RSSC within the TNC's AOR.

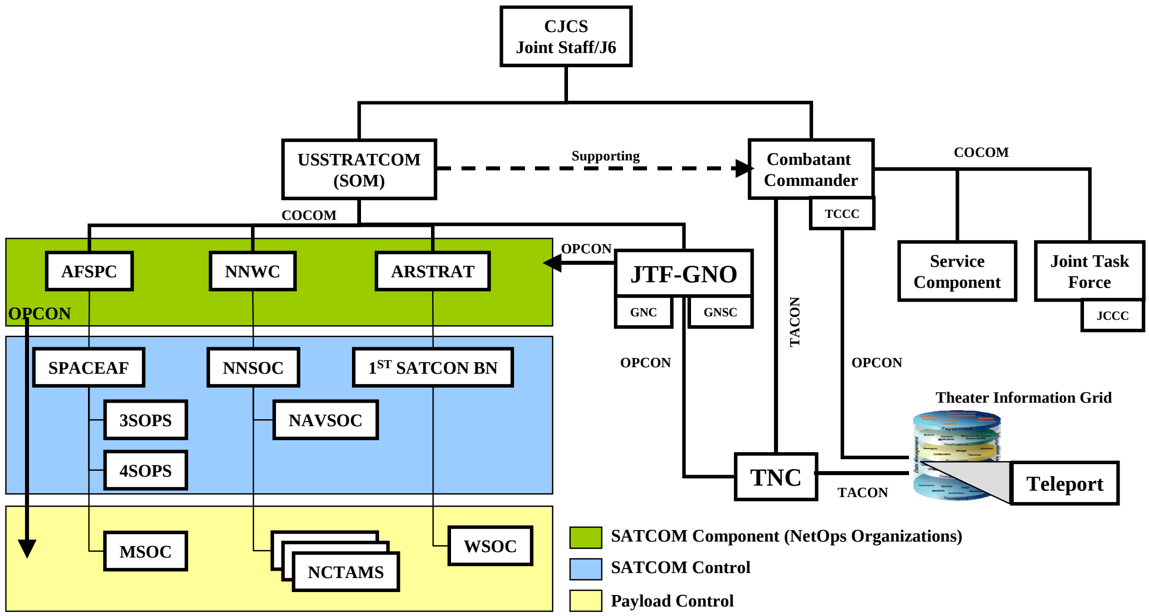


Figure A-A-4: Teleport Organizational Relationships Chart (OV-4)

A.7 Operational Activity Model (OV-5)

a. Product Purpose. The Operational Activity Model describes the operations that are normally conducted in the course of achieving a mission, or a business goal. The Operational Activity Model describes operational activities, the information exchanged between operational nodes (which conduct those activities), and the information exchanged with other activities outside the scope of the architecture (i.e., external exchanges); that are needed to affect the execution of a mission or goal as stated by the architecture purpose.

b. Product Description. The Operational Activity Model (OV-5) describes the applicable activities associated with the DOD Teleport System architecture and the data/information exchanged between internal and external activities to perform the capabilities depicted in the previous architectural views. It captures the activities performed in an operational mission and the inputs, controls, outputs, and mechanisms.

The OV-5 Hierarchy Chart shown below depicts the external and internal activities that must be performed in order to support the Teleport system. Day-to-Day control and Management of communications resources is preformed by those external NetOps management and control entities and governed by Reference (e) and (u).

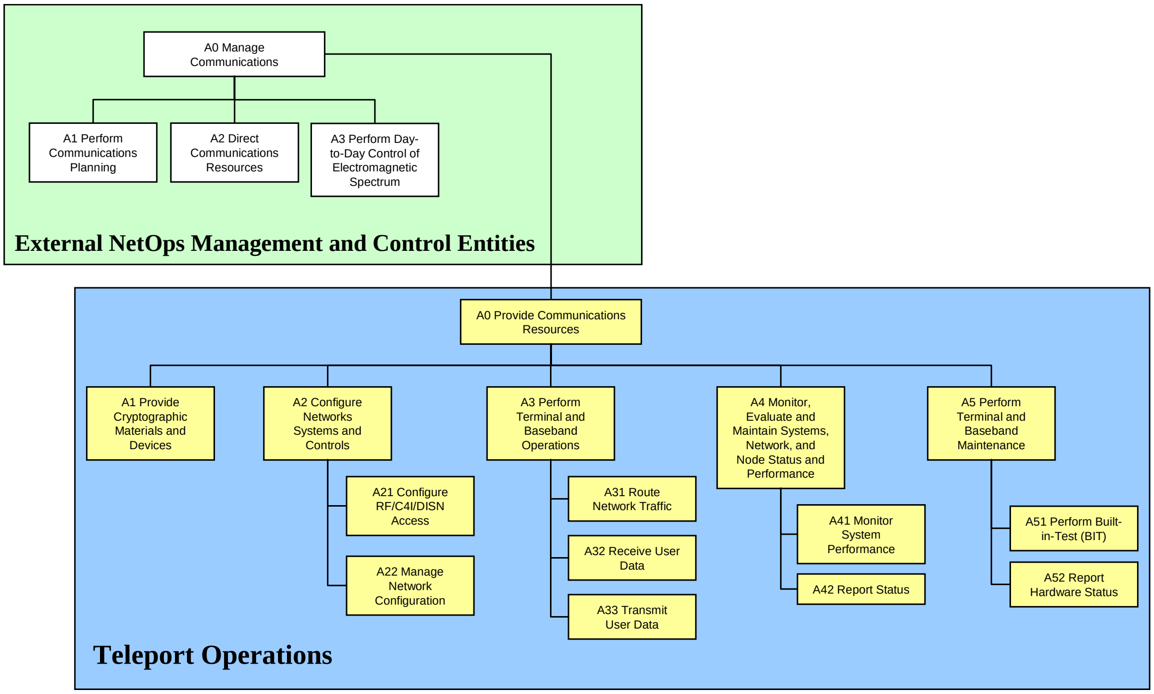


Figure A-A-5a: Teleport Activity Model (OV-5) Hierarchy

Figure A-A-5b shows the inputs and outputs required by the activities support the Teleport system. Those activities highlighted reflect identical activities of the Network Centric Operations and Warfare Reference Model (NCOW RM).

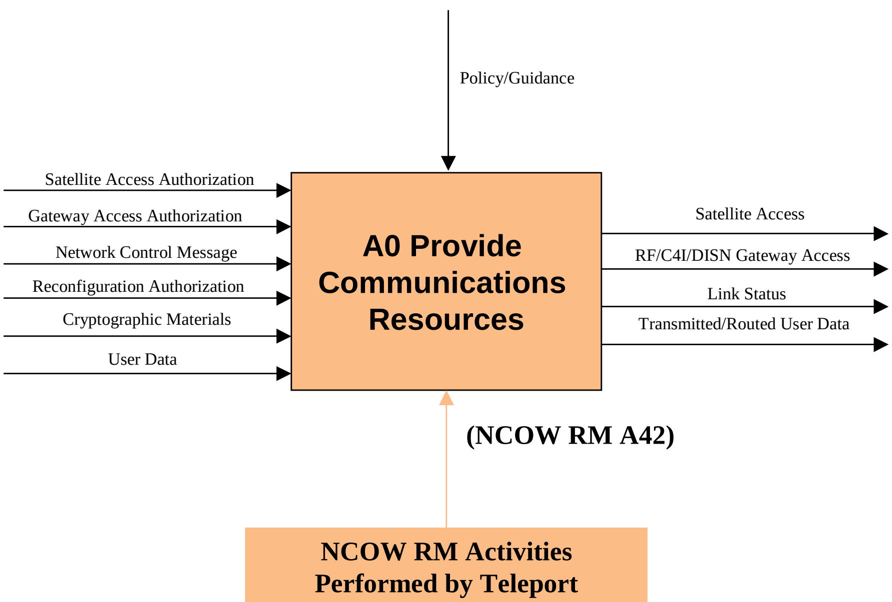


Figure A-A-5b: Teleport Activity Model (OV-5) (Context Diagram)

Figures A-A-5c, A-A-5d, A-A-5e, and A-A5f depict further decomposition and those activities, inputs, and outputs required for the Teleport system.

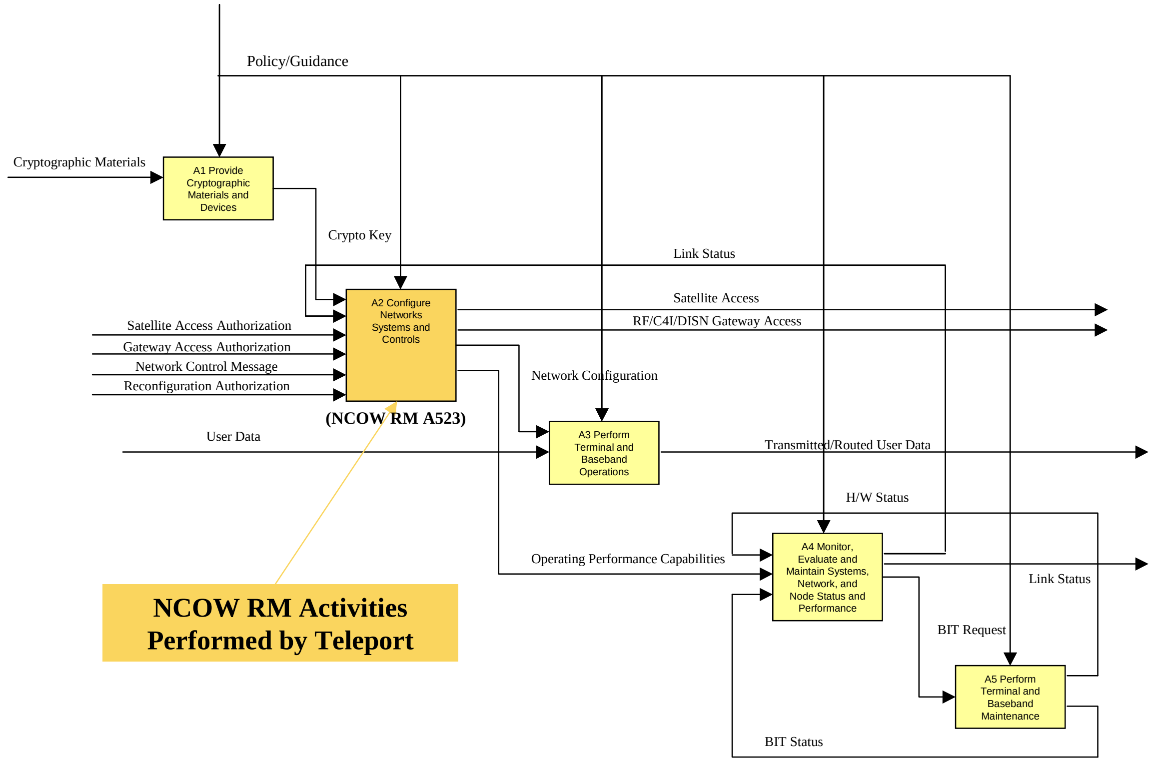


Figure A-A-5c: Teleport Activity Model (OV-5) Level 2 (Activities A1 and A5)

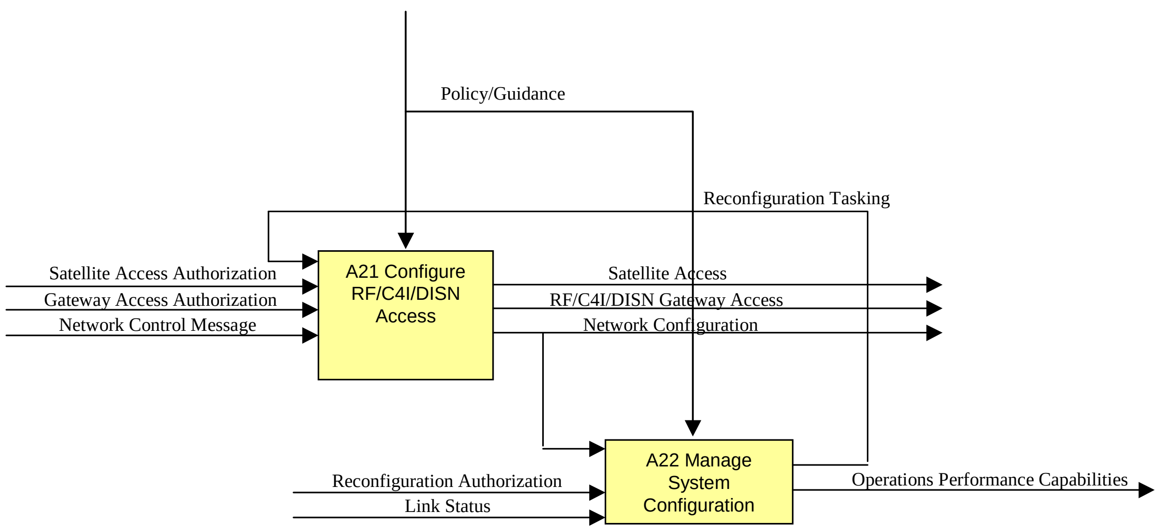


Figure A-A-5d: Teleport Activity Model (OV-5) Level 3 (Activities A21 – A22)

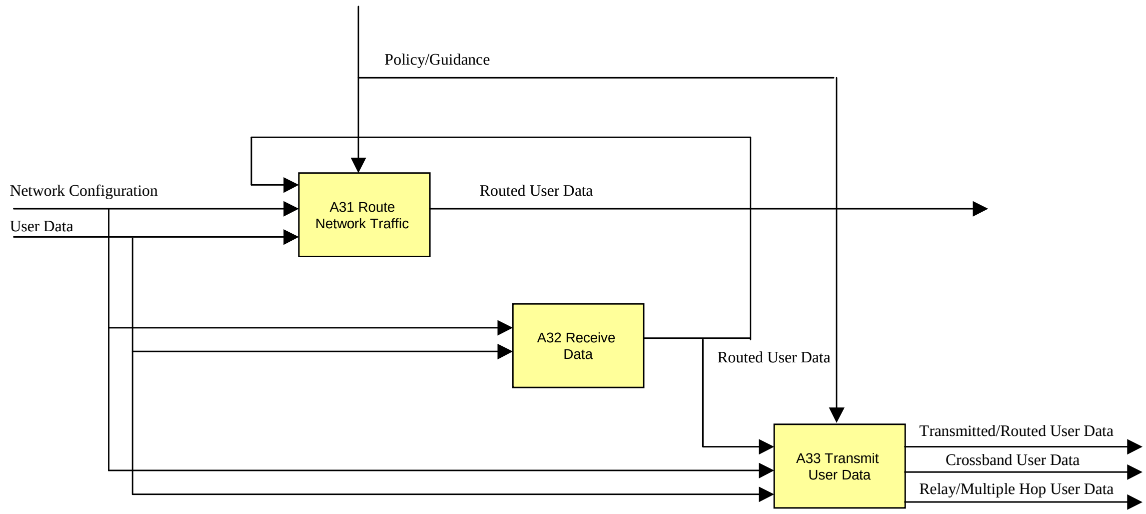


Figure A-A-5e: Teleport Activity Model (OV-5) Level 3 (Activities A31 – A33)

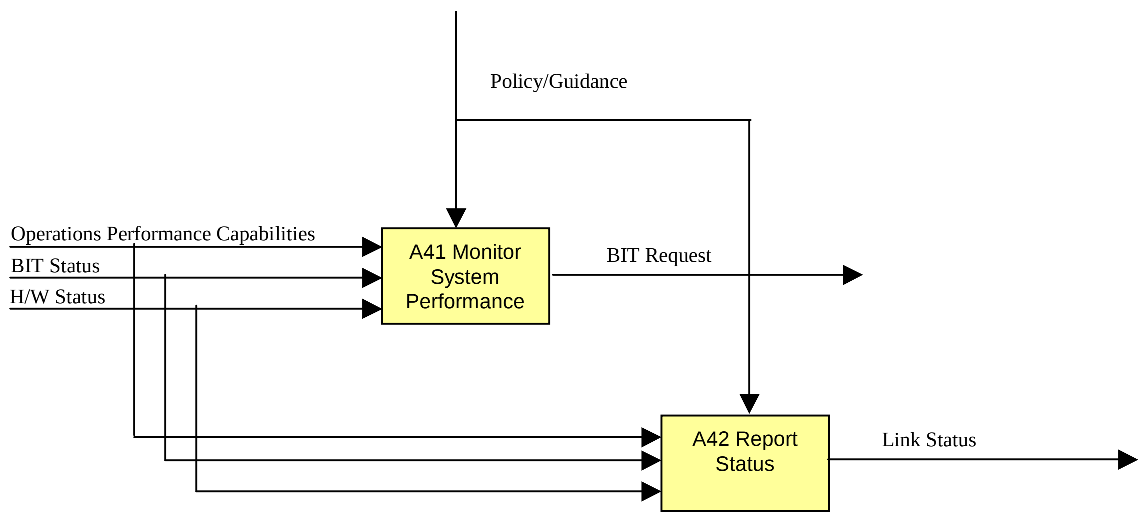


Figure A-A-5f: Teleport Activity Model (OV-5) Level 3 (Activities A41 – A42)

A.8 Operational Event Trace Description (OV-6c)

a. Product Purpose. The Operational Event Trace Description (OV-6c) provides a time ordered description of the information exchanges between participating operations nodes as a result of a particular scenario.

b. Product Description. The following Teleport OV-6c figures depict the operational procedures and information exchanges required for access to the Teleport system via wideband, narrowband, and protected band systems.

COCOM	GNC/TNC	CONEX	GNC/TNC	CSB	Distant Component	Comm NOC	Teleport	Time

COCOM	GNC/TNC	CONEX	GNC/TNC	CSB	Distant Component	Comm NOC	Teleport	Time
Approve SAR/GAR	Process SAR	Process GAR	Process GAR					21 Days
Submit RFS*	Coordinate with CSB*			Process RFS*				45 Days*
	Send SAA	Coordinate with TNC	Send GAA		Process SAA/GAA		Process SAA/GAA	1 Day
					Configure SATCOM Access	Provide Access Coord.	Configure SATCOM and DISN/DCGS/ NCTAMS Access	1 Day
							Total	23–47* Days

*Commercial access

Table A-A-4a: Wideband Teleport Access Operational Event Trace Description (OV-6c)

COCOM	GNC/TNC	SOM	CONEX	GNC/TNC	Distant Component	Teleport	Time
Approve SAR/GAR	Process SAR	Coord. SAR	Process GAR	Process GAR			30 Days
	Send SAA		Coordinate with TNC	Send GAA	Process SAA/GAA	Process SAA/GAA	1 Day
					Configure SATCOM Access	Configure SATCOM & DISN/DCGS/ NCTAMS Access	1 Day
						Total	32 Days

Table A-A-4b: Narrowband Teleport Access Operational Event Trace Description (OV-6c)

COCOM	GNC/TNC	SOM	CONEX	GNC/TNC	Distant Component	MSOC	Teleport	Time
Approve SAR/GAR	Process SAR	Coord. SAR	Process GAR	Process GAR				30 Days
	Send SAA		Coordinate with TNC	Send GAA	Process SAA/GAA		Process SAA/GAA	1 Day
					Configure SATCOM Access	Provide Access Coord.	Configure SATCOM & DISN/DCGS/ NCTAMS Access	1 Day
							Total	32 Days

Table A-A-4c: Protected Band Teleport Access Operational Event Trace Description (OV-6c)

A.9 Systems Functionality Description (SV-4)

a. Product Purpose. The purpose of the System Functionality Description (SV-4) is to document systems functionality hierarchies and systems functions, and the data flows between them. The figures shown below describe the flow of data among systems functions and are the systems view counterpart to the OV-5 Operational Activity Model.

b. Product Description. The DOD Teleport System Functionality Hierarchy Model (SV-4) is shown in Figure A-A-6. This SV-4 documents the system functional decomposition for the Teleport system.

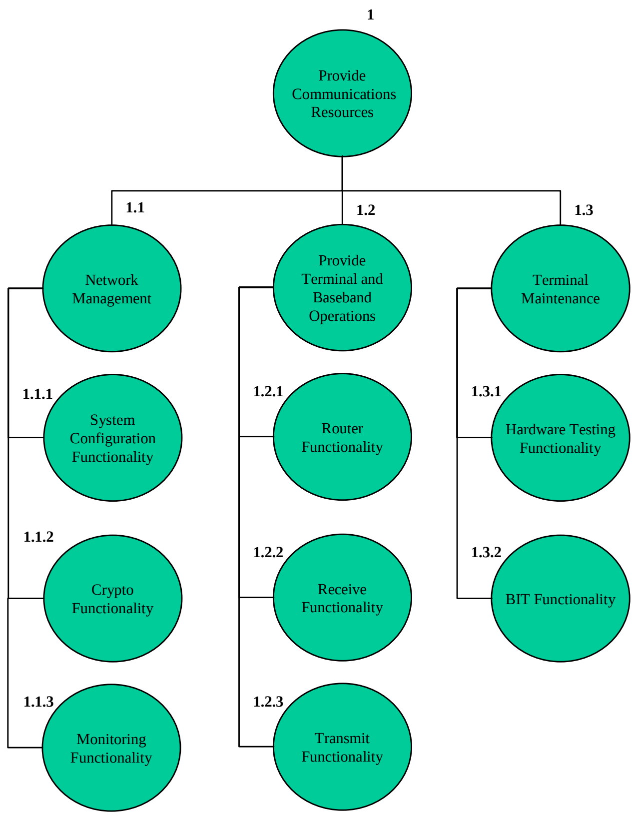


Figure A-A-6: Teleport System Functionality Description (Hierarchy Model) (SV-4)

A.10 Operational Activity to System Functionality Traceability Matrix (SV-5)

a. Product Purpose. The Operational Activity to System Traceability Matrix (SV-5) maps the operational activities to system functions, and essentially identifies the transformation of an operational need into a purposeful action performed by a system.

b. Product Description. Table A-A-5 below documents the Teleport system functions (SV-4) and correlates them to the activities depicted in Operational Activity Model (OV-5).

	Operational Activities													
System Functions	A1	A2	A21	A22	A3	A31	A32	A33	A4	A41	A42	A5	A51	A52
1.1		X												
1.1.1		X	X	X										
1.1.2	X	X	X	X										
1.1.3									X	X	X			X
1.2					X									
1.2.1						X								
1.2.2							X							
1.2.3								X						
1.3												X		
1.3.1													X	
1.3.2													X	

Table A-A-5: Teleport Operational Activity to System Functionality Matrix (SV-5)

A.11 Systems Data Exchange Matrix (SV-6)

a. Product Purpose. The Systems Data Exchange Matrix (SV-6) specifies characteristics and requirements of the data exchanged between systems. This product focuses on automated information and data exchanges that will be implemented in systems. The systems data exchanges express the relationship across the three basic data elements of an SV (system functions, systems, and data flow) and focus on the specific aspects of the data flow.

b. Product Description. The below figures show the relationships of the nodes and the automated the data exchanges required to support the Teleport mission areas. These figures correspond the Teleport OV-2s.

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	GNC/TNC	TCCC	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
2	GNC/TNC	JCCC	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
3	GNC/TNC	DOD Teleport	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
4	GNC/TNC	SYSCON	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
5	GNC/TNC	Deployed User	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
6	GNC/TNC	TCCC	DISN Directive	GAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
7	GNC/TNC	JCCC	DISN Directive	GAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
8	GNC/TNC	DOD Teleport	DISN Directive	GAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
9	GNC/TNC	SYSCON	DISN Directive	GAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
10	GNC/TNC	Deployed User	DISN Directive	GAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
11	NCTAMS	DOD Teleport	C4I Directive	C4I Service Activation	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
12	NCTAMS	Deployed User	C4I Directive	C4I Service Activation	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
13	DCGS	DOD Teleport	Non-DISN Directive	DISN Configuration Management	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
14	DCGS	SYSCON	Non-DISN Directive	DISN Configuration Management	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
15	DCGS	Deployed User	Non-DISN Directive	DISN Configuration Management	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS

Table A-A-6a: Configure Network and Systems Control (SV-6) for Reachback and Hub/Spoke Operations

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	GNC/TNC	TCCC	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
2	GNC/TNC	JCCC	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
3	GNC/TNC	NAVSOC	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
4	GNC/TNC	DOD Teleport	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
5	GNC/TNC	NCTAMS	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
6	GNC/TNC	SYSCON	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS
7	GNC/TNC	Deployed User	RF Directive	SAA	SIPRNET	DATA	TEXT	S	One-Time, priority message prior to mission activation	NLT 2 Hours Prior	10-100	DMS

Table A-A-6b. Configure Network System Controls (SV-6) for Crossbanding and M-Hop Operations

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	GNC/TNC	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
2	TCCC	DOD Teleport	RF, DISN, C4I Access	RF, DISN, C4I Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
3	GNC/TNC	DOD Teleport	DISN Access	DISN Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
4	NCTAMS	DOD Teleport	C4I Access	C4I Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
5	DCGS	DOD Teleport	C4I Access	C4I Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
6	SYSCON	DOD Teleport	RF, DISN, C4I Access	RF, DISN, C4I Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
7	Deployed User	DOD Teleport	RF, DISN, C4I Access	RF, DISN, C4I Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
8	JCCC	DOD Teleport	RF, DISN, C4I Access	RF, DISN, C4I Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
9	Comm NOC	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
10	Comm NOC	JCCC	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
11	Comm NOC	SYSCON	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS

Table A-A-6c. Provide Communication Resources (SV-6) for Reachback and Hub/Spoke Operations

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	GNC/TNC	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
2	TCCC	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
3	NCTAMS	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
4	NCTAMS	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
5	Deployed User	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
6	JCCC	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
7	JCCC	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS
8	Comm NOC	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
9	Comm NOC	DOD Teleport	RF Access	RF Access Coordination	DSN/ SIPRNET	VOICE/ DATA	VOICE/ TEXT	S	Multiple, prior to mission activation	30 SEC	<56/ 1-10	Secure Voice/ DMS
10	SYSCON	DOD Teleport	RF Access	RF Access Coordination	SIPRNET	DATA	TEXT	S	Multiple, prior to mission activation	30 SEC	1-10	DMS

Table A-A-6d. Provide Communication Resources (SV-6) for Crossbanding and M-Hop Operations

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	DOD Teleport	GNC/TNC	RF or DISN Gateway	RF or DISN Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
2	DOD Teleport	TCCC	RF, DISN, or C4I Gateway	RF, DISN, or C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
3	DOD Teleport	TNC	RF or DISN Gateway	RF or DISN Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
4	DOD Teleport	NCTAMS	C4I Gateway	C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
5	DOD Teleport	DCGS	C4I Gateway	C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
6	DOD Teleport	SYSCON	RF, DISN, or C4I Gateway	RF, DISN, or C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
7	DOD Teleport	Deployed User	RF, DISN, or C4I Gateway	RF, DISN, or C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
8	DOD Teleport	JCCC	RF, DISN, or C4I Gateway	RF, DISN, or C4I Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
9	DOD Teleport	Comm NOC	RF Gateway	RF Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS

Table A-A-6e. Perform Day-to-Day Control of Electromagnetic Spectrum (SV-6) for Reachback and Hub/Spoke Operations

IER#	SENDER	RECEIVER	IER EVENT	CONTENT	MEDIA	INFO CHAR	FORMAT	SECURITY CLASS	FREQUENCY	TIMELINESS	FILE SIZE (Kbytes)	APPLICATION
1	DOD Teleport	GNC/TNC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
2	DOD Teleport	TCCC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
3	DOD Teleport	GNC/TNC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
4	DOD Teleport	Deployed User	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
5	DOD Teleport	NCTAMS	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
6	DOD Teleport	NAVSOC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
7	DOD Teleport	JCCC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
8	DOD Teleport	Comm NOC	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS
9	DOD Teleport	SYSCON	RF Gateway	RF Gateway Status	SIPRNET	DATA	TEXT	S	One-time change in status notification	30 SEC	10-100	DMS

Table A-A-6f. Perform Day-to-Day Control of Electromagnetic Spectrum (SV-6) for Crossbanding and M-Hop Operations

A12 Technical Standards Profile (TV-1)

a. Product Purpose. The Technical Standards Profile (TV-1) provides the mandated and emerging technical systems-implementation standards upon which engineering specifications are based, common building blocks are established, and product lines are developed.

b. Product Description. The TV-1 collects the various mandated legislative, policy, and other systems standards rules that constrain the choices that can be made in a design and implementation of architecture.

The Teleport TV-1 was developed using the Joint C4I Program Assessment Tool *DOD Information Technology Standards Registry (DISR) Online* and can be found at http://disronline.disa.mil. Below was printed from DISROnline.

Technical View (TV)

Printed by DISRonline on 2004-10-15

System Profile: DoD Teleport II
System Description: TV-1/TV-2 standards for Teleport derived from questionnaire profiling method
Created by: Jhkroese
Last Updated: 2004-09-28

IT Profile: Access Control

IT Description:	Security Access Control standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method
Last Updated:	2004-09-28

Service Area	Standard ID & Title of Standard	Status
Access Control	ACP 120A - Allied Communications Publication 120, Common Security Protocol (CSP), Rev A, 7 May 1998.	Mandated [SUNSET]
Access Control	FIPS Pub 140-2 - Security Requirements for Cryptographic Modules, 25 May 2001.	Mandated
Access Control	FIPS Pub 180-1 - Secure Hash Algorithm-1, April 1995.	Mandated
Access Control	FIPS Pub 186-2 - Digital Signature Standard (DSS) Digital Signature Algorithm (DSA), 27 January 2000.	Mandated
Access Control	FIPS Pub 197 - Advance Encryption Standard (AES), 26 November 2001.	Mandated
Access Control	FIPS Pub 46-3:1999 - Data Encryption Standard, 25 October 1999.	Mandated [SUNSET]
Access Control	FORTEZZA Application - FORTEZZA Application Implementers Guide, MD4002101-1.52, 5 March 1996.	Mandated [SUNSET]
Access Control	FORTEZZA CIPG 1.52 - FORTEZZA Cryptologic Interface Programmers Guide (CIPG), MD4000501-1.52, 30 January 1996.	Mandated [SUNSET]
Access Control	FORTEZZA ICD - FORTEZZA Interface Control Document, Revision P1.5, 22 December 1994.	Mandated [SUNSET]
Access Control	IETF RFC 2630 - Cryptographic Message Syntax, June 1999.	Mandated [SUNSET]
Access Control	IETF RFC 2632 - S/MIME Version 3, Certificate Handling, June 1999.	Mandated
Access Control	IETF RFC 2633 - S/MIME Version 3, Message Specification, June 1999.	Mandated
Access Control	IETF RFC 2634 - Enhanced Security Services for S/MIME, June 1999.	Mandated
Access Control	IETF RFC 2743 - Generic Security Service Application Program Interface, Version 2, Update 1, January 2000.	Emerging
Access Control	ITU-T Rec. X.411 - Message Handling Systems (MHS) - Message Transfer System: Abstract Service Definition Procedures, 1999. (ISO/IEC 10021-4:1999)	Mandated [SUNSET]
Access Control	ITU-T Rec. X.481 - Message Handling Systems (MHS) - Security Information Objects for Access Control, 2000. (ISO/IEC 15816-12:2000)	Mandated [SUNSET]
Access Control	ITU-T Rec. X.509:2000 - The Directory: Public-Key and Attribute Certificate Frameworks, 2001, with Technical Corrigendum 1:2002, and Technical Corrigendum 2:2002. (ISO/IEC 9594-8.2001)	Mandated
Access Control	SDN.801 - Access Control Concept and Mechanisms, Revision C, 12 May 1999.	Mandated [SUNSET]
Access Control	SKIPJACK/KEA - SKIPJACK and KEA Algorithm Specification, Version 2.0, NIST, 29 May 1998.	Mandated [SUNSET]

IT Profile:	Application-specific data
IT Description:	Application data exchange standards for Teleport derived from the questionnaire profiling method

Last Updated: 2004-09-15

Service Area	Standard ID & Title of Standard	Status
Application-specific Data Interchange	ITU-R TF.460-5 - Standard-frequency and Time-signal Emissions, 1997.	Mandated
Application-specific Data Interchange	NADSI - NATO Advanced Data Storage Interface, STANAG 4575, Edition 1, Ratification Draft.	Emerging
Application-specific Data Interchange	NTSDS DID & CSD - NTSDS Database Implementation Description & Core Schema Definition, Version 1.2a, 19 September 1997.	Mandated
Application-specific Data Interchange	NTSDS SSD - NTSDS Supplemental Schema Definition, Version 1.1, 24 September 1997.	Mandated
Application-specific Data Interchange	OMG ptc/03-07-07 - Data Distribution Service for Real-Time Systems Specification, Version 1.0, July 2003.	Mandated

---

IT Profile: Audio data

IT Description: Audio data exchange standards for Teleport derived from the questionnaire profiling method

Last Updated: 2004-09-15

Service Area	Standard ID & Title of Standard	Status
Audio Data Interchange	A/D Conversion - Analog-to-Digital Conversion of Voice by 1200 Bit/Second Mixed Excitation Linear Prediction (MELP).	Emerging
Audio Data Interchange	ISO/IEC 11172-3 - Encoding of moving pictures and associated audio for digital storage media at up to about 1.5 Megabits per second (Mbit/s) - Part 3 (Audio Layer-3 only); with Technical Corrigendum 1:1996.	Mandated
Audio Data Interchange	ISO/IEC 11172-3 - Encoding of moving pictures and associated audio for digital storage media at up to about 1.5 Megabits per second (Mbit/s) - Part 3 (Audio Layer-3 only); with Technical Corrigendum 1:1996.	Mandated
Audio Data Interchange	ISO/IEC 13818-3:1998 - Generic coding of moving pictures and associated audio information - Part 3: Audio (MPEG-2), 1998.	Mandated
Audio Data Interchange	MIL-STD-3005 - Analog-to-Digital Conversion of Voice by 2400 Bit/Second Mixed Excitation Linear Prediction (MELP), 20 December 1999.	Mandated

---

IT Profile: Backplanes & Busses

IT Description: Backplane and system bus standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-13

Service Area	Standard ID & Title of Standard	Status
Backplanes and Busses	ANSI/VITA 1 - American National Standard for VME64, 1994.	Mandated
Backplanes and Busses	ANSI/VITA 1.1 - VME64 Extensions, 1997.	Mandated
Backplanes and Busses	DVI - Digital Visual Interface (DVI), Digital Display Working Group (DDWG) Revision 1.0, 2 April 1999.	Emerging
Backplanes and Busses	IEEE 1155 - VME bus Extensions for Instrumentation (VXI), 1992.	Mandated
Backplanes and Busses	PC/104_Spec - PC/104 Specification, V2.4, August 2001.	Mandated
Backplanes and Busses	PC/104plus - PC/104 PLUS Specification, V1.2, August 2001.	Mandated

IT Profile:	Database management
IT Description:	Database management standards for Teleport derived from questionnaire profiling method
Last Updated:	2004-09-13

Service Area	Standard ID & Title of Standard	Status
Database Management System	ANSI X3.135.10 - Database languages -SQL - Part 10: Object Language Bindings (SQL/OLB), 1998.	Mandated
Database Management System	ISO/IEC 9075 - Database Languages - SQL, with amendment 1, 1996, as moified by FIPS PUB 127-2:1993, Database Language for Relational DBMSs. (Entry Level SQL).	Mandated
Database Management System	ISO/IEC 9075-1 - Database languages - SQL - Part 1: Framework (SQL/Framework), 1999.	Emerging
Database Management System	ISO/IEC 9075-2 - Database languages - SQL - Part 2: Foundation (SQL/Foundation), 1999.	Emerging
Database Management System	ISO/IEC 9075-3 - Database Languages - SQL - Part 3: Call-Level Interface (SQL/CLI), 1995.	Mandated
Database Management System	ISO/IEC 9075-3:1999 - Database languages - SQL - Part 3: Call-Level Interface (for SQL3), 1999.	Emerging
Database Management System	ISO/IEC 9075-4 - Database languages - SQL - Part 4: Persistent Stored Modules (SQL/PSM), 1999.	Emerging
Database Management System	ISO/IEC 9075-5 - Database languages - SQL - Part 5: Host Language Bindings (SQL/Bindings).	Emerging
Database Management System	ISO/IEC 9579:2000 - Remote Database Access for SQL with security enhancement.	Emerging

IT Profile:	Email
IT Description:	Electronic mail standards for Teleport derived from questionnaire profiling method
Last Updated:	2004-09-13

Service Area	Standard ID & Title of Standard	Status
Platform Communications Services	ACP 123:1997 - Common Messaging Strategy and Procedures, Edition A, 15 August 1997.	Mandated [SUNSET]
Platform Communications Services	ACP 123A:2001 - Common Messaging Strategy and Procedures, Edition A, U.S. Supplement No. 1, 26 June 2001	Mandated [SUNSET]
Platform Communications Services	IETF RFC 1870 - Simple Mail Transfer Protocol Services Extension for Message Size Declaration, November 1995	Mandated
Platform Communications Services	IETF RFC 2231 - MIME Parameter Value and Encoded Word Extensions: Character Sets, Languages, and Continuations, November 1997.	Emerging
Platform Communications Services	IETF RFC 2646 - The Text/Plain Format Parameter, August 1999.	Emerging
Platform Communications Services	IETF RFC 2821 - Simple Mail transfer Protocol, April 2001.	Mandated
Platform Communications Services	IETF RFC 2822 - Internet Message Format, April 2001.	Mandated
Platform Communications Services	IETF RFC 3023 - XML Medi Types, April 2001.	Emerging

Platform Communications Services	IETF RFCs 2045-2049 - Multipurpose Internet Mail Extensions (MIME) Parts 1-5, November 1996.	Mandated

---

IT Profile: Imagery Data

IT Description: Still imagery data standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-09-13

Service Area	Standard ID & Title of Standard	Status
Still Imagery Data Interchange	ISO/IEC 12087-5:1998 - Information technology - Computer graphics and image processing - Image Processing and Interchange (IPI) Functional specification - Part 5: Basic Image Interchange Format (BIIF), 1 December 1998, with Technical Corrigendum 1:2001.	Emerging
Still Imagery Data Interchange	ISO/IEC 15444-1:2001 - JPEG 2000 Image Coding System - Part 1: Core coding system, 20 December 2001, with Amendments 1 and 2, 29 January 2002.	Mandated
Still Imagery Data Interchange	ISO/IEC 15444-2:2001 - JPEG 2000 Image Coding System - Part 2, July 2001	Emerging
Still Imagery Data Interchange	MIL-STD-188-196(1) - Bi-Level Image Compression for the National Imagery Transmission Format Standard, 18 June 1993 with Notice 1, 27 June 1996.	Mandated
Still Imagery Data Interchange	MIL-STD-188-199(1) - Vector Quantization Decompression for the National Imagery Transmission Format Standard, 27 June 1994 with Notice 1, 27 June 1996.	Mandated
Still Imagery Data Interchange	MIL-STD-2500B(2) - National Imagery Transmission Format (Version 2.1) for the National Imagery Transmission Format Standard, 22 August 1997 with Notice 1, 2 October 1998, and Notice 2, 1 March 2001.	Mandated
Still Imagery Data Interchange	NITF Extensions 2.1 - The Compendium of Controlled Extensions (CE) for the National Imagery Transmission Format (NITF), Version 2.1, 16 November 2000.	Mandated

---

IT Profile: Messaging

IT Description: Military messaging standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-10-13

Service Area	Standard ID & Title of Standard	Status
Military Messaging	ANSI/IEEE 754 - Binary Floating-Point Arithmetic, March 21, 1985.	Mandated
Military Messaging	MIL-STD-6040:2002 - United States Message Text Format (USMTF), 31 March 2002.	Mandated

---

IT Profile: Motion Imagery

IT Description: Motion Imagery standards for the Teleport TV-1/TV-2 profile derived from the questionnaire profiling method

Last Updated: 2004-09-28

Service Area	Standard ID & Title of Standard	Status

Motion Imagery Data Interchange	MISP v2.0 - Motion Imagery Standards Profile, version 2.0, 29 November 2001.	Mandated

IT Profile:	Network Technologies
IT Description:	Networking standards for Teleport derived from the questionnaire profiling method
Last Updated:	2004-09-15

Service Area	Standard ID & Title of Standard	Status
Platform Communications Services	ACP 123:1997 - Common Messaging Strategy and Procedures, Edition A, 15 August 1997.	Mandated [SUNSET]
Platform Communications Services	ACP 123A:2001 - Common Messaging Strategy and Procedures, Edition A, U.S. Supplement No. 1, 26 June 2001	Mandated [SUNSET]
Network Technologies	af-AIC-0178.000 - ATM-Multiprotocol Label Switching (MPLS) Network Interworking Version 1.0, August 2001.	Emerging
Network Technologies	af-ilmi-0065.000 - Integrated Local Management Interface Specification, Version 4.0, September 1996.	Mandated [SUNSET]
Network Technologies	af-phy-0015.000 - ATM Physical Medium Dependent Interface for 155 Mbps over Twisted Pair Cable, September 1994.	Mandated [SUNSET]
Network Technologies	af-phy-0016.000 - DS-1 Physical Layer Specification, September 1994.	Mandated [SUNSET]
Network Technologies	af-phy-0043.000 - A Cell-based Transmission Convergence Sublayer for Clear Channel Interfaces, November 1995.	Mandated [SUNSET]
Network Technologies	af-phy-0046.000 - 622.08 Mbps Physical Layer Specification, January 1996.	Mandated [SUNSET]
Network Technologies	af-phy-0054.000 - DS3 Physical Layer Interface Specification, January 1996.	Mandated [SUNSET]
Network Technologies	af-phy-0064.000 - E1 Physical Layer Interface Specification, September 1996.	Mandated [SUNSET]
Network Technologies	af-phy-0086.000 - Inverse Multiplexing for ATM (IMA) Specification Version 1.0, July 1997.	Mandated [SUNSET]
Network Technologies	af-phy-0086.001 - Inverse Multiplexing for ATM (IMA) Specification Version 1.1, March 1999.	Emerging
Network Technologies	af-saa-0124.000 - Gateway for H.323 Media Transport Over ATM, July 1999.	Emerging
Network Technologies	af-sig-0061.000 - ATM UNI Signaling Specification, Version 4.0, July 1996.	Mandated [SUNSET]
Network Technologies	af-sig-0076.000 - Addendum to UNI Signaling V4.0 for ABR parameter negotiation, January 1997.	Emerging
Network Technologies	af-tm-0121.000 - Traffic Management Specification Version 4.1, March 1999.	Emerging
Network Technologies	af-vtoa-0078.000 - Circuit Emulation Service Interoperability Specification, Version 2.0, January 1997.	Mandated [SUNSET]
Network Technologies	af-vtoa-0113.000 - ATM Trunking Using AAL2 for Narrowband Services, February 1999.	Emerging
Network Technologies	af-vtoa-0119.000 - Low Speed Circuit Emulation Service (LSCES), May 1999.	Emerging
Network Technologies	ANSI T1.105 - Synchronous Optical Network (SONET) Basic Description Including Multiplex Structure, Rates and Formats, 1995.	Mandated
Network Technologies	ANSI T1.107 - Digital Hierarchy - Formats Specifications, 1995.	Mandated

Network Technologies	ANSI T1.117 - Digital Hierarchy - Optical Interface Specifications (Single Mode - Short Reach), 1991 (Reaffirmed 1997).	Mandated
Network Technologies	CCSDS 713.0-B-1/ISO 15891:2000 - Protocol specification for space communications - Network protocol, 5 October 2000.	Emerging
Network Technologies	CCSDS 713.5-B-1/ISO 15892:2000 - Protocol specification for space communications - Security protocol, 5 October 2000.	Emerging
Network Technologies	CCSDS 714.0-B-1/ISO 15893:2000 - Protocol specification for space communications - Transport protocol, 5 October 2000.	Emerging
Network Technologies	CCSDS 717.0-B-1/ISO 15894:2000 - Protocol specification for space communications - File protocol, 5 October 2000.	Emerging
Network Technologies	DoD ATM - DoD ATM Addressing Plan, 17 April 1998.	Mandated [SUNSET]
Network Technologies	GR-253 - Synchronous Optical Network (SONET) Transport Systems: Common Generic Criteria, Rev01, Bellcore, December 1997.	Mandated
Network Technologies	IEEE 802.10:1998 - Interoperable LAN/MAN Security (SILS), 17 September 1998.	Emerging
Network Technologies	IEEE 802.10a:1999 - Supplement to Standard for Interoperable LAN/MAN Security (SILS) - Security Architecture Framework (Clause 1), 22 March 1999.	Emerging
Network Technologies	IEEE 802.10c - Interoperable LAN/MAN Security (SILS) Key Management (Clause 3), 17 April 1998.	Emerging
Network Technologies	IETF RFC 1305 - Network Time Protocol (Version 3) Specification, Implementation, and Analysis, March 1992.	Mandated
Network Technologies	IETF RFC 1332 - PPP Internet Protocol Control Protocol (IPCP), May 1992.	Mandated
Network Technologies	IETF RFC 1570 - PPP LCP Extensions, 11 January 1994.	Mandated
Network Technologies	IETF RFC 1738 - Uniform Resource Locators (URL), 20 December 1994.	Mandated
Network Technologies	IETF RFC 1771 - A Border Gateway Protocol 4 (BGP-4), 21 March 1995.	Mandated
Network Technologies	IETF RFC 1772 - Application of the Border Gateway Protocol in the Internet, 21 March 1995.	Mandated
Network Technologies	IETF RFC 1812 - Requirements for IP Version 4 Routers, 22 June 1995.	Mandated
Platform Communications Services	IETF RFC 1870 - Simple Mail Transfer Protocol Services Extension for Message Size Declaration, November 1995	Mandated
Network Technologies	IETF RFC 1889 - RTP: A Transfer Protocol for Real-Time Applications, January 1996.	Emerging
Network Technologies	IETF RFC 1989 - PPP Link Quality Monitoring (LQM), 16 August 1996.	Mandated
Network Technologies	IETF RFC 1990 - PPP Multi-link Protocol, 16 August 1996.	Emerging
Network Technologies	IETF RFC 1994 - PPP Challenge Handshake Authentication Protocol (CHAP), 30 August 1996.	Mandated
Platform Communications Services	IETF RFC 1995 - Incremental Zone Transfer in DNS, August 1996.	Emerging
Platform Communications Services	IETF RFC 1996 - A Mechanism for Prompt Notification of Zone Changes (DNS NOTIFY), August 1996.	Emerging
Network Technologies	IETF RFC 2126 - ISO Transport Service on Top of TCP (ITOT), March 1997.	Emerging
Network Technologies	IETF RFC 2131 - Dynamic Host Configuration Protocol, March 1997.	Mandated

Network Technologies	IETF RFC 2132 - DHCP Options and BOOTP Vendor Extensions, March 1997.	Mandated
Platform Communications Services	IETF RFC 2136 - Dynamic Updates in the Domain Name System, April 1997.	Mandated
Platform Communications Services	IETF RFC 2231 - MIME Parameter Value and Encoded Word Extensions: Character Sets, Languages, and Continuations, November 1997.	Emerging
Network Technologies	IETF RFC 2236 - Internet Group Management Protocol, Version 2 (IGMPv2), November 1997.	Mandated
Network Technologies	IETF RFC 2396 - Uniform Resource Identifiers (URI), Generic Syntax, August 1998.	Mandated
Network Technologies	IETF RFC 2464 - Transmission of Ipv6 Packet Over Ethernet Networks, December 1998.	Emerging
Network Technologies	IETF RFC 2472 - IPv6 Over PPP, December 1998.	Mandated
Network Technologies	IETF RFC 2545 - Use of BGP-4 Multiprotocol Extensions for IPv6 Inter-Domain Routing, March 1999.	Mandated
Network Technologies	IETF RFC 2581 - TCP Congestion Control, April 1999.	Mandated
Network Technologies	IETF RFC 2616 - Hypertext Transfer Protocol - HTTP 1.1, June 1999.	Mandated
Platform Communications Services	IETF RFC 2646 - The Text/Plain Format Parameter, August 1999.	Emerging
Network Technologies	IETF RFC 2702 - Requirements for Traffic Engineering over MPLS September 1999	Mandated
Network Technologies	IETF RFC 2732 - Format for Literal IPv6 Addresses in URLs, December 1999.	Emerging
Network Technologies	IETF RFC 2740 - OSPF for IPv6, December 1999.	Mandated
Network Technologies	IETF RFC 2784 - Generic Routing Encapsulation (GRE) March 2000	Mandated
Network Technologies	IETF RFC 2794 - Mobile IP Network Access Identification Extension for IPv4, March 2000.	Emerging
Platform Communications Services	IETF RFC 2821 - Simple Mail transfer Protocol, April 2001.	Mandated
Platform Communications Services	IETF RFC 2822 - Internet Message Format, April 2001.	Mandated
Network Technologies	IETF RFC 2858 - Multiprotocol Extensions for BGP-4, June 2000.	Mandated
Network Technologies	IETF RFC 2917 - Core MPLS IP VPN Architecture September 2000	Mandated
Network Technologies	IETF RFC 3015 - Megaco Protocol Version 1.0, November 2000.	Emerging
Platform Communications Services	IETF RFC 3023 - XML Medi Types, April 2001.	Emerging
Network Technologies	IETF RFC 3241 - Robust Header Compression (ROHC) over PPP, April 2002.	Emerging
Video Teleconferencing	IETF RFC 3261 - Session Initiation Protocol, June 2002.	Emerging
Network Technologies	IETF RFC 3315 - Dynamic Host Configuration Protocol for IPv6 (DHCPv6), July 2003.	Emerging
Network Technologies	IETF RFC 3344 - IP Mobility Support for IPv4, August 2002.	Emerging
Video Teleconferencing	IETF RFC 3435 - Media Gateway Control Protocol (MGCP) Version 1.0, January 2003.	Emerging
Platform Communications Services	IETF RFCs 2045-2049 - Multipurpose Internet Mail Extensions (MIME) Parts 1-5, November 1996.	Mandated
Platform Communications Services	IETF Standard 13/RFC 1034/RFC 1035 - Domain Name System, November 1987.	Mandated

Platform Communications Services	IETF Standard 13/RFC 1034/RFC 1035 - Domain Name System, November 1987.	Mandated
Network Technologies	IETF Standard 3/RFC 1122/RFC 1123 - Requirements for Internet Host, October 1989.	Mandated
Network Technologies	IETF Standard 33/RFC 1350 - Trivial FTP (TFTP), Revision 2, July 1992.	Mandated
Network Technologies	IETF Standard 5 - Internet Protocol, September 1981. With RFC's 791/950/919/922/792/1112.	Mandated
Network Technologies	IETF Standard 51/RFC 1661/RFC 1662 - Point-to-Point Protocol (PPP), July 1994.	Mandated
Network Technologies	IETF Standard 54/RFC 2328 - Open Shortest Path First Routing Version 2, April 1998.	Mandated
Network Technologies	IETF Standard 6/RFC 768 - User Datagram Protocol, 28 August 1980.	Mandated
Network Technologies	IETF Standard 7/RFC 793 - Transmission Control Protocol, September 1981.	Mandated
Network Technologies	IETF Standard 7/RFC 793 - Transmission Control Protocol, September 1981.	Mandated
Network Technologies	IETF Standard 8/RFC 854/RFC 855 - TELNET Protocol, May 1983.	Mandated
Network Technologies	IETF Standard 9/RFC 959 - File Transfer Protocol, October 1985, with the following FTP commands mandated for reception: Store unique (STOU), Abort (ABOR), and Passive (PASV).	Mandated
Network Technologies	ISO/IEC 8802-3:2000 - Local and Metropolitan Area Networks - Specific Requirements - Part 3: Carrier Sense Multiple Access with Collision Detection (CSMA/CD) Access Method and Physical Layer Specifications, 2000 (IEEE 802.3, 2000 Edition).	Mandated
Video Teleconferencing	ITU-T H.323 v2 - Packet-based Multimedia Communications Systems, (Version 2), February 1998.	Mandated
Network Technologies	ITU-T I.363.1 - B-ISDN ATM Adaptation Layer Specification: Type 1 ATM Adaptation Layer (AAL1), August 1996.	Mandated [SUNSET]
Network Technologies	ITU-T I.363.5 - B-ISDN ATM Adaptation Layer Specification: Type 5 ATM Adaptation Layer (AAL5), August 1996.	Mandated [SUNSET]
Platform Communications Services	ITU-T X.500 - The Directory: Overview of Concepts, Models, and Services - Data Communication Networks Directory, 1993.	Mandated [SUNSET]
Platform Communications Services	ITU-T X.500:2001 - The Directory: Overview of Concepts, Models, and Services - Data Communication Networks Directory, February 2001.	Emerging
Network Technologies	MIL-STD-2045-47001C - Connectionless Data Transfer Application Layer Standard, 22 March 2002.	Mandated [SUNSET]
Network Technologies	TIA/EIA 232-F - Interface Between Data Terminal Equipment and Data Circuit Terminating Equipment Employing Serial Binary Data Interchange, October 1997.	Mandated
Network Technologies	TIA/EIA 530-A - High Speed 25-Position Interface for Data Terminal Equipment and Data Circuit Terminating Equipment, Including Alternative 26-Position Connector, December 1998.	Mandated
Network Technologies	TIA/EIA/IS 787 - Common ATM Satellite Interface Interoperability Specification (CASI), July 1999.	Emerging

---

IT Profile: Operating System Services

IT Description: Operating system standards for Teleport TV-1/TV-2 derived from the

	questionnaire profiling method	
Last Updated:	2004-10-14	
Service Area	**Standard ID & Title of Standard**	**Status**
Operating System Services	C808 - Networking Services (XNS), Issue 5.2, Open Group Technical Standard, ISBN-1-85912-241-8, January 2000.	Emerging
Operating System Services	IEEE 1003.1d - Portable Operating System Interface (POSIX) Part 1: System Application Program Interface (API) - Amendment d: Additional Real-Time Extensions [C Language], 1999.	Emerging
Operating System Services	IEEE 1003.1j:2000 - Portable Operating System Interface (POSIX) - Part 1: System Application Program Interface (API) - Amendment j: Advanced Real-Time Extensions [C Language], 2000.	Emerging
Operating System Services	IEEE 1003.2d - Portable Operating System Interface for Computer Environments (POSIX) - Part 2: Shell and Utilities - Amendment 1: Batch Environment, 1994.	Mandated
Bindings/Object Code Linking	ISO/IEC 14519 - POSIX Ada Language Interfaces -Binding for System Application Program Interface (API) - Real-Time Extensions, 1999.	Mandated
Operating System Services	ISO/IEC 9945-1 - Portable Operating System Interface (POSIX) - Part 1: System Application Program Interface (API) [C language], 1996 (Mandated Services).	Mandated
Operating System Services	ISO/IEC 9945-1:Real-time - Portable Operating System Interface (POSIX) - Part 1: System Application Program Interface (API) [C language], 1996 (Real-time Optional Services).	Mandated
Operating System Services	ISO/IEC 9945-1:Thread - Portable Operating System Interface (POSIX) - Part 1: System Application Program Interface (API) [C language], 1996 (Thread Optional Services).	Mandated
Operating System Services	ISO/IEC 9945-2 - Portable Operating System Interface (POSIX) - Part 2: Shell and Utilities, 1993.	Mandated
Operating System Services	P1003.1q d8 - Portable Operating System Interface (POSIX) Part 1: System Application Program Interface (API) - Amendment x: Tracing [C Language], Draft 8, April 2000.	Emerging
Operating System Services	P1003.21 v3 - Portable Operating System Interface (POSIX) - Part 1: Real-Time Distributed Systems Communication Application Program Interface (API) [Language-Independent], V3.0, October 1999.	Emerging
Operating System Services	UNIX Version 3 - Single UNIX Specification Version 3 (SUS v3), The Open Group, 2002.	Emerging
User Interface Services	Win32 APIs - Window Management and Graphics Device Interface, Volume 1 Microsoft Win32 Programmers Reference Manual, 1993, Microsoft Press.	Mandated

---

IT Profile:	Platform Comm Services	
IT Description:	Platform communications standards for Teleport derived from the questionnaire profiling method	
Last Updated:	2004-09-15	
Service Area	**Standard ID & Title of Standard**	**Status**
Platform Communications Services	ACP 123:1997 - Common Messaging Strategy and Procedures, Edition A, 15 August 1997.	Mandated [SUNSET]
Platform Communications Services	ACP 123A:2001 - Common Messaging Strategy and Procedures, Edition A, U.S. Supplement No. 1, 26 June 2001	Mandated [SUNSET]

Platform Communications Services	IETF RFC 1870 - Simple Mail Transfer Protocol Services Extension for Message Size Declaration, November 1995	Mandated
Platform Communications Services	IETF RFC 1995 - Incremental Zone Transfer in DNS, August 1996.	Emerging
Platform Communications Services	IETF RFC 1996 - A Mechanism for Prompt Notification of Zone Changes (DNS NOTIFY), August 1996.	Emerging
Platform Communications Services	IETF RFC 2136 - Dynamic Updates in the Domain Name System, April 1997.	Mandated
Platform Communications Services	IETF RFC 2231 - MIME Parameter Value and Encoded Word Extensions: Character Sets, Languages, and Continuations, November 1997.	Emerging
Platform Communications Services	IETF RFC 2646 - The Text/Plain Format Parameter, August 1999.	Emerging
Platform Communications Services	IETF RFC 2821 - Simple Mail transfer Protocol, April 2001.	Mandated
Platform Communications Services	IETF RFC 2822 - Internet Message Format, April 2001.	Mandated
Platform Communications Services	IETF RFC 3023 - XML Medi Types, April 2001.	Emerging
Platform Communications Services	IETF RFC 3377 - Lightweight Directory Access Protocol (v3): Technical Specification; September 2002	Mandated
Platform Communications Services	IETF RFCs 2045-2049 - Multipurpose Internet Mail Extensions (MIME) Parts 1-5, November 1996.	Mandated
Platform Communications Services	IETF Standard 13/RFC 1034/RFC 1035 - Domain Name System, November 1987.	Mandated
Platform Communications Services	ITU-T X.500 - The Directory: Overview of Concepts, Models, and Services - Data Communication Networks Directory, 1993.	Mandated [SUNSET]
Platform Communications Services	ITU-T X.500:2001 - The Directory: Overview of Concepts, Models, and Services - Data Communication Networks Directory, February 2001.	Emerging

IT Profile: QOS

IT Description: Networking QOS standards for Teleport derived from the questionnaire profiling method

Last Updated: 2004-09-15

Service Area	Standard ID & Title of Standard	Status
Transport-Oriented (quality of service)	IEEE 802.1p - LAN Layer 2 Qos/Cos Protocol for Traffic Prioritization. Sep 21,1995	Emerging
Transport-Oriented (quality of service)	IEEE 802.1q - Local and Metropolitan Area Networks: Virtual Bridge Local Area Networks, 1998.	Emerging
Transport-Oriented (quality of service)	IETF RFC 2205 - Resource ReSerVation Protocol RSVP Version 1 Functional Specification, September 1997.	Emerging
Transport-Oriented (quality of service)	IETF RFC 2207 - RSVP Extensions for IPSEC Data Flows, September 1997.	Emerging
Transport-Oriented (quality of service)	IETF RFC 2210 - The Use of RSVP with IETF Integrated Services, September 1997.	Emerging
Transport-Oriented (quality of service)	IETF RFC 2474 - Definition of the Differentiated Services Field (DS Field) in the IPv4 and IPv6 Headers, December 1998.	Mandated
Transport-Oriented (quality of service)	IETF RFC 3031 - Multi-protocol Label Switching Architecture, January 2001.	Emerging

Transport-Oriented (quality of service)	IETF RFC 3168 - The Addition of Explicit Congestion Notification (ECN) to IP, September 2001.	Emerging
Transport-Oriented (quality of service)	IETF RFC 3175 - Aggregation of RSVP for IPv4 and IPv6 Reservations, September 2001.	Emerging
Transport-Oriented (quality of service)	ISO/IEC 15802-3 - Local and Metropolitan Area Networks - Common Specifications - Part 3: Media Access Control (MAC) Bridges, 1998.	Emerging

---

IT Profile: SATCOM

IT Description: SATCOM standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-09-28

Service Area	Standard ID & Title of Standard	Status
Satellite Communications	ISO 11754 (CCSDS 101.0-B-4) - Space Data and Information Transfer System - Telemetry Channel Coding, 1994.	Mandated
Satellite Communications	MIL-STD-1582D(2) - EHF LDR Uplinks and Downlinks, 30 September 1996; with Notice of Change 1, 14 February 1997, and Notice of Change 2, 17 February 1999.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-136A(2) - EHF MDR Uplinks and Downlinks, 8 June 1998; with Notice of Change 1, 1 July 1999, and Notice of Change 2, 30 October 2000.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-164A - Interoperability of SHF Satellite Communications Earth Terminals, 15 April 2002.	Mandated
Satellite Communications	MIL-STD-188-165A - Interoperability of SHF Satellite Communications PSK Modems (FDMA Operations), 15 April 2002.	Mandated
Satellite Communications	MIL-STD-188-166 - Interface Standard, Interoperability and Performance Standards for SHF SATCOM Link Control.	Emerging
Satellite Communications	MIL-STD-188-167 - Interface Standard, Message Format for SHF SATCOM Link Control.	Emerging
Satellite Communications	MIL-STD-188-168 - Interoperability Standard for SHF Satellite Communications Baseband Equipment, 3 October 2002.	Mandated
Satellite Communications	MIL-STD-188-170 - Interoperability and Performance Standard for SHF Satellite Communications Anti-Jamming Modems.	Emerging
Satellite Communications	MIL-STD-188-181B(1) - Interoperability Standard for Single Access 5-Khz and 25-Khz UHF Satellite Communications Channels, 20 March 1999, with Notice of Change 1, 16 October 2001.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-182A(3) - Interoperability Standard for 5-Khz UHF DAMA Terminal Waveform, 31 March 1997, with Notice of Change 1, 9 September 1998; Notice of Change 2, 22 January 1999; and Notice of Change 3, 4 June 1999.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-182B - Interoperability and Performance Standard for UHF SATCOM DAMA Orderwire Messages and Protocols.	Emerging
Satellite Communications	MIL-STD-188-183A(2) - Interoperability Standard for 25-Khz TDMA/DAMA Terminal Waveform, 20 March 1998; with Notice of Change 1, 9 September 1998; and Notice of Change 2, 4 June 1999.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-183B - Interoperability and Performance Standard for Multiple Accessing 5-KHz and 25 KHz UHF SATCOM Channels.	Emerging

Satellite Communications	MIL-STD-188-184(1) - Interoperability and Performance Standard for the Data Control Waveform, 20 August 1993, with Notice of Change 1, 9 September 1998.	Mandated [SUNSET]
Satellite Communications	MIL-STD-188-184A - Interoperability and Performance Standard for the Data Control Waveform.	Emerging
Satellite Communications	MIL-STD-188-185(2) - DoD Interface Standard, Interoperability of UHF MILSATCOM DAMA Control System, 29 May 1996, with Notice of Change 1, 1 December 1997; and Notice of Change 2, 9 September 1998.	Mandated [SUNSET]

IT Profile: Security - Architectures

IT Description: Security architecture standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Architectures and Applications	Application-level Firewall - Basic - U.S. DoD Application-level Firewall Protection Profile for Basic Robustness Environments, Version 1.0, June 2000.	Emerging
Architectures and Applications	Application-level Firewall -Medium:2000 - U.S. DoD Application-level Firewall Protection Profile for Medium Robustness Environments, Version 1.0, 28 June 2000.	Emerging
Architectures and Applications	draft-ietf-idwg-beep-idxp-04.txt - Intrusion Detection Exchange Protocol (IDXP), 11 September 2001.	Emerging
Access Control	FIPS Pub 140-2 - Security Requirements for Cryptographic Modules, 25 May 2001.	Mandated
Access Control	FIPS Pub 180-1 - Secure Hash Algorithm-1, April 1995.	Mandated
Access Control	FIPS Pub 46-3:1999 - Data Encryption Standard, 25 October 1999.	Mandated [SUNSET]
Architectures and Applications	IDS Analyzer:2000 - Intrusion Detection System Analyzer Protection Profile, Draft 3, 15 September 2000.	Emerging
Architectures and Applications	IDS Scanner:2000 - Intrusion Detection System Scanner Protection Profile, Draft 3, 15 September 2000.	Emerging
Architectures and Applications	IDS Sensor:2000 - Intrusion Detection System Sensor Protection Profile, Draft 3, 15 September 2000.	Emerging
Architectures and Applications	IETF RFC 2314 - PKCS #10, Certification Request Syntax, Version 1.5, March 1998.	Emerging
Architectures and Applications	IETF RFC 2315 - Public Key Cryptography Standard (PKCS) #7, Cryptographic Message Syntax, Version 1.5, March 1998.	Emerging
Architectures and Applications	IETF RFC 2459 - Internet X.509 Public Key Infrastructure Certificate and CRL Profile, January 1999, as profiled by TWG-98-07.	Emerging
Architectures and Applications	IETF RFC 2559 - Internet X.509 Public Key Infrastructure Operational Protocols: LDAPv2, April 1999.	Emerging
Architectures and Applications	IETF RFC 2587 - Internet X.509 Public Key Infrastructure LDAPv2 Schema, June 1999.	Emerging
Access Control	IETF RFC 2743 - Generic Security Service Application Program Interface, Version 2, Update 1, January 2000.	Emerging
Access Control	ITU-T Rec. X.509:2000 - The Directory: Public-Key and Attribute Certificate Frameworks, 2001, with Technical Corrigendum 1:2002, and Technical Corrigendum 2:2002. (ISO/IEC 9594-8.2001)	Mandated

Architectures and Applications	RSA Labs PKCS #11:1999 - Cryptographic Token Interface Standard, Version 2.10, December 1999.	Emerging
Architectures and Applications	RSA Labs PKCS #12:1999 - Personal Information Exchange Syntax Standard, Version 1.0, RSA, 24 June 1999.	Emerging
Architectures and Applications	RSA Labs PKCS #15:2000 - Cryptographic Token Information Format Standard, Version 1.1, RSA, 6 June 2000.	Emerging
Architectures and Applications	Traffic Filtering Firewall - Low Risk - U.S. Government Traffic Filter Firewall Protection Profile for Low Risk Environments, Version 1.1, April 1999.	Emerging
Architectures and Applications	Traffic Filtering Firewall - Medium Robustness 1.4 - U.S. DoD Traffic Filter Firewall Protection Profile for Medium Robustness Environments, Version 1.4, 1 May 2000.	Emerging
Architectures and Applications	TWG-98-07:2002 - TWG-98-07, DoD Certificate Policy, Version 6, 31 May 2002.	Emerging
Architectures and Applications	VPN Protection Profile - Virtual Private Network Protection Profile for Protecting Sensitive Information, Version 1.0, 26 February 2000.	Emerging

---

IT Profile: Security - Authentication

IT Description: Authentication standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Authentication	IETF RFC 1510 - The Kerberos Network Authentication Service, Version 5, 10 September 1993.	Mandated
Authentication	IETF RFC 2138 - Remote Authentication Dial in User Service (RADIUS), April 1997.	Emerging

---

IT Profile: Security - Crypto key mgt

IT Description: Crypto key management standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Network Technologies	IEEE 802.10a:1999 - Supplement to Standard for Interoperable LAN/MAN Security (SILS) - Security Architecture Framework (Clause 1), 22 March 1999.	Emerging
Cryptographic Key Management	IETF RFC 2104 - HMAC: Keyed-Hashing for Message Authentication, February 1997.	Mandated
Cryptographic Key Management	IETF RFC 2401 - Security Architecture for the Internet Protocol, November 1998.	Mandated
Cryptographic Key Management	IETF RFC 2402 - IP Authentication Header, November 1998.	Mandated
Cryptographic Key Management	IETF RFC 2404 - The Use of HMAC-SHA-1-96 Within ESP and AH, November 1998.	Mandated
Cryptographic Key Management	IETF RFC 2406 - IP Encapsulating Security Payload (ESP), November 1998.	Mandated
Cryptographic Key Management	IETF RFC 2407 - The Internet IP Security Domain of Interpretation for ISAKMP, Internet Draft, November 1998.	Mandated
Cryptographic Key	IETF RFC 2408 - Internet Security Association and Key	Mandated

Management	Management Protocol (ISAKMP), November 1998.	
Cryptographic Key Management	IETF RFC 2409 - The Internet Key Exchange (IKE), November 1998.	Mandated
Cryptographic Key Management	IETF RFC 2420 - The PPP Triple-DES Encryption Protocol (3DESE), September 1998.	Emerging

---

IT Profile: Security - IS System Management

IT Description: Information system security management standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Information System Security Management	ISO/IEC 15408 - Evaluation Criteria for IT Security (parts 1 through 3), 1 December 1999 .	Mandated

---

IT Profile: Security - Protocols

IT Description: Security protocol standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Access Control	ACP 120A - Allied Communications Publication 120, Common Security Protocol (CSP), Rev A, 7 May 1998.	Mandated [SUNSET]
Security Protocols	draft-ietf-secsh-architecture-13.txt - Secure Shell (SSH) Protocol Architecture,23 September 2002.	Emerging
Security Protocols	IETF RFC 2228 - File Transfer Protocol, October 1997.	Emerging
Security Protocols	IETF RFC 2246 - The Transport Layer Security (TLS) Protocol Version 1.0, January 1999.	Mandated
Access Control	IETF RFC 2633 - S/MIME Version 3, Message Specification, June 1999.	Mandated
Security Protocols	SSL - Secure Sockets Layer (SSL) Protocol, Version 3.0, 18 November 1996.	Mandated [SUNSET]

IT Profile: System Mgt Services

IT Description: Network management standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
System Management Services	ANSI T1.204:1997 - OAM&P - Lower Layer Protocols for TMN Interfaces Between Operations Systems and Network Elements, 1997.	Mandated [SUNSET]
System Management Services	ANSI T1.208:1997 - OAM&P - Upper Layer Protocols for TMN Interfaces Between Operations Systems and Network Elements, 1997.	Mandated [SUNSET]
System Management Services	CIM 2.2 - Common Information Model (CIM) Version 2.2, Distributed Management Task Force, Inc., 14 June 1999.	Emerging
System Management Services	CIM HTTP - Specification for CIM Operations over HTTP Version 1.0, Distributed Management Task Force, Inc., 11 August 1999.	Emerging

System Services	Management	CIM Schema 2.5 - Common Information Model (CIM) Version 2.2, Distributed Management Task Force, Inc., 12 June 2001.	Emerging
System Services	Management	CIM XML - Specification for the Representation of CIM in XML Version 2.0, Distributed Management Task Force, Inc., 20 July 1999.	Emerging
System Services	Management	DMI 2.0 - Desktop Management Interface v2.0s Specification, Distributed Management Task Force, Inc., 24 June 1998.	Emerging
System Services	Management	IETF RFC 1471 - Definitions of Managed Objects for the Link Control Protocol of the Point-To-Point Protocol, June 1993.	Emerging
System Services	Management	IETF RFC 1472 - Definitions of Managed Objects for the Security Protocol of the Point-to-Point Protocol, 8 June 1993.	Emerging
System Services	Management	IETF RFC 1473 - Definitions of Managed Objects for the IP Network Control Protocol of the Point-to-Point Protocol, June 1993.	Emerging
System Services	Management	IETF RFC 1474 - Definitions of Managed Objects for the Bridge Network Control Protocol of the Point-to- Point Protocol, 8 June 1993.	Emerging
System Services	Management	IETF RFC 1611 - DNS Server MIB Extensions, 17 May 1994.	Emerging
System Services	Management	IETF RFC 1612 - DNS Resolver MIB Extensions, 17 May 1994.	Emerging
System Services	Management	IETF RFC 1657 - Definitions of Managed Objects for the Fourth Version of the Border Gateway Protocol (BGP-4) using SMIv2, 21 July 1994.	Emerging
System Services	Management	IETF RFC 1850 - Open Shortest Path First (OSPF) Version 2 Management Information Base, November 1995.	Mandated
System Services	Management	IETF RFC 2006 - The Definitions of Managed Objects for IP Mobility Support using SMIv2, 22 October 1996.	Emerging
System Services	Management	IETF RFC 2011 - SNMPv2 Management Information Base for the Internet Protocol, November 1996.	Emerging
System Services	Management	IETF RFC 2012 - SNMPv2 Management Information Base for the Transmission Control Protocol (TCP), 12 November 1996.	Emerging
System Services	Management	IETF RFC 2013 - SNMPv2 Management Information Base for the User Datagram Protocol (UDP), 12 November 1996.	Emerging
System Services	Management	IETF RFC 2021 - Remote Network Monitoring Management Information Base Version 2, using SMIv2, 16 January 1997.	Emerging
System Services	Management	IETF RFC 2515 - Definitions of Managed Objects for ATM Management, February 1999.	Emerging
System Services	Management	IETF RFC 2571 - An Architecture for Describing SNMP Management Frameworks, April 1999.	Emerging
System Services	Management	IETF RFC 2572 - Message Processing and Dispatching for the Simple Network Management Protocol (SNMP), April 1999.	Emerging
System Services	Management	IETF RFC 2573 - SNMP Applications, April 1999.	Emerging
System Services	Management	IETF RFC 2574 - User-based Security Model (USM) for Version 3 of the Simple Network Management Protocol (SNMPv3), April 1999.	Emerging
System Services	Management	IETF RFC 2575 - View-based Access Control Model (VACM) for the Simple Network Management Protocol (SNMP), April 1999.	Emerging
System Services	Management	IETF RFC 2605 - Directory Server Monitoring MIB, June 1999.	Emerging
System Services	Management	IETF RFC 2788 - Network Services Monitoring MIB, March 2000.	Emerging

System Management Services	IETF RFC 2789 - Mail Monitoring MIB, March 2000.	Emerging
System Management Services	IETF RFC 2790 - Host Resources MIB, March 2000.	Mandated
System Management Services	IETF RFC 3060 - Policy Core Information Model 6 Version 1 Specification, February 2000.	Emerging
System Management Services	IETF Standard 15/RFC 1157 - Simple Network Management Protocol (SNMP), May 1990.	Mandated
System Management Services	IETF Standard 16/RFC 1155/RFC 1212 - Structure of Management Information (SMI), May 1990.	Mandated
System Management Services	IETF Standard 17/RFC 1213 - Management Information Base, March 1991.	Mandated
System Management Services	IETF Standard 50/RFC 1643 - Definitions of Managed Objects for the Ethernet-like Interface Types, July 1994.	Mandated
System Management Services	IETF Standard 59/RFC 2819 - Remote Network Monitoring Management Information Base, November 1995.	Mandated
System Management Services	ITU-T M.3400:2000 - TMN Management Functions, 2000.	Mandated [SUNSET]

IT Profile: Teleport - Multimedia processing

IT Description: Multimedia processing standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-09-13

Service Area	Standard ID & Title of Standard	Status
Multimedia Processing	ISMA 1.0:2001 - Internet Streaming Media Alliance Standard, ISMA 1.0	Emerging

IT Profile: Transaction Processing

IT Description: Transaction processing and electronic records management standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-09-13

Service Area	Standard ID & Title of Standard	Status
Transaction Processing	DoD 5015.2-STD:2002 - Design Criteria Standard for Electronic Records Management Software Applications, 19 June 2002 (Sections 2.2.1-2.2.1.1 only).	Emerging

IT Profile: User Interface Services

IT Description: User application interface standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-09-13

Service Area	Standard ID & Title of Standard	Status
User Interface Services	Win32 APIs - Window Management and Graphics Device Interface, Volume 1 Microsoft Win32 Programmers Reference Manual, 1993, Microsoft Press.	Mandated

IT Profile: VTC

IT Description: VTC standards for Teleport derived from questionnaire profiling method

Last Updated: 2004-09-13

Service Area	Standard ID & Title of Standard	Status
Video Teleconferencing	IETF RFC 3261 - Session Initiation Protocol, June 2002.	Emerging
Video Teleconferencing	IETF RFC 3435 - Media Gateway Control Protocol (MGCP) Version 1.0, January 2003.	Emerging
Video Teleconferencing	ITU-T G.711 - Pulse Code Modulation (PCM) of Voice Frequencies, November 1988.	Mandated
Video Teleconferencing	ITU-T G.722 - 7 KHz Audio-Coding within 64 kbit/s, November 1988.	Mandated
Video Teleconferencing	ITU-T G.728 - Coding of Speech at 16 kbits/s using Low-Delay Code Excited Linear Prediction (LD-CELP)., September 1992.	Mandated
Video Teleconferencing	ITU-T H.225.0 - Call Signaling Protocols and Media Stream Packetization for Packet-based Multimedia Communications Systems, February 1998.	Mandated
Video Teleconferencing	ITU-T H.245 - Control Protocol for Multimedia Communications, February 1998.	Mandated
Video Teleconferencing	ITU-T H.248 - Gateway Control Protocol, June 2000.	Emerging
Video Teleconferencing	ITU-T H.261 - Video CODEC for Audiovisual Services at p x 64 Kbps, March 1993	Mandated [SUNSET]
Video Teleconferencing	ITU-T H.264 - Advanced Video Coding, July 2002. (ITU-T H.264/ISO/ IEC FCD 14496-10)	Emerging
Video Teleconferencing	ITU-T H.323 v2 - Packet-based Multimedia Communications Systems, (Version 2), February 1998.	Mandated
Video Teleconferencing	ITU-T H.323:2000 - Packet-based Multimedia Communications Systems, November 2000.	Emerging
Video Teleconferencing	ITU-T T.120 - Transmission Protocols for Multimedia Data, July 1996.	Mandated
Video Teleconferencing	ITU-T T.122:1998 - Multipoint Communications Service - Service Definition, February 1998.	Mandated
Video Teleconferencing	ITU-T T.123:1999 - Network - Specific Data Protocol Stacks Multimedia Conferencing, May 1999.	Mandated
Video Teleconferencing	ITU-T T.124:1998 - Generic Conference Control, February 1998	Mandated
Video Teleconferencing	ITU-T T.125:1998 - Multipoint Communications Service Protocol Specification, February 1998.	Mandated
Video Teleconferencing	ITU-T T.126:1997 - Multipoint Still Image and Annotation Protocol, July 1997.	Mandated
Video Teleconferencing	ITU-T T.127 - Multipoint Binary File Transfer Protocol, August 1995.	Mandated
Video Teleconferencing	ITU-T T.128 - Multipoint Application Sharing, February 1998.	Mandated
Video Teleconferencing	ITU-T T.81 - Digital Compression and Coding of Continuous-tone Still Images - Requirements and Guidelines, September 1992.	Mandated
Video Teleconferencing	ITU-T T.82 - Coded Representation of Picture and Audio Information - Progressive Bi-level Image Compression, March 1993.	Mandated

---

IT Profile: Web Services

IT Description: Web service standards for Teleport TV-1/TV-2 derived from the questionnaire profiling method

Last Updated: 2004-10-14

Service Area	Standard ID & Title of Standard	Status
Network Technologies	IETF RFC 1738 - Uniform Resource Locators (URL), 20 December 1994.	Mandated
Network Technologies	IETF RFC 2396 - Uniform Resource Identifiers (URI), Generic Syntax, August 1998.	Mandated
Network Technologies	IETF RFC 2616 - Hypertext Transfer Protocol - HTTP 1.1, June 1999.	Mandated

A13 Level of Information System Interoperability (LISI) Profile

The LISI provides the DOD with a framework to measure and assess information systems interoperability in context with discrete levels of information-exchange sophistication. The LISI profile was created using the Joint C4I Program Assessment Tool *InspeQtor* and can be found at http://lisi.ncr.smil.mil.

APPENDIX B: REFERENCES

a. "Operational Requirements Document (ORD) for Department of Defense (DOD) Teleport", S/N 567-06-00, 31 July 2000

b. "Operational Requirements Document (ORD) for Department of Defense (DOD) Teleport Generation Two Update", JROCM 116-04, 9 July 2004.

c. CJCSI 3170.01 Series, "Joint Capabilities Integration and Development System."

d. CJCSI 6212.01 Series, "Interoperability and Supportability of Information Technology and National Security Systems."

e. CJCSI 6250.01 Series, "Satellite Communications."

f. "DOD Architecture Framework", Version 1.0 Final Draft, 30 August 2003.

g. DOD Directive 5000.1, 12 May 2003, "The Defense Acquisition System."

h. DOD Instruction 5000.2, 12 May 2003, "Operation of the Defense Acquisition System."

i. DOD Directive 8500.1, 24 October 2002, "Information Assurance."

j. DOD Instruction 8500.2, 2 February 2003, "Information Assurance (IA) Implementation."

k. DOD Directive 8100.1, 19 September 2002, "Global Information Grid (GIG) Overarching Policy."

l. DOD Directive 4630.5 Series, "Interoperability and Supportability of Information Technology (IT) and National Security Systems (NSS)."

m. DOD Directive 4630.8 Series, "Procedures for Interoperability and Supportability of Information Technology and National Security Systems (NSS)."

n. "Phase Three System Security Authorization Agreement (SSAA) for the DOD Teleport System, Generation One – IOC1", April 2004.

o. "DOD Global Information Grid (GIG) Teleport System Generation One (IOC1) Interface Control Document", September 2002.

p. "DOD Global Information Grid (GIG) Teleport System Generation One (IOC1) System View (SV) to Technical View (TV) Bridge and TV-1 and TV-2", September 2002.

q. Cryptographic Modernization (CM) Capstone Requirements Document (CRD), JROCM 009-01, 20 January 2004.

r. Joint Staff (VJ6) Memorandum, "Satellite Throughput and Defense Information Services Network Requirements for DOD Teleport Generation Two (Gen 2)", 11 September 2003.

s. Global Information Grid (GIG) Architecture Version 2.0, August 2003.

t. Department of Defense (DOD) Information Technology Standards Registry (DISR), 15 July 2004.

u. Joint Concept of Operations for Global Information Grid NetOps, 5 May 2004.

v. DOD Instruction 8551.1, "Ports, Protocols, and Services Management", 13 August 2004.

w. National Security Telecommunications and Information Systems Security Advisory Memorandum (NSTISSAM) Computer Security (COMPUSEC) #11, "Transition from the Trust Computer System Evaluation Criteria to the International Common Criteria for Information Technology Security Evaluation", 11 March 1999.

x. National Security Telecommunications and Information Systems Security Policy (NSTISSP) No. 11, "National Information Assurance Acquisition Policy", January 2000.

y. General Accounting Office, "Defense Acquisitions: Knowledge of Software Suppliers Needed to Manage Risks", May 2004.

APPENDIX C: ACRONYMS

AOR	Area of Responsibility
BIT	Built in Test
C4ISP	Command, Control, Communications, Computers, and Intelligence Support Plan
CDD	Capabilities Development Document
CNR	Combat Net Radios
CMP	Configuration Management Plan
COCOM	Combatant Command
CSB	Commercial SATCOM Branch
DCGS	Distributed Common Ground System
DISA	Defense Information Systems Agency
DISN	Defense Information System Network
DISR	DOD Information Technology Standards Registry
DOD	Department of Defense
DRSN	Defense Red Switched Network
DSCS	Defense Satellite Communications System
DSN	Defense Switched Network
EHF	Extremely High Frequency
FOC	Final Operating Capability
GAA	Gateway Access Authorization
GAR	Gateway Access Request
GIG	Global Information Grid
GMF	Ground Mobile Forces
GNC	Global NetOps Center
GNSC	Global NetOps Support Center
GNOSC	Global Network Operations and Security Center
GSSC	Global Satellite Communications Support Center
IA	Information Assurance
INMS	Integrated Network Management System
IOC	Initial Operating Capability
ISP	Information Support Plan
JCCC	Joint Communications Control Center
JFTOC	Joint Fleet Tactical Operations Center
JNMS	Joint Network Management System
JTRS	Joint Tactical Radio System
JTF	Joint Task Force
JTF-GNO	Joint Task Force – Global Network Operations

JWICS	Joint Worldwide Intelligence Communications System
KIP	Key Interface Profile
KPP	Key Performance Parameter
M&C	Management and Control
MILSATCOM	Military Satellite Communications
MSE	Mobile Subscriber Equipment
MUOS	Mobile User Objective System
NAVSOC	Naval Satellite Operations Center
NCOW RM	Net-centric Operations Warfare Reference Model
NCTAMS	Naval Computer and Telecommunications Area Master Station
NTCS	Naval Computer and Telecommunications Station
NIPRNET	Nonclassified Internet Protocol Router Network
NR KPP	Net Ready Key Performance Parameter
NOC	Network Operations Center
ORD	Operational Requirements Document
OV	Operational View
RF	Radio Frequency
RFS	Request for Service
RNOSC	Regional Network Operations and Security Center
RSSC	Regional Satellite Communications Support Center
SAA	Satellite Access Authorization
SAR	Satellite Access Request
SATCOM	Satellite Communications
SCOC	Systems Control and Operations Concept
SHF	Super High Frequency
SIPRNET	Secure Internet Protocol Router Network
SOM	SATCOM Operations Manager
SV	Systems View
SYSCON	Systems Control Center
TCCC	Theater Communications Control Center
TNC	Theater NetOps Center
TV	Technical View
UHF	Ultra High Frequency
VTC	Video Teleconference Center